

PD-LMI package User Manual

Version v1.2.2

July 30, 2026

Abstract

The PD-LMI package is a MATLAB/YALMIP package for modelling and certifying parameter-dependent and differentiable parameter-dependent linear matrix inequalities (PD-LMIs and DPD-LMIs) over axis-aligned parameter and rate boxes. It represents coefficient-backed known matrices and continuous decision matrices as cell-local tensor-product Bernstein polynomials on an independently partitioned physical parameter grid. This finite representation supports exact cellwise coefficient algebra and differentiation, while Direct, Pólya, Putinar, SparseFullBox, and FullBox constructions turn the remaining continuum sign condition into inspectable YALMIP constraints.

The intended workflow is direct: define the tensor grid and coefficient-backed known data with `pdmatrix`; create continuous polynomial decisions with `pdvar`; use `rhodiff` when scheduling-rate terms are present; form an affine residual comparison; optionally replace the default Direct certificate by one of the four supported alternatives; call `toYalmip`; then solve with YALMIP and recover values only after checking the solver status. The manual develops each step from runnable examples through the underlying tensor convolution, degree alignment, rate-row, and Gram-assembly mathematics.

The representation keeps four ideas distinct: the tensor grid exactly partitions the parameter box; the chosen piecewise-polynomial decision space is a finite modelling restriction; rate-vertex reduction is exact because rate dependence is affine; and each finite certificate is a sufficient condition over the parameter continuum.

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Version History

The entries below identify the current tagged GitHub Release and retain the final documentation snapshot or latest release from each completed minor line.

v1.2.0 — July 29, 2026 — current and latest GitHub Release

Adds direction-wise Bernstein degrees to `pdbase`, `pdmat`, and `pdvar`; scalar-or-vector elevation and componentwise degree algebra; exact multivariate `rhodiff` alignment with degree-zero-axis support; and scalar-or-vector Pólya, Putinar, SparseFullBox, and FullBox certificate parameters. Public degrees and orders are stored as row vectors, tensor counts use the product of per-axis cardinalities, and SparseFullBox retains a scalar bandwidth with anisotropic endpoint normalization.

v1.1.6 documentation — July 28, 2026 — final v1.1 snapshot

Unifies tensor-grid, cell, Bernstein-degree, residual, target, and Gram notation across the printable and Web manuals; states SparseFullBox directly through the paper notation $Q_J^{(w)}$ for symmetric block-band Gram matrices of bandwidth w ; expands the Bernstein convolution and Gram/SOS foundations; adds the Math Concepts navigation and interactive `pdmat` addition, multiplication, and elevation explorers; and fits the Welcome three-dimensional grid at every supported rotation. The MATLAB API is unchanged.

v1.0.0 — July 19, 2026

Marks the first stable public release of the documented package workflow. It consolidates installation and test-suite verification, the `pdbase/pdmat/pdvar/pdlmi` object model, continuous arbitrary-degree decision storage and rate-vertex differentiation, MATLAB-style matrix operations and indexing, and direct, Pólya, Putinar, and full-box finite certificate assembly. The 1.0 documentation also makes the affine-expression, tensor-box, certificate-sufficiency, and solver-ownership boundaries explicit; this milestone does not add new runtime behavior beyond the implementation covered by the current code and tests.

v0.4.7 — July 18, 2026

Completed the v0.4 manual and Web explorer refresh around the LPV-to-cellwise-Bernstein-to-solver workflow, including installer guidance, the public API reference, mathematical cross-references, and storage-transformation examples. Released by commit `c6a2e25`.

v0.3.6 — July 16, 2026

Completed the public rename to `pdbase/pdmat/pdvar/pdlmi`, corrected forward Bernstein coordinates, added FullBox and Putinar certificates with independent SOS validation, hardened API boundaries, and defined entry-wise inequality behavior. Released by commit `fe74a65`.

v0.2.12 — July 13, 2026

Completed the `pdmat/pdvar/pdlmi` layers, rate-aware differentiation, display, plotting, structural algebra, direct finite assembly, Pólya selection, and deterministic reference examples and tests. Released by commit `98b63e5`.

v0.1.0 — July 3, 2026

Introduced the original grid-and-storage foundation and the first package manual, including Bernstein traversal, validation, and helper contracts. Released by commit `63c06ee`.

Chapter 1

From a Parameter Box to Finite Constraints

Parameter-dependent LMIs require a matrix inequality to hold throughout a continuous parameter domain, rather than at a single operating point. The package represents the known data and decision functions on a parameter box with cellwise Bernstein polynomials, then converts the resulting continuum condition into finite constraints for YALMIP. An LPV induced- L_2 -gain problem provides the first example; the sections that follow develop the general differentiable parameter-dependent LMI (DPD-LMI) and its finite certificate construction. The supporting algebra is developed in Appendices A and B, and the API contracts are collected in Chapters 4–6.

1.1 LPV Induced- L_2 -Gain Motivation

Consider the LPV plant

$$\begin{aligned}\dot{x}(t) &= A(\boldsymbol{\rho}(t))x(t) + B_w(\boldsymbol{\rho}(t))w(t), \\ z(t) &= C_z(\boldsymbol{\rho}(t))x(t) + D_{zw}(\boldsymbol{\rho}(t))w(t),\end{aligned}\tag{1.1}$$

where $x \in \mathbb{R}^{n_x}$, $w \in \mathbb{R}^{n_w}$, $z \in \mathbb{R}^{n_z}$, and the measurable scheduling trajectory satisfies $(\boldsymbol{\rho}(t), \dot{\boldsymbol{\rho}}(t)) \in \mathcal{P} \times \mathcal{R}$. For $\gamma > 0$, the zero-state induced- L_2 objective is to certify

$$\|z\|_2 < \gamma \|w\|_2$$

for every nonzero admissible disturbance and every admissible scheduling trajectory.

A differentiable matrix $P(\boldsymbol{\rho}) = P(\boldsymbol{\rho})^\top$ supplies the scaled bounded-real DPD-LMI [1]

$$P(\boldsymbol{\rho}) \succ 0, \tag{1.2}$$

$$\begin{bmatrix} \dot{P}(\boldsymbol{\rho}, \dot{\boldsymbol{\rho}}) + \text{He}(P(\boldsymbol{\rho})A(\boldsymbol{\rho})) & P(\boldsymbol{\rho})B_w(\boldsymbol{\rho}) & C_z(\boldsymbol{\rho})^\top \\ B_w(\boldsymbol{\rho})^\top P(\boldsymbol{\rho}) & -\gamma I_{n_w} & D_{zw}(\boldsymbol{\rho})^\top \\ C_z(\boldsymbol{\rho}) & D_{zw}(\boldsymbol{\rho}) & -\gamma I_{n_z} \end{bmatrix} \prec 0 \tag{1.3}$$

for every $(\boldsymbol{\rho}, \dot{\boldsymbol{\rho}}) \in \mathcal{P} \times \mathcal{R}$, where $\text{He}(X) = X + X^\top$ and

$$\dot{P}(\boldsymbol{\rho}, \dot{\boldsymbol{\rho}}) = \sum_{s=1}^{\ell} \dot{\rho}_s \frac{\partial P}{\partial \rho_s}(\boldsymbol{\rho}).$$

The first inequality makes $x^\top P(\boldsymbol{\rho})x$ positive definite, and the block inequality supplies the dissipation condition for the gain bound. This application is one instance of the general derivative-dependent residual below. The strict signs are analysis notation: the package uses non-strict semidefinite comparisons and inserts no hidden numerical margin.

1.2 General Problem and Assumptions

Let $y = (y_1, \dots, y_N)$ collect scalar parameter-dependent decision functions. The general task is to satisfy the single residual

$$\begin{aligned} \mathcal{F}(\boldsymbol{\rho}, \dot{\boldsymbol{\rho}}; y) &= F_0(\boldsymbol{\rho}) + \sum_{k=1}^N F_k(\boldsymbol{\rho}) y_k(\boldsymbol{\rho}) \\ &+ \sum_{k=1}^N \sum_{s=1}^{\ell} T_{k,s}(\boldsymbol{\rho}) \dot{\rho}_s \frac{\partial y_k}{\partial \rho_s}(\boldsymbol{\rho}) \preceq 0, \quad \forall (\boldsymbol{\rho}, \dot{\boldsymbol{\rho}}) \in \mathcal{P} \times \mathcal{R}, \end{aligned} \quad (1.4)$$

The package fixes the abstract sets and decision class in (1.4) through the following assumptions.

- The scheduling set and rate set are axis-aligned boxes; general polytopic or curved scheduling domains are outside the implemented model:

$$\mathcal{P} = \prod_{s=1}^{\ell} [\underline{\rho}_s, \bar{\rho}_s], \quad \mathcal{R} = \prod_{s=1}^{\ell} [\underline{v}_s, \bar{v}_s], \quad \mathcal{V}_{\mathcal{R}} = \prod_{s=1}^{\ell} \{\underline{v}_s, \bar{v}_s\}.$$

- The functions F_0 , F_k , and $T_{k,s}$ are known matrix functions of compatible dimensions, and the y_k are scalar decision functions. A matrix-valued decision is represented by vectorizing its independent scalar entries and assigning one component y_k to each entry, so symmetry can be encoded without treating a whole matrix as one scalar decision. Setting every $T_{k,s} = 0$ removes the derivative term and recovers an ordinary derivative-free PD-LMI.
- The theoretical statement uses differentiable y_k . The package searches instead over continuous piecewise-Bernstein functions that are differentiable within each physical cell. Their values agree on shared faces, but their one-sided derivatives may differ; derivative data and the resulting residual conditions remain cell-local at those interfaces.
- Known data used by a certificate have cell-local polynomial coefficient evidence, and the assembled residual remains affine in the underlying YALMIP decisions. Supported numeric or known parameter-dependent factors may multiply a decision expression, but products of two decision-bearing expressions are outside the affine model.
- Dependence on $\dot{\boldsymbol{\rho}}$ is affine. Consequently, checking the 2^{ℓ} vertices in $\mathcal{V}_{\mathcal{R}}$ is exactly equivalent to checking the complete rate box $\mathcal{R}$. This exact reduction is separate from the two finite restrictions over the scheduling variables.
- MATLAB and YALMIP use non-strict semidefinite comparisons. A theoretical strict condition must therefore be modeled with an explicit user-selected margin, for example $-\mathcal{F}(\boldsymbol{\rho}, \dot{\boldsymbol{\rho}}; y) - \varepsilon I \succeq 0$, and failure of a finite certificate is not proof that (1.4) is infeasible.

1.3 Gridding the Parameter Domain: a Separate Two-Dimensional Example

For each parameter direction $s = 1, \dots, \ell$, choose an independent, possibly nonuniform axis grid

$$\mathcal{G}_s = \{\rho_s^{(1)}, \dots, \rho_s^{(k_s)}\}, \quad \underline{\rho}_s = \rho_s^{(1)} < \dots < \rho_s^{(k_s)} = \bar{\rho}_s.$$

Their Cartesian product

$$\mathcal{G} = \mathcal{G}_1 \times \dots \times \mathcal{G}_\ell$$

contains $\prod_{s=1}^\ell k_s$ physical nodes. The multi-index $\mathbf{c} = (c_1, \dots, c_\ell)$, with $c_s \in \{1, \dots, k_s - 1\}$, defines the location of a physical cell $\mathbf{c}$ and its width along s :

$$\mathcal{H}_{\mathbf{c}} = \prod_{s=1}^\ell [\rho_s^{(c_s)}, \rho_s^{(c_s+1)}], \quad h_s^{\mathbf{c}} = \rho_s^{(c_s+1)} - \rho_s^{(c_s)} > 0.$$

Thus $\mathcal{G}$ determines $\prod_s (k_s - 1)$ physical cells whose union is exactly $\mathcal{P}$. The grid does not sample or approximate the box geometry. Gridding is nevertheless a modeling approximation because it restricts the infinite-dimensional search for y to the selected finite-dimensional space of continuous piecewise-polynomial decision functions.

To make this general tensor geometry visible, take the following nonuniform two-dimensional example

$$\mathcal{G}_1 = \{0, 0.4, 1\}, \quad \mathcal{G}_2 = \{0, 0.25, 1\}.$$

Their tensor grid $\mathcal{G} = \mathcal{G}_1 \times \mathcal{G}_2$ has nine physical nodes and exactly partitions $[0, 1]^2$ into four physical cells. The cell indices are $(1, 1)$, $(1, 2)$, $(2, 1)$, and $(2, 2)$.

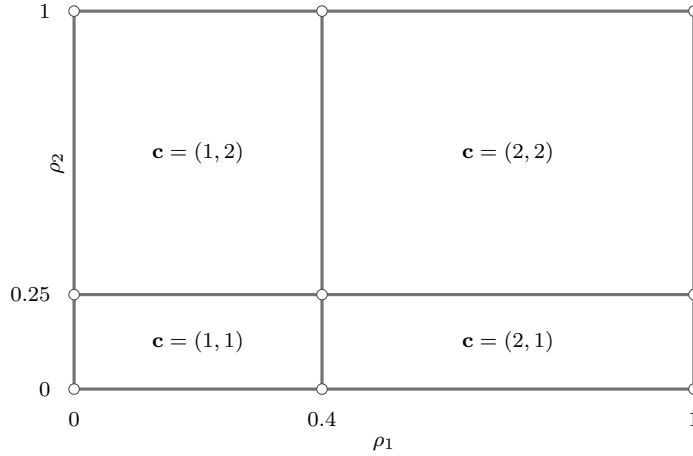


Figure 1.1: A nonuniform two-dimensional tensor grid with nine physical nodes and four physical cells. Every point of the box belongs to at least one cell; the lines partition the domain rather than sample it.

The gridding above fixes nine physical nodes and four physical cells. Within each cell, it has its own local coordinate $\alpha \in [0, 1]^\ell$ and carries a separate finite set of Bernstein coefficients. For example, a tensor-product polynomial of modeled degree $m = 3$ in each direction has sixteen local labels $\mathbf{i} = (i_1, i_2) \in \{0, 1, 2, 3\}^2$. These labels identify coefficients inside the cell; they are not

sixteen additional physical sample points. In general, a degree- m tensor-product basis has $(m+1)^\ell$ local labels per cell. Shared-face coefficient identification makes the represented decision function continuous across neighboring cells. Section A.7 gives the complete traversal convention.

1.4 Bernstein Basis on Each Cell

Fix any physical cell $\mathcal{H}_{\mathbf{c}}$. Its forward affine map from the unit box to the physical cell is

$$\phi_{\mathbf{c}} : [0, 1]^\ell \longrightarrow \mathcal{H}_{\mathbf{c}}, \quad \phi_{\mathbf{c},s}(\alpha_s) = \rho_s^{(c_s)} + h_s^{\mathbf{c}} \alpha_s.$$

Hence $\boldsymbol{\alpha} = \phi_{\mathbf{c}}^{-1}(\boldsymbol{\rho})$, with $\alpha_s = (\rho_s - \rho_s^{(c_s)})/h_s^{\mathbf{c}}$. For any cellwise quantity X , write its pullback as $X^{(\mathbf{c})} = X \circ \phi_{\mathbf{c}}$. The normalized tensor-product Bernstein basis of modeled degree m is

$$B_{\mathbf{i}}^m(\boldsymbol{\alpha}) = \prod_{s=1}^{\ell} \binom{m}{i_s} (1 - \alpha_s)^{m-i_s} \alpha_s^{i_s}, \quad \mathbf{i} \in \{0, \dots, m\}^\ell,$$

and a cell-local polynomial is represented by

$$X^{(\mathbf{c})}(\boldsymbol{\alpha}) = \sum_{\mathbf{i} \in \{0, \dots, m\}^\ell} B_{\mathbf{i}}^m(\boldsymbol{\alpha}) X^{(\mathbf{c})}[\mathbf{i}]. \quad (1.5)$$

The notation is illustrated by the following one-dimensional known-data example. Consider the scalar polynomial

$$a(\rho) = 1 + \rho + \rho^2$$

on the two-cell grid $\mathcal{G}_1 = \{0, 0.5, 1\}$. In this one-dimensional case, $\mathcal{H}_c = [\rho_1^{(c)}, \rho_1^{(c+1)}]$, $h_1^c = \rho_1^{(c+1)} - \rho_1^{(c)}$, and $\phi_c(\alpha) = \rho_1^{(c)} + h_1^c \alpha$, so $\alpha = \phi_c^{-1}(\rho) = (\rho - \rho_1^{(c)})/h_1^c$. Thus c is the one-dimensional physical-cell index, while the Bernstein label remains a separate subscript. Specifying **Degree=2** instructs `pdmat` to retain exact function evaluation together with validated degree-two Bernstein coefficient data.

```
grid = {[0 0.5 1]};
a = pdmat(grid, @(rho) 1 + rho + rho.^2, Degree=2);

physicalCells = a.cells()
localLabels = a.lbls()
cell1Coefficients = cell2mat(a.coeffs(1))
cell2Coefficients = cell2mat(a.coeffs(2))
cell1Table = bernsteinTable(a, 1, "oneLine");
disp(cell1Table)
```

```
physicalCells = 2×1 double
    1
    2

localLabels = 3×1 double
    0
    1
    2

cell1Coefficients = 1×3 double
    1.0000    1.2500    1.7500
```

```
cell2Coefficients = 1×3 double
    1.7500    2.2500    3.0000
```

CellSubscript	Expression
-----	-----
{[1]}	"(1-alpha)^2*1 + 2(1-alpha)alpha*1.25 + alpha^2*1.75"

Fix the first physical cell $\mathcal{H}_1 = [0, 0.5]$. Its local coordinate is $\alpha^{(1)}(\rho) = 2\rho$, and (1.5) becomes

$$a^{(1)}(\alpha) = B_0^2(\alpha) 1 + B_1^2(\alpha) 1.25 + B_2^2(\alpha) 1.75, \quad \alpha = \alpha^{(1)}(\rho).$$

This equality represents $1 + \rho + \rho^2$ throughout the complete cell; the three coefficients are not samples at three new physical grid points. The upper-face coefficient 1.75 of cell 1 equals the lower-face coefficient of cell 2, reflecting continuity of this known polynomial at the shared physical node $\rho = 0.5$. Sections A.3, A.6 and A.8 develop the corresponding higher-degree, elevation, and multiplication rules.

1.5 From a Continuous Cell Condition to Direct

Fix any physical cell $\mathcal{H}_{\mathbf{c}}$ and suppress the already enumerated rate-row index. After all local operations have been aligned to the assembled residual degree $\mathbf{M}$, the pullback of the residual has the general form

$$\mathcal{F}^{(\mathbf{c})}(\boldsymbol{\alpha}) = \sum_{\mathbf{i} \in \prod_s \{0, \dots, M_s\}} B_{\mathbf{i}}^{\mathbf{M}}(\boldsymbol{\alpha}) \mathcal{C}^{(\mathbf{c})}[\mathbf{i}] \prec 0.$$

Here $\boldsymbol{\alpha} = \phi_{\mathbf{c}}^{-1}(\boldsymbol{\rho})$, $B_{\mathbf{i}}^{\mathbf{M}}$ is the direction-wise degree- $\mathbf{M}$ tensor-product Bernstein basis, $\mathbf{i}$ is a zero-based local label, and $\mathcal{C}^{(\mathbf{c})}[\mathbf{i}]$ is the corresponding residual coefficient. Define the sign-normalized positive target and its coefficients by

$$S^{(\mathbf{c})}(\boldsymbol{\alpha}) = -\mathcal{F}^{(\mathbf{c})}(\boldsymbol{\alpha}) = \sum_{\mathbf{i} \in \prod_s \{0, \dots, M_s\}} B_{\mathbf{i}}^{\mathbf{M}}(\boldsymbol{\alpha}) C^{(\mathbf{c})}[\mathbf{i}], \quad C^{(\mathbf{c})}[\mathbf{i}] = -\mathcal{C}^{(\mathbf{c})}[\mathbf{i}].$$

The theoretical target is $S^{(\mathbf{c})}(\boldsymbol{\alpha}) \succ 0$ on the unit box. The Direct certificate imposes the finite non-strict family

$$C^{(\mathbf{c})}[\mathbf{i}] \succeq 0 \quad \text{for every local Bernstein label } \mathbf{i}.$$

Because every $B_{\mathbf{i}}^{\mathbf{M}}$ is nonnegative on the unit box and the basis functions sum to one,

$$[C^{(\mathbf{c})}[\mathbf{i}] \succeq 0 \text{ for all } \mathbf{i}] \implies S^{(\mathbf{c})}(\boldsymbol{\alpha}) \succeq 0 \text{ for all } \boldsymbol{\alpha} \in [0, 1]^\ell.$$

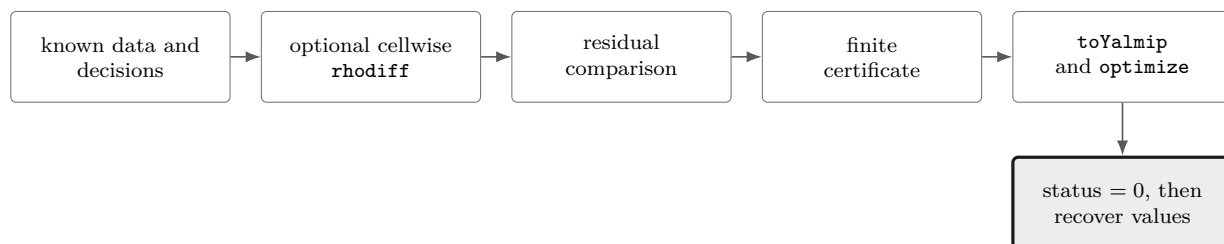
Repeating this implication for every physical cell and every stored rate row produces finitely many YALMIP constraints that suffice for the continuous sign condition. The implication is not a converse: a positive matrix polynomial may have an indefinite Bernstein coefficient, so failure of Direct does not prove that the continuous condition is infeasible. MATLAB and YALMIP comparisons are non-strict; to model the strict theoretical target, the user must impose an explicit margin such as $S^{(\mathbf{c})} - \varepsilon I \succeq 0$.

1.5.1 Opt-in alternatives

Direct is the default finite certificate. The package also provides opt-in [Pólya](#), [Putinar](#), [SparseFullBox](#), and [FullBox](#) alternatives. Their orders, Gram constructions, support rules, and validation boundaries belong to the detailed `pdlmi` chapter and Appendix B; they are intentionally not part of this opening overview. Each alternative remains a sufficient certificate for the parameter continuum rather than a sampling method or a solver call.

1.6 The Complete Path to YALMIP

The full workflow now fits in one panorama: represent known data, create continuous decision functions, differentiate them cell by cell when rate terms are present, assemble and compare a residual, select a finite certificate, export it, solve, validate the status, and only then recover numerical values.



```
yalmip('clear')
grid = {[0 0.5 1]};
rateBounds = [-0.2 0.2];

a = pdmat(grid, @(rho) 1 + rho + rho.^2, ...
    Degree=2, RateBounds=rateBounds);
P = pdvar(1, grid, Degree=2, RateBounds=rateBounds);
dP = rhodiff(P);

residual = P - 0.1 * dP - a;
certificate = residual >= 1e-6;
F = certificate.toYalmip();
opts = sdpsettings('solver', 'mosek', 'verbose', 0);
sol = optimize(F, [], opts);
assert(sol.problem == 0, sol.info)
solvedP = value(P);
```

Here `rhodiff` evaluates the affine derivative term at the vertices of the stored rate box; that vertex reduction is exact. The comparison creates a Direct `pdlmi` by default, and an opt-in certificate selector could be inserted before `toYalmip`. Solver settings and objectives remain ordinary YALMIP choices. Only `sol.problem == 0` supports recovery of `solvedP`; any infeasible, numerical, unknown, or otherwise nonzero status must be handled before retaining decisions or objectives. A failed sufficient certificate is not proof that the original continuum condition is infeasible.

1.7 Manual Outline



Chapter [2](#) covers installation and verification. The `pdbase` chapter explains shared backend geometry, after which Chapters [4](#) and [5](#) develop known data and decision functions. Chapter [6](#) gives the complete certificate-selection and YALMIP-export contracts, the worked examples assemble application-specific residuals and objectives, and Appendices [A](#) and [B](#) derive the Bernstein and certificate mathematics.

Chapter 2

Install and Verify the Package

2.1 One-Call Installation

Start MATLAB in the checked-out repository and run the root-level installer once. The installer adds only the repository root to the end of the MATLAB path; MATLAB resolves the package's class, package, and private directories from that root. It does not recursively scan the checkout, download dependencies, or edit `startup.m`.

```
report = install_pd_lmi();
```

The returned structure records the package root, the YALMIP root discovered on the current MATLAB path, the working SDP solver, the paths added by this run, and whether the path was persisted:

```
report.PackageRoot  
report.YALMIPRoot  
report.Solver  
report.AddedPaths  
report.Persisted
```

YALMIP must already be on the active MATLAB path. The installer validates `yalmip`, `sdpvar`, `sdpsettings`, `optimize`, and `getavailablesolvers`; it then probes available SDP solvers in the fixed commercial-first order MOSEK, COPT, SeDuMi, SDPT3, and LMILAB using a tiny bounded SDP. A failed probe falls through to the next candidate. If any precondition, shadow check, solver probe, or path persistence operation fails, the installer restores the original MATLAB path and stops with one of these stable errors:

```
install_pd_lmi:MissingYALMIP  
install_pd_lmi:IncompleteYALMIP  
install_pd_lmi:NoWorkingSDPSolver  
install_pd_lmi:PathConflict  
install_pd_lmi:PathSaveFailed
```

Repeated successful calls are idempotent: when the repository root is already present, the installer adds no duplicate path entry and returns an empty `AddedPaths`.

2.2 Validate the Installation

After the installer has selected and recorded a working solver, run the current MATLAB test entry point from the repository root:

```
results = tests.run_all();
```

The entry point covers installer rollback and persistence behavior, helper utilities, `pdbase`, `pdmatrix`, `pdvar`, and `pdlmi`. YALMIP-backed tests and the solver smoke tests use the same automatic commercial-first solver policy as the installer; their diagnostics identify the actual solver used.

2.3 Run a Minimal Solver Example

The installer already rejects shadowed package classes before it persists the path. To inspect the selected solver and exercise the solver-facing assembly boundary, use the report together with a small known-data example:

```
report.Solver
A = pdmat({[0 1]}, {1, 3}, Degree=1);
T = bernsteinTable(A, "oneLine");
disp(T)
```

CellSubscript	Expression
[1]	$(1-\alpha)*1 + \alpha*3$

```
yalmip('clear')
P = pdvar(2, {[0 1]}, "symmetric", Degree=4);
C = P >= 0;
F = toYalmip(C);
Cputinar = C.applyPutinar();
Fputinar = toYalmip(Cputinar);
Csp = C.applySparseFullBoxPreorder(2);
Fsp = toYalmip(Csp);
Cbox = C.applyFullBoxPreorder();
Fbox = toYalmip(Cbox);
```

`F` is the direct coefficient certificate; `Fputinar`, `Fsp`, and `Fbox` are the default Putinar, width-two SparseFullBox, and FullBox certificates for the same stored residual. Any one can be passed to ordinary YALMIP calls. Use the selected solver from the report when an explicit solver setting is useful:

```
opts = sdpsettings('solver', report.Solver, 'verbose', 0);
sol = optimize(F, objective, opts);
```

2.4 Navigate the Reference

Category	Entries
Backend	pdbase , constructor , cells , lbls , coeffs , ncell , ncoeff , npar , shape protocols , shared matrix operations
Known data	pdmat , construction , evaluate , bernsteinTable , disp/display , plot , operators , matrix methods
Decision variables	pdvar , construction , rhodiff , affine/mixed algebra , matrix methods , comparisons
LMIs	pdlmi , construction , direct coefficient assembly , applyPolya , applyPutinar , applySparseFullBoxPreorder , applyFullBoxPreorder , toYalmip , solver examples

Chapter 3

pdbase: Grid and Storage Backend

`pdbase` is the developer-facing backend parent for `pdmatrix` and `pdvar`. Ordinary users should model known data with `pdmatrix` and decision expressions with `pdvar`. This section documents `pdbase` because it owns the shared storage, traversal, inspection, matrix-shape, unary-transformation, reduction, and reordering implementations inherited by both public modeling classes. Central implementation does not remove these methods from `pdmatrix` or `pdvar`; their class-specific syntax, examples, validation, and limitations remain documented in sections 4.10 and 5.8.

Its traversal contract is the concrete form of the notation in section A.7: `cells()` enumerates one-based physical indices `c`, `lbls()` enumerates zero-based local labels `i`, and `coeffs(c)` returns $C^{(c)}[i]$ in that label order. If a leaf contains derivative data, rows additionally enumerate the already constructed vertices of $\mathcal{R}$; rows never change $\mathcal{H}_c$ or `i`. Public `elevate/elevVals` and the protected `bernElev/bernProd/bernTbl/mapVals/matSubs/mergeGrid` mechanisms preserve this separation while owning componentwise degree alignment, ordered convolution, tables, payload mapping, matrix subscripting, and common-grid refinement.

3.1 API Map

Task	Function or field
Create backend object	<code>pdbase</code>
Inspect cells/labels	<code>cells</code> , <code>lbls</code> , <code>coeffs</code>
Evaluate a stored function	<code>evaluate</code>
Differentiate a rate box	<code>rhodiff</code>
Count sizes and query shape	<code>ncell</code> , <code>ncoeff</code> , <code>npar</code> , <code>size/height/width/length/ndims/numel</code>
Transform matrix payloads	unary signs, transpose, diagonal, reshape, reductions, reordering, repetition, and concatenation dispatch
Elevate stored coefficients	<code>elevate</code> , <code>elevVals</code>
Backend algebra mechanisms	<code>elevLocalValues</code> , <code>bernElev</code> , <code>bernProd</code> , <code>bernTbl/mapVals/matSubs</code> , <code>mergeGrid</code>
Read storage fields	<code>GridInfo</code> , <code>LocalValues</code> , metadata fields

3.2 Construct pdbase Objects

Syntax

```
obj = pdbase(gridVectors, matrixSize, degree)
obj = pdbase(gridVectors, matrixSize, degree, localValues)
obj = pdbase(..., IsContinuous=tf)
obj = pdbase(..., ContainsDecision=tf)
obj = pdbase(..., HasRateDependence=tf)
obj = pdbase(..., RateBounds=rb)
obj = pdbase(..., SourceSummary=txt)
obj = pdbase(..., ValidationMode=mode)
```

Arguments

- **gridVectors** is a cell vector with one strictly increasing node vector per parameter, such as {[0 1 2], [10 20]}.
- **matrixSize** is a positive two-element matrix size, such as [2 2].
- **degree** is one finite nonnegative integer scalar shorthand or an ℓ -element direction-wise vector. It is stored as a 1-by- ℓ row $\mathbf{m} = (m_1, \dots, m_\ell)$; a scalar expands uniformly.
- **localValues** is the nested physical-cell storage. When omitted, **pdbase** creates a zero coefficient-backed object.
- **IsContinuous**, **ContainsDecision**, and **HasRateDependence** are logical metadata scalars. They default to false; subclasses set them to describe their own payload contract.
- **RateBounds** is empty or an $\ell \times 2$ matrix of finite lower and upper rate bounds, such as **RateBounds**=[-1 1; -2 2]. Nonempty bounds also mark the object as rate-dependent.
- **SourceSummary** is inspectable origin text and defaults to "coefficient-backed".
- **ValidationMode** is transient scalar text "fast" or "strict", case-insensitively, with default "fast". For a raw public **pdbase** construction in fast mode, repeated coefficient structure is checked only in the first supplied physical cell; malformed coefficient counts, payloads, or rate-row layouts in later supplied cells can therefore remain undetected. Strict mode audits every supplied cell. Grid, metadata, and options are validated globally in both modes, and no mode property is stored. The public **pdmat** and **pdvar** constructors normalize their source data before applying their class-specific guarantees; their fast-mode contract must not be inferred from this developer-facing raw-storage exception.

Examples and output

This backend example constructs a two-parameter parent object. The object has two physical cells because the first parameter has two intervals and the second parameter has one interval.

```
obj = pdbase({[0 1 2], [10 20]}, [2 2], 1);
objectClass = class(obj)
matrixSize = obj.MatrixSize
degree = obj.Degree
cellCount = obj.ncell()
coefficientCount = obj.ncoeff()
parameterCount = obj.npar()
objectSize = size(obj)
```

```
objectClass = 'pdbase'
matrixSize = 1×2 double
             2     2

degree = 1×2 double
         1     1
cellCount = 2
coefficientCount = 4
parameterCount = 2
objectSize = 1×2 double
             2     2
```

The printed `ncoeff` value is $\prod_s (m_s + 1) = (1 + 1)(1 + 1) = 4$, one local Bernstein coefficient for each tensor-product label in the two parameter directions.

Remarks

- Malformed grids and node vectors raise `pdbase:InvalidGrid` or `pdbase:InvalidGridVector`.
- Invalid matrix sizes, degrees, rate bounds, or validation modes raise `pdbase:InvalidMatrixSize`, `pdbase:InvalidDegree`, `pdbase:InvalidRateBounds`, or `pdbase:InvalidValidationMode`.
- The nested tree must contain one leaf per physical cell and $\prod_s (m_s + 1)$ coefficient columns per leaf. Violations raise `pdbase:InvalidLocalValues` or `pdbase:InvalidCoefficientCell`.
- Payload matrices must have the declared size and be finite, real, and numeric or supported real YALMIP expressions. Malformed payloads raise `pdbase:InvalidCoefficientPayload`.
- Rate-affine payloads and `HasRateDependence=true` require compatible nonempty `RateBounds`.

3.3 Inspect Grid and Storage Properties

Every `pdbase` property below is publicly readable through dot access but has private set access. Users inspect `obj.Degree` or `obj.GridInfo.Vectors`; construction and package algebra create new objects instead of assigning these properties.

Property	Meaning
GridInfo	Grid vectors, node counts, and grid bounds.
MatrixSize	Size of each matrix coefficient or matrix expression.
Degree	Direction-wise local Bernstein degree stored as a 1-by- ℓ row.
LocalValues	Nested cell tree indexed by physical-cell subscript; each leaf stores coefficient cells.
IsContinuous	Metadata flag; subclasses are responsible for boundary sharing if continuity is true.
ContainsDecision	True when coefficient payloads include YALMIP decision variables.
HasRateDependence	True when the object carries rate metadata or rate-vertex rows.
RateBounds	Parameter-rate lower and upper bounds used by <code>rhodiff</code> and <code>pdlmi</code> .
SourceSummary	Short origin label such as <code>coefficient-backed</code> , <code>decision</code> , or <code>derivative</code> .

```
obj = pdbase([0 1 2], [10 20], [2 3], 1);
degree = obj.Degree
matrixSize = obj.MatrixSize
firstGrid = obj.GridInfo.Vectors{1}
source = obj.SourceSummary
```

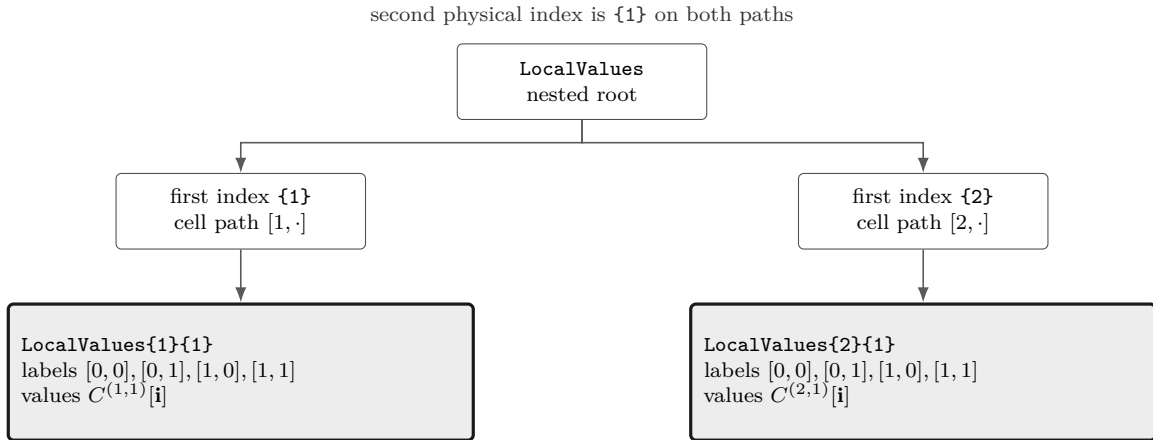


Figure 3.1: Nested cell-local storage for a two-parameter, degree-one object on $\{[0 \ 1 \ 2], [10 \ 20]\}$. There are two physical cells, $[1, 1]$ and $[2, 1]$. Each leaf is a flat local coefficient cell in `1bls` order: $[0, 0], [0, 1], [1, 0], [1, 1]$.

The nesting follows physical-cell subscripts, not the Bernstein labels. Start with `LocalValues{i1}`, then take one cell index per remaining parameter; this selects physical cell $[i_1, i_2, \dots, i_\ell]$. Only after that selection does the leaf's flat cell array enumerate the $\prod_s (m_s + 1)$ local Bernstein coefficients. Thus `coeffs(obj, [2 1])` returns the lower leaf in Fig 3.1, while `1bls(obj)` explains the order

inside that leaf.

3.4 Inspect cells, lbls, and coeffs

Syntax

```
subs = cells(obj)
subs = obj.cells()

labels = lbls(obj)
labels = obj.lbls()

c = coeffs(obj, cellSubs)
c = obj.coeffs(cellSubs)
```

Arguments

- `obj` is a `pdbase` object or a `pdmatrix`/`pdvar` subclass instance using the shared cell-local storage convention.
- `cellSubs` selects one physical cell for `coeffs`. Supply a numeric row vector such as `[2 1]` or a cell array of scalar subscripts such as `{2,1}`.
- `cells` returns physical hypercube subscripts in the shared traversal order;
- `lbls` returns local Bernstein labels in the shared coefficient order.
- `c` is the flat coefficient cell array for the selected physical cell, ordered consistently with `lbls(obj)`. Rate-dependent leaves return one row per active rate vertex in the stored row order.

Remarks

`coeffs` requires one positive integer physical-cell index per parameter, within the corresponding cell count. Invalid numeric or cell-array selectors raise `pdbase:InvalidCellSubs`.

Examples and output

The label order is shared by `pdmatrix`, `pdvar`, and `pdlmi`. In this two-parameter degree-one object, each physical cell stores four coefficient entries with labels `[0,0]`, `[0,1]`, `[1,0]`, and `[1,1]`.

```
obj = pdbase({[0 1 2], [10 20]}, [1 1], 1);
labels = lbls(obj)
physicalCells = cells(obj)
c = coeffs(obj, [2 1]);
coefficientCount = numel(c)
```

```

labels = 4×2 double
    0     0
    0     1
    1     0
    1     1

physicalCells = 2×2 double
    1     1
    2     1

coefficientCount = 4

```

3.5 evaluate Stored Coefficients

Syntax

```

val = evaluate(obj, point)
val = obj.evaluate(point)

```

Arguments

point is a finite real scalar for one parameter or a finite real vector with one coordinate per parameter, inside the physical grid bounds. The method locates one physical cell using right-cell ownership at interior grid nodes, converts the point to local coordinates, and reconstructs the tensor-product Bernstein polynomial. One ordinary coefficient row returns one numeric matrix or symbolic YALMIP expression with `size(obj)`; rate-row storage returns a 1-by- N cell array in stored row order.

Examples and output

```

obj = pdbase([0 2]), [1 1], 2, {[1, 3, 9]});
valueAtOne = obj.evaluate(1)
rateObj = pdbase([0 2]), [1 1], 1, ...
    {[1, 3; 10, 14]}, HasRateDependence=true, ...
    RateBounds=[-1 1]);
rateRows = cell2mat(rateObj.evaluate(0.5))

```

```

valueAtOne = 4
rateRows = 1×2 double
    1.5000    11.0000

```

Remarks

Malformed points raise `class(obj) + ":InvalidPoint"` and out-of-bounds points raise `class(obj) + ":PointOutOfBounds"`. `pdmat` overrides the function-backed path so exact stored handles remain exact, while coefficient-backed `pdmat` and all `pdvar` data use this row-aware backend reconstruction. Evaluation does not change the object, select a certificate, or assign YALMIP variables.

3.6 Differentiate Stored Coefficients with rhodiff

Syntax

```
D = rhodiff(A)
D = rhodiff(A, rateBounds)
```

Arguments

rhodiff differentiates coefficient-backed storage cell by cell and returns the same dynamic class as **A**. The no-argument form uses **A.RateBounds**; if those bounds are empty, explicit **rateBounds** is required. Explicit bounds supply missing metadata, but when **A.RateBounds** is already stored they must match it exactly. On physical cell **c**, the partial coefficient along a nonzero-degree axis *s* is

$$\left(\partial_s A^{(c)}\right)[\mathbf{i}] = \frac{m_s}{h_s^c} \left(A^{(c)}[\mathbf{i} + \mathbf{e}_s] - A^{(c)}[\mathbf{i}]\right), \quad i_s = 0, \dots, m_s - 1.$$

For one parameter, an input of degree $m > 0$ returns degree $m - 1$, while degree zero remains degree zero. For more than one parameter, a nonzero-axis partial first has degree $\mathbf{m} - \mathbf{e}_s$, then is elevated exactly to the common tensor degree $\mathbf{m} = \mathbf{A.Degree}$ before the rate-weighted partials are added; hence **D.Degree=A.Degree**. A zero-degree axis contributes the zero partial and still remains part of that common degree vector. Each physical-cell leaf has one row per rate-box vertex, in the order produced by applying **helper.combRows** to the parameter-wise bound rows. The result has **IsContinuous=false**, **HasRateDependence=true**, and derivative source metadata. Sections [A.9](#) and [A.9.1](#) derive the degree reduction, common-degree restoration, and row ordering.

Examples and output

```
A = pdbase({[0 2]}, [1 1], 1, {[1, 5]}, ...
    RateBounds=[-2 3]);
D = rhodiff(A);
derivativeClass = class(D)
derivativeDegree = D.Degree
derivativeRows = cell2mat(D.coeffs(1))
```

```
derivativeClass = 'pdbase'
derivativeDegree = 0
derivativeRows = 2×1 double
    -4
     6
```

Remarks

- Function-only storage raises **pdbase:FunctionOnlyDiff**; an object that already has explicit rate rows raises **pdbase:InvalidDiff**.
- Missing, malformed, or inconsistent bounds raise **pdbase:MissingRateBounds**, **pdbase:InvalidRateBounds**, or **pdbase:RateBoundsMismatch**, respectively. Repeated differentiation is

rejected because the first derivative already contains rate rows.

See also. Class-specific `pdmatrix` and `pdvar` differentiation is documented in sections [4.6](#) and [5.5](#).

3.7 Inspect Counts and Shape

Syntax

```
n = ncell(obj)
n = obj.ncell()

n = ncoeff(obj)
n = obj.ncoeff()

n = npar(obj)
n = obj.npar()

sz = size(obj)
d = size(obj, dim)
[m, n] = size(obj)
n = height(obj)
n = width(obj)
n = length(obj)
n = ndims(obj)
n = numel(obj)
idx = end(obj, k, n)
n = numArgumentsFromSubscript(obj, S, ctx)
```

Arguments

- `obj` is a `pdbase` object or subclass instance.
- `dim` is a positive integer matrix dimension for `size`; dimensions after row and column are singleton, and multiple outputs split the stored two-dimensional shape with trailing outputs equal to one.
- `ncell` returns $\prod_s (k_s - 1)$, the number of physical hypercube cells.
- `ncoeff` returns $\prod_s (\text{obj.Degree}(s) + 1)$, the number of local Bernstein coefficients stored per physical cell.
- `npar` returns the parameter dimension ℓ .
- `size`, `height`, `width`, `length`, `ndims`, and one-argument `numel` report the matrix payload: respectively its two-element shape, row count, column count, largest dimension, the value two, and the product of the two dimensions.
- `end` reports the payload boundary used by inherited two-index syntax; one-index use returns the product of the dimensions, dimensions after two return one.

- `numArgumentsFromSubscript` always returns one so dot access and subclass indexing do not create comma-separated expansion even though `numel` follows payload semantics.

Examples and output

For a two-parameter ($\ell = 2$) degree-one backend object, the first grid has two physical intervals and the second has one. The matrix payload is 2-by-3, independently of the grid shape. Thus $n_{\text{cell}} = (3 - 1)(2 - 1) = 2$ and each cell stores $n_{\text{coeff}} = (1 + 1)^2 = 4$ entries, ordered by the explicit labels $[0, 0], [0, 1], [1, 0], [1, 1]$.

```
obj = pdbase({[0 1 2], [10 20]}, [2 3], 1);
physicalCellCount = obj.ncell()
coefficientCount = obj.ncoeff()
parameterCount = obj.npar()
matrixSize = size(obj)
rowCount = size(obj, 1)
columnCount = size(obj, 2)
thirdDimension = size(obj, 3)
rowCountByName = height(obj)
columnCountByName = width(obj)
longestDimension = length(obj)
elementCount = numel(obj)
dimensionCount = ndims(obj)
```

```
physicalCellCount = 2
coefficientCount = 4
parameterCount = 2
matrixSize = 1×2 double
             2     3

rowCount = 2
columnCount = 3
thirdDimension = 1
rowCountByName = 2
columnCountByName = 3
longestDimension = 3
elementCount = 6
dimensionCount = 2
```

Remarks

`pdbase` is a scalar value container whose overloaded `numel(obj)` describes the payload rather than a MATLAB object array. MATLAB may call `numel` with subscripts while resolving indexing; that internal form returns one. Direct `pdbase` does not expose matrix `subsref/subsasgn`; `end` and `numArgumentsFromSubscript` support the derived `pdmat`/`pdvar` implementations documented in sections 4.11 and 5.9.

3.8 Apply Shared Matrix Operations

The methods in this section are implemented once by `pdbase`. Each operation maps the same MATLAB matrix transformation over every local coefficient, every active rate row, and both terms of legacy rate-affine payloads while preserving the physical grid, degree, cell/label order, and metadata. A direct backend input returns a direct `pdbase`; the `pdmat` and `pdvar` reconstruction hooks retain their respective classes and class-specific semantics. Function-only `pdmat` data have no Bernstein coefficients to map and raise `pdmat:FunctionOnlyAlgebra`.

3.8.1 Unary Signs, Transpose, Vectorization, Diagonals, And Reshape

Syntax

```
B = +A
B = -A
T = transpose(A)
H = ctranspose(A)
T = A.'
H = A'
v = vec(A)
d = diag(A)
d = diag(A, k)
D = diag(v)
D = diag(v, k)
t = trace(A)
R = reshape(A, m, n)
R = reshape(A, [m n])
S = squeeze(A)
```

Description

Unary plus returns the same value; unary minus, transpose, conjugate transpose, vectorization, diagonal extraction/construction, trace, reshape, and squeeze map coefficient payloads without changing the parameter representation. `vec` uses MATLAB column-major order. `trace` returns a 1-by-1 object and sums the main diagonal even for a row or column vector. `diag` accepts a finite integer offset; for a matrix, the selected diagonal must be nonempty because the package cannot store an empty payload. `reshape` supports exactly two positive dimensions, permits at most one inferred `[]` dimension, and preserves the element count. `squeeze` is the identity because the storage contract remains two-dimensional.

Examples and output

```
A = pdbase({[0 1]}, [2 2], 1, ...
    {[[1 2; 3 4], [10 20; 30 40]]});
T = A.';
v = vec(A);
d = diag(A);
t = trace(A);
R = reshape(A, 1, 4);
transposeCoefficients = T.coeffs(1);
vectorCoefficients = v.coeffs(1);
diagonalCoefficients = d.coeffs(1);
traceCoefficients = t.coeffs(1);
transposeSize = size(T)
transposeFirst = transposeCoefficients{1}
vectorFirst = vectorCoefficients{1}
diagonalFirst = diagonalCoefficients{1}
traceFirst = traceCoefficients{1}
reshapeSize = size(R)
```

```
transposeSize = 1×2 double
    2    2
transposeFirst = 2×2 double
    1    3
    2    4
vectorFirst = 4×1 double
    1
    3
    2
    4
diagonalFirst = 2×1 double
    1
    4
traceFirst = 5
reshapeSize = 1×2 double
    1    4
```

Remarks

Malformed or empty diagonal selections raise `class(A) + ":InvalidDiag"`. Unsupported reshape forms, multiple inferred dimensions, nonpositive dimensions, or an element-count mismatch raise `class(A) + ":InvalidReshape"`. These identifiers therefore appear as `pdbase:*`, `pdmat:*`, or `pdvar:*` according to the public input class.

3.8.2 Triangular Parts And Reductions

Syntax

```
L = tril(A)
L = tril(A, k)
U = triu(A)
U = triu(A, k)
S = sum(A)
S = sum(A, dim)
S = sum(A, vecdim)
S = sum(A, "all")
M = mean(A)
M = mean(A, dim)
M = mean(A, vecdim)
M = mean(A, "all")
C = cumsum(A)
C = cumsum(A, dim)
C = cumsum(A, direction)
C = cumsum(A, dim, direction)
```

Description

`tril` and `triu` retain entries on and below or above the finite integer offset `k`. `sum` and `mean` accept one nonempty vector of unique positive integer dimensions or the case-insensitive text `"all"`; dimensions above two are singleton no-ops. Without a dimension, each follows the first nonsingleton matrix dimension. `cumsum` accepts an optional positive integer scalar dimension and an optional case-insensitive `"forward"` or `"reverse"` direction. A direction without a dimension uses the first nonsingleton matrix dimension; a dimension above two leaves every matrix payload unchanged.

Examples and output

```
A = pdbase({[0 1]}, [2 2], 1, ...
    {[[1 2; 3 4], [10 20; 30 40]]});
L = tril(A);
S = sum(A, [1 2]);
M = mean(A, "all");
C = cumsum(A, 2, "reverse");
lowerCoefficients = L.coeffs(1);
sumCoefficients = S.coeffs(1);
meanCoefficients = M.coeffs(1);
cumsumCoefficients = C.coeffs(1);
lowerFirst = lowerCoefficients{1}
sumFirst = sumCoefficients{1}
meanFirst = meanCoefficients{1}
reverseCumsumFirst = cumsumCoefficients{1}
```

```
lowerFirst = 2×2 double
    1     0
    3     4
sumFirst = 10
meanFirst = 2.5000
reverseCumsumFirst = 2×2 double
    3     2
    7     4
```

Remarks

Invalid triangular offsets raise `class(A) + ":InvalidTriangularPart"`. Malformed reduction forms raise `class(A) + ":InvalidSum"`, `class(A) + ":InvalidMean"`, or `class(A) + ":InvalidCumsum"`. Missing-value, native-output, or other MATLAB reduction flags are not implemented.

3.8.3 Reordering, Rotation, And Repetition

Syntax

```
F = flip(A)
F = flip(A, dim)
F = flipud(A)
F = fliplr(A)
R = rot90(A)
R = rot90(A, k)
Q = repmat(A, m)
Q = repmat(A, m, n)
Q = repmat(A, [m n])
```

Description

`flip`, `flipud`, and `fliplr` reverse entries inside each coefficient matrix and never reverse a physical parameter grid. `flip(A)` follows MATLAB's first-nonsingleton behavior; `flip(A,dim)` requires a positive integer scalar dimension. `rot90` applies a finite integer number of counterclockwise quarter-turns, reducing the count modulo four. `repmat` supports a positive scalar interpreted as `[m m]`, two positive counts, or a two-element count vector; only the payload shape changes.

Examples and output

```
A = pdbase({[0 1]}, [2 2], 1, ...
    {[[1 2; 3 4], [10 20; 30 40]]});
F = flipud(A);
R = rot90(A);
Q = repmat(A, [1 2]);
flippedCoefficients = F.coeffs(1);
rotatedCoefficients = R.coeffs(1);
flippedFirst = flippedCoefficients{1}
rotatedFirst = rotatedCoefficients{1}
repeatedSize = size(Q)
```

```
flippedFirst = 2×2 double
     3     4
     1     2
rotatedFirst = 2×2 double
     2     4
     1     3
repeatedSize = 1×2 double
     2     4
```

Remarks

Invalid flip dimensions, rotation counts, or repetition forms raise `class(A) + ":InvalidFlip"`, `class(A) + ":InvalidRot90"`, or `class(A) + ":InvalidRepmat"`. N-dimensional repetition is outside the two-dimensional payload contract.

3.8.4 Concatenation Dispatch

Syntax

```
C = horzcat(A, B, ...)  
C = [A, B, ...]  
C = vertcat(A, B, ...)  
C = [A; B; ...]  
C = cat(dim, A, B, ...)
```

Description

`pdbase` owns the inherited `horzcat`/`vertcat` entry points, which dispatch compatible derived objects to the class-specific `cat` implementation. `pdmat` and `pdvar` therefore retain coefficient-wise concatenation, grid/degree alignment, numeric promotion, validation, and result construction as documented in sections 4.10.7 and 5.8.7. Direct `pdbase` operands are scalar backend containers rather than MATLAB object arrays and cannot be concatenated.

Examples and output

```
A = pdmat({[0 1]}, {1, 2}, Degree=1);  
B = [A, A];  
resultClass = class(B)  
resultSize = size(B)
```

```
resultClass = 'pdmat'  
resultSize = 1×2 double  
           1     2
```

Remarks

Any direct `pdbase` operand in `cat`, `horzcat`, or `vertcat` raises `pdbase:Unsupported Concatenation`. Derived classes support only matrix dimensions one and two and report their own compatibility errors. `blkdiag` remains class-specific to `pdmat` and `pdvar`; it is not a `pdbase` implementation.

3.9 Transform Bernstein Storage

The following entries separate whole-object elevation from the protected helpers used to align degrees, convolve coefficients, and refine grids. They change coefficient representations or temporary storage trees; they do not change the meanings of `ncell`, `ncoeff`, `npar`, or `size` described above.

3.9.1 elevate

Syntax

```
out = obj.elevate(degreeIncrement)
```

Description

`degreeIncrement` is one finite nonnegative integer scalar shorthand or an ℓ -element vector. A scalar expands uniformly; the vector adds direction-wise. `out` is the same dynamic class as `obj`; it contains elevated coefficient evidence at the componentwise sum `obj.Degree + degreeIncrement` while preserving the represented polynomial, physical cells, metadata, YALMIP decision variables, rate-row order, and subclass properties. If $\mathbf{m} = \text{obj.Degree}$, $\mathbf{M} = \mathbf{m} + \text{degreeIncrement}$, and $U_{\mathbf{m}}^{(c)}[\mathbf{i}] = (\prod_s \binom{m_s}{i_s}) C^{(c)}[\mathbf{i}]$, the backend computes

$$\hat{C}^{(c)}[\mathbf{k}] = \frac{(U_{\mathbf{m}}^{(c)} *_{\ell} K_{\mathbf{M}-\mathbf{m}})[\mathbf{k}]}{\prod_s \binom{M_s}{k_s}}, \quad K_{\mathbf{M}-\mathbf{m}}[\mathbf{r}] = \prod_s \binom{M_s - m_s}{r_s}.$$

Thus elevation is an ℓ -dimensional discrete convolution with a fixed separable kernel followed by binomial normalization. The source object is unchanged, and elevation introduces no new decision freedom. Section A.6 derives the formula and its exactness.

Examples and output

```
A = pmat({[0 1]}, {1, 3}, Degree=1);
B = A.elevate(1);
sourceDegree = A.Degree
elevatedDegree = B.Degree
elevatedCoefficients = cell2mat(B.coeffs(1))
sameValue = [A.evaluate(0.25), B.evaluate(0.25)]
```

```
sourceDegree = 1
elevatedDegree = 2
elevatedCoefficients = 1x3 double
    1    2    3
sameValue = 1x2 double
    1.5000    1.5000
```

Remarks

Invalid increments raise `pdbase:InvalidDegreeIncrement`. Function-only `pdmatrix` data have no coefficient evidence and raise `pdbase:MissingCoefficientEvidence`; constructing the function with a verified explicit `Degree` supplies the needed evidence. `elevate` returns an object, whereas `elevVals` returns only an elevated `LocalValues`-shaped tree.

3.9.2 elevVals

Syntax

```
vals = obj.elevVals(degreeIncrement)
```

Arguments

- `degreeIncrement` is a finite nonnegative integer scalar shorthand or an ℓ -element vector. It is added componentwise to `obj.Degree`; scalar zero or an all-zero vector returns coefficient evidence at the current degree.

Description

`vals` is a nested `LocalValues`-shaped cell tree. Each physical cell contains coefficients of degree `obj.Degree + degreeIncrement`, with the represented polynomial, physical-cell tree, and any rate-row ordering preserved. The method does not modify `obj` or its stored coefficient evidence.

Examples and output

```
A = pdmat({[0 1]}, {1, 3}, Degree=1);  
elevatedTree = A.elevVals(1);  
elevated = cell2mat(elevatedTree{1})
```

```
elevated = 1×3 double  
    1     2     3
```

Remarks

An invalid increment raises `pdbase:InvalidDegreeIncrement`. A function-only `pdmatrix` has no stored Bernstein coefficient evidence, so this method raises `pdbase:MissingCoefficientEvidence`.
Relationship to backend and public algebra. Unlike protected `bernElev`, which elevates one flat cell coefficient family for internal alignment, `elevVals` is a public whole-object query returning elevated `LocalValues`. Public coefficient algebra uses `bernElev` internally when compatible operands need degree alignment; it does not mutate either operand.

3.9.3 elevLocalValues

Internal/protected mechanism. This method elevates a temporary nested coefficient tree after public algebra has remapped operands to a common physical grid. It is documented as backend context, not as a supported user call form.

Syntax

```
vals = pdbase.elevLocalValues(vals, fromDeg, toDeg, grid)
```

Each leaf must be a cell array with one column per direction-wise degree-**fromDeg** Bernstein label. The method elevates each rate-vertex row independently and componentwise to **toDeg**; it never mixes rows. The required **grid** supplies the temporary physical-cell layout. Equal source and target degree vectors return the input tree unchanged. A malformed leaf raises **pdbase:InvalidCoefficientCell**. Public users encounter this behavior through grid merging, assignment, concatenation, and arithmetic rather than by calling the protected method directly.

Examples and output

```
A = pdmat({[0 1]}, @(rho) rho, Degree=1);
B = pdmat({[0 .5 1]}, @(rho) 2*rho, Degree=2);
C = A + B;
mergedGrid = C.GridInfo.Vectors{1}
resultDegree = C.Degree
cell1Coefficients = cell2mat(C.coeffs(1))
cell2Coefficients = cell2mat(C.coeffs(2))
AGrid = A.GridInfo.Vectors{1}
ADegree = A.Degree
BGrid = B.GridInfo.Vectors{1}
BDegree = B.Degree
```

```
mergedGrid = 1×3 double
            0    0.5000    1.0000
```

```
resultDegree = 2
```

```
cell1Coefficients = 1×3 double
            0    0.7500    1.5000
```

```
cell2Coefficients = 1×3 double
            1.5000    2.2500    3.0000
```

```
AGrid = 1×2 double
            0     1
```

```
ADegree = 1
```

```
BGrid = 1×3 double
            0    0.5000    1.0000
```

```
BDegree = 2
```

The example exercises the protected mechanism through public **pdmat** addition; users do not call protected **elevLocalValues** directly.

3.9.4 bernElev

Internal/protected mechanism. This signature documents the backend algorithm used by public algebra; it is not a supported user call form.

Syntax

```
out = pdbase.bernElev(coeffs, fromDeg, toDeg, nPar)
```

Arguments

-
- `coeffs` is a flat coefficient-cell family in Bernstein label order.
 - `fromDeg` and `toDeg` are scalar shorthand or `nPar`-element nonnegative integer degree vectors, with `toDeg` $\geq$ `fromDeg` componentwise.
 - `nPar` is the positive integer parameter count.
 - `out` contains $\prod_s (\text{toDeg}(s) + 1)$ elevated coefficients in the same label order.

Examples and output

```
A = pdmat({[0 1]}, {1, 2}, Degree=1);  
B = pdmat({[0 1]}, {1, 2, 3}, Degree=2);  
C = A + B;  
degree = C.Degree
```

```
degree = 2
```

The example invokes the protected helper through public addition; users do not call `bernElev` directly. Invalid degree values raise `pdbase:InvalidDegree`; a target below the source degree raises `pdbase:InvalidDegreeElevation`; a coefficient count inconsistent with the source degree and parameter dimension raises `pdbase:InvalidCoefficientCell`.

3.9.5 bernProd

Internal/protected mechanism. This signature documents the backend algorithm used by public matrix multiplication; it is not a supported user call form.

Syntax

```
out = bernProd(obj, lhs, lhsDeg, rhs, rhsDeg)
```

Arguments

- `lhs` and `rhs` are flat coefficient-cell families whose matrix payload inner dimensions are compatible for multiplication.
- `lhsDeg` and `rhsDeg` are their scalar shorthand or direction-wise local Bernstein degrees.
- `out` is the binomial-normalized coefficient convolution at the componentwise degree sum `lhsDeg + rhsDeg`; payload products follow MATLAB matrix multiplication rules.

For one parameter, the protected helper implements

$$C[k] = \sum_{i+j=k} \frac{\binom{m}{i} \binom{n}{j}}{\binom{m+n}{k}} A[i]B[j].$$

It enumerates ordered payload products, groups them by the anti-diagonal $i + j = k$, and only then applies the binomial normalization. For tensor labels, it performs the ℓ -dimensional ordered convolution

$$C[\mathbf{k}] = \sum_{\mathbf{i}+\mathbf{j}=\mathbf{k}} \left(\prod_{s=1}^{\ell} \frac{\binom{m_s}{i_s} \binom{n_s}{j_s}}{\binom{m_s+n_s}{k_s}} \right) A[\mathbf{i}]B[\mathbf{j}].$$

See the public `pdmat` multiplication entry in section 4.8.2, the `pdvar` known-data boundary in section 5.7, and the full derivation in section A.8.

Examples and output

```
A = pdmat({[0 1]}, {1, 2}, Degree=1);
P = A * A;
degree = P.Degree
coefficients = cell2mat(P.coeffs(1))
```

```
degree = 2
coefficients = 1×3 double
               1     2     4
```

The example invokes the protected helper through public matrix multiplication; users do not call `bernProd` directly. Even for scalar data, the middle coefficient comes from two ordered contributions rather than from one sample.

3.9.6 `bernTbl`, `mapVals`, and `matSubs`

Internal/protected mechanisms. These object-aware utilities are owned by `pdbase`; they are not functions in the public `+helper` namespace and are not supported user call forms.

Syntax

```
T = bernTbl(obj, errId, valFcn, exprFcn, rateVerts, ...)
mapped = mapVals(vals, fcn, grid)
[rows, cols] = matSubs(subs, sz, errId)
```

Description

`bernTbl` builds the detailed and one-line inspection tables used by `pdmatrix.bernsteinTable` and `pdvar.bernsteinTable`. It traverses `cells`, `lbls`, and `coeffs`, accepts at most one physical-cell selector and the case-insensitive `"oneLine"` option, and preserves the caller's error namespace. `mapVals` applies a coefficient mapping to every payload in every nested physical-cell leaf while preserving tree and label order; the owning class still validates matrix, symbolic, and rate semantics. `matSubs` normalizes exactly two row/column selectors, accepting `:`, finite positive integer vectors, or matching logical vectors, and rejects linear indexing, empty/out-of-range numeric indices, mismatched logical selectors, and array growth.

Examples and output

```
A = pdmat({[0 1]}, {[1 2; 3 4]}, [10 20; 30 40]), Degree=1);
T = A.bernsteinTable("oneLine");
column = A(:, 2);
tableHeight = height(T)
columnSize = size(column)
```

```
tableHeight = 1
columnSize = 1×2 double
           2     1
```

Remarks

Users reach `bernTbl` through `bernsteinTable`, `mapVals` through inherited unary operations, and `matSubs` through `pdmatrix/pdvar` indexing and assignment. Invalid table selectors and rate-row mismatches raise the caller-owned `errId`; indexing uses the class-specific identifier supplied by `subsref` or `subsasgn`. The seven genuine cross-class helpers remain documented in Chapter 7.

3.9.7 mergeGrid

Internal/protected mechanism. This signature documents the backend grid-refinement algorithm used by public algebra; it is not a supported user call form.

Syntax

```
grid = obj.mergeGrid(errId, other1, other2, ...)
```

Arguments

- `errId` is the error identifier used when parameter counts or physical bounds are incompatible.
- `other1`, `other2`, ... are coefficient-backed operands, including function-backed operands constructed with an explicit `Degree`. They must have equal parameter counts and matching first and last nodes on every parameter axis.
- `grid` has the sorted union of the interior nodes. Public algebra re-expresses operands on these cells before degree elevation, coefficient-wise addition, or product convolution.

This protected helper is invoked by public algebra methods rather than called directly by users. Public addition, subtraction, concatenation, and assignment then apply a common four-stage alignment: form this sorted union grid, use exact subdivision to re-express every operand on its physical cells, elevate direction-wise to the componentwise maximum degree, and only then operate on matching local coefficients. Grid alignment changes neither the represented functions nor the outer parameter box. Section A.6 gives the degree step, while section A.7 fixes the physical-cell and tensor-label ordering.

Examples and output

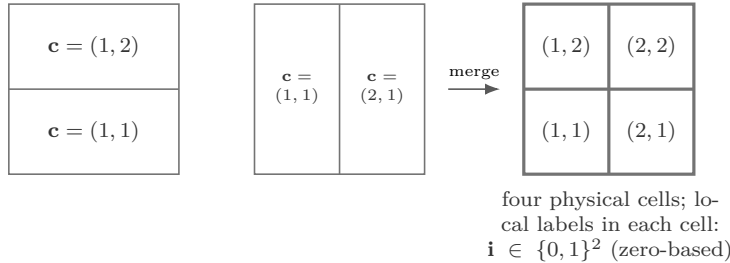
```
gridA = {[0 1], [0 0.5 1]};
gridB = {[0 0.5 1], [0 1]};
A = pdmat(gridA, @(rho1, rho2) rho1 + 2*rho2, Degree=[1 1]);
B = pdmat(gridB, @(rho1, rho2) 2*rho1 - rho2, Degree=[1 1]);
C = A + B;
rho1Grid = C.GridInfo.Vectors{1}
rho2Grid = C.GridInfo.Vectors{2}
```

```
rho1Grid = 1×3 double
    0    0.5000    1.0000
```

```
rho2Grid = 1×3 double
    0    0.5000    1.0000
```

The example invokes the protected helper through public addition; users do not call `mergeGrid` directly. Here $\ell = 2$. The physical-cell index $\mathbf{c} = (c_1, c_2)$ is one-based, while the degree-one local Bernstein label $\mathbf{i} = (i_1, i_2) \in \{0, 1\}^2$ is zero-based inside every cell.

$$\mathcal{G}_A = \{[0, 1], [0, 0.5, 1]\} \quad \mathcal{G}_B = \{[0, 0.5, 1], [0, 1]\} \quad \mathcal{G} = \{[0, 0.5, 1], [0, 0.5, 1]\}$$



Each axis uses the sorted union of its input nodes. The result is the unique minimal same-bound tensor grid containing both inputs.

3.10 Boundaries and Related APIs

pdmat, pdvar, cells, coeffs, lbls, evaluate, ncell, ncoeff, npar, size, height, width, length, ndims, numel, end, numArgumentsFromSubscript, unary +/-, transpose, ctranspose, trace, vec, diag, reshape, squeeze, tril, triu, sum, mean, cumsum, flip, flipud, fliplr, rot90, repmat, cat, horzcat, vertcat, elevate, elevVals, elevLocalValues, bernElev, bernProd, bernTbl, mapVals, matSubs, mergeGrid.

Chapter 4

pdmat: Known Parameter-Dependent Matrices

`pdmat` stores known finite real matrix data on a tensor grid. Objects can be coefficient-backed, function-only, or function-backed with explicit Bernstein coefficient evidence. Coefficient-backed objects participate in algebra. Function-only objects evaluate exactly through the stored handle, but do not expose coefficient evidence for `bernsteinTable` or coefficient algebra. Shared matrix-shape, unary, structural, reduction, reordering, and repetition methods are implemented centrally by `pdbase` and remain complete public `pdmat` behavior; this chapter documents their known-data syntax, numeric examples, validations, and boundaries.

For coefficient-backed data, one storage leaf is the pullback $A^{(\mathbf{c})} = A \circ \phi_{\mathbf{c}}$, with local expansion $A^{(\mathbf{c})}(\boldsymbol{\alpha}) = \sum_{\mathbf{i}} B_{\mathbf{i}}^{\mathbf{m}}(\boldsymbol{\alpha}) A^{(\mathbf{c})}[\mathbf{i}]$ and $\boldsymbol{\alpha} = \phi_{\mathbf{c}}^{-1}(\boldsymbol{\rho})$. Construction chooses the source of those coefficients; `evaluate` reconstructs the matrix value; `elevate` changes the coefficient representation without changing the polynomial; and ordered multiplication applies the binomial-weighted convolution in section A.8. A function-backed object keeps exact handle evaluation separate from its finite coefficient evidence, while a function-only object has no coefficient algebra to certify.

4.1 API Map

Task	Entry
Construct data	<code>pdmat</code>
Evaluate	<code>evaluate</code>
Inspect/display	<code>bernsteinTable</code> , <code>disp</code> , <code>display</code>
Inspect/elevate backend storage	<code>cells/lbls/coeffs</code> , <code>counts</code> , <code>elevVals</code> , and <code>elevate</code>
Plot	<code>plot</code>
Differentiate	<code>rhodiff</code>
Arithmetic	<code>+</code> , <code>-</code> , unary <code>+/-</code> , <code>*</code>
Matrix/index methods	shape and structural methods, indexing and assignment
Compare/certify	logical <code>==</code> and known-data <code><=/>=</code>

4.2 Construct pdmat Objects

Syntax

```
A = pdmat(gridVector, source)
A = pdmat(gridVector, source, Degree=degree)
A = pdmat(gridVector, source, RateBounds=rb)
A = pdmat(gridVectors, source)
A = pdmat(gridVectors, source, Degree=degree, RateBounds=rb)
A = pdmat(..., ValidationMode=mode)
```

Arguments

- `gridVector` is a numeric vector used as a one-parameter grid shorthand, such as `[0 1 2]`.
- `gridVectors` is a cell vector with one numeric grid vector per parameter, such as `{[0 1], [10 20]}`.
- `source` may be:
 1. A function handle, e.g., `@(rho) [rho rho^2]` is a one-parameter function with two matrix entries.
 2. A global cell grid of numeric Bernstein coefficients, e.g., `{[1 2], [3 4]}` is a two-cell degree-one grid with two 1×1 coefficients per cell.
 3. A explicit nested `LocalValues`, e.g., `{1, 2, 3}` is scalar degree-two data on one cell.
- `Degree=degree` accepts one finite nonnegative integer scalar shorthand or an ℓ -element vector and is stored as a 1-by- ℓ row $\mathbf{m} = (m_1, \dots, m_\ell)$. A scalar expands uniformly. An explicitly supplied scalar on a multidimensional grid emits `pdmat:ScalarDegreeExpansion` once; omitted degree inference, function-only defaults, and one-dimensional forms remain warning-free. For function handles, omitting `Degree` creates a function-only object; supplying `Degree` validates and stores Bernstein coefficient evidence while retaining the exact function handle for `evaluate`.
- `RateBounds=rb` is an ℓ -by-2 finite real matrix of lower and upper scheduling-rate bounds and is accepted with every source family. With an ordinary one-row source it is metadata used by `rhodiff`. An explicit nested leaf may instead contain either one ordinary row or all 2^ℓ rate-vertex rows, uniformly across every physical cell; their deterministic order is induced by applying `helper.combRows` to `num2cell(rb,2).'`.
- `ValidationMode=mode` accepts scalar text `"fast"` or `"strict"`, case-insensitively, and defaults to `"fast"`. It controls only repeated post-normalization structural checks and is not stored. Public source validation remains complete in both modes.

Examples and output

The scalar-grid shorthand is the shortest coefficient-backed form. The three numeric cells below are the global Bernstein coefficients for two physical cells with degree one.

```
A = pdmat([0 1 2], {10, 20, 30}, Degree=1);
disp(A)
matrixSize = size(A)
degree = A.Degree
sourceSummary = A.SourceSummary
```

```
pdmat [1 1] over 1-D grid, degree 1, source coefficient-backed, rate rows false
```

```
matrixSize = 1×2 double
    1     1
```

```
degree = 1
sourceSummary = "coefficient-backed"
```

For tensor grids, pass one grid vector per parameter. The source cell array matches the global coefficient grid implied by the two parameter directions.

```
B = pdmat([0 1], [10 20], {1 2; 3 4}, Degree=[1 1]);
disp(B)
secondGridVector = B.GridInfo.Vectors{2}
labels = B.lbls()
```

```
pdmat [1 1] over 2-D grid, degree [1 1], source coefficient-backed, rate rows false
```

```
secondGridVector = 1×2 double
    10    20
```

```
labels = 4×2 double
    0     0
    0     1
    1     0
    1     1
```

Use explicit nested `LocalValues` when each physical cell must be written directly. Here each cell stores two 1×2 row-vector coefficients.

```
localVals = {[[1 0], [0 1]], {[0 1], [0 2]}};
C = pdmat([0 1 2], localVals, Degree=1);
disp(C)
matrixSize = size(C)
secondCellCoefficients = cell2mat(C.coeffs(2))
```

```
pdmat [1 2] over 1-D grid, degree 1, source coefficient-backed, rate rows false
```

```
matrixSize = 1×2 double
    1     2
```

```
secondCellCoefficients = 1×4 double
    0     1     0     2
```

When `source` is a function handle and `Degree` is omitted, the object keeps the exact evaluator but

does not claim coefficient evidence.

```
F = pdmat({[0 1]}, @(rho) [rho rho^2]);
disp(F)
sourceSummary = F.SourceSummary
evaluatedValue = F.evaluate(0.5)
```

pdmat [1 2] over 1-D grid, degree 1, source function, rate rows false

```
sourceSummary = "function"
evaluatedValue = 1×2 double
    0.5000    0.2500
```

Add **Degree** for a polynomial function when later coefficient inspection or algebra should use Bernstein evidence. The exact function handle is still used for point evaluation.

```
G = pdmat({[0 1]}, @(rho) rho.^2, Degree=2);
disp(G)
sourceSummary = G.SourceSummary
firstCellCoefficients = cell2mat(G.coeffs(1))
```

pdmat [1 1] over 1-D grid, degree 2, source function-bernstein, rate rows false

```
sourceSummary = "function-bernstein"
firstCellCoefficients = 1×3 double
    0    0    1
```

These forms all create finite known-data objects, but only the coefficient- backed and function-Bernstein forms can participate in coefficient algebra.

Explicit rate rows attach one complete coefficient row to each vertex of the rate box. This one-parameter example has two rate vertices and therefore two rows in its only physical-cell leaf.

```
A = pdmat([0 1], {{1, 3; 10, 14}}, ...
    Degree=1, RateBounds=[-1 2]);
disp("rateBounds =")
disp(A.RateBounds)
disp("rateRowValues =")
disp(cell2mat(A.coeffs(1)))
```

```
rateBounds =
    -1     2

rateRowValues =
     1     3
    10    14
```

Remarks

`FunctionHandle` retains the supplied function handle for function-only and function-Bernstein sources; it is empty for cell-backed sources. Its set access is private.

- Constructor option errors use `pdmat:InvalidOptions`, `pdmat:UnsupportedOption`, or `pdmat:UnknownOption`; grid errors use `pdmat:InvalidGrid` or `pdmat:InvalidGridVector`.
- Source and function validation uses `pdmat:InvalidSource`, `pdmat:InvalidFunctionHandle`, `pdmat:InvalidFunctionOutput`, or `pdmat:InvalidData`.
- Degree, rate, and storage failures use `pdmat:InvalidDegree`, `pdmat:InvalidRateBounds`, `pdmat:InvalidLocalValues`, `pdmat:InvalidCoefficientCell`, or `pdmat:InvalidCoefficientPayload`. Explicit multidimensional scalar degree expansion is accepted with warning `pdmat:ScalarDegreeExpansion`.
- Invalid validation modes raise `pdmat:InvalidValidationMode`. With explicit `Degree`, nonrepresentable functions raise `pdmat:NonBernsteinPolynomial`; invalid validation probes raise `pdmat:InvalidFunctionValue`.
- Mismatched shared faces are retained as discontinuous local data and emit warning `pdmat:DiscontinuousLocalValues`.

4.3 Inspect Object Properties

`pdmat` adds the read-only `FunctionHandle` property to the inherited `pdbase` properties in section 3.3. All use private set access. `FunctionHandle` contains the supplied handle for function and function-Bernstein sources and is empty for cell-backed sources. The inherited properties describe the grid, matrix size, Bernstein degree, cell-local evidence, continuity, and origin.

```
A = pdmat({[0 1]}, @(rho) rho.^2, Degree=2);
degree = A.Degree
source = A.SourceSummary
hasFunction = ~isempty(A.FunctionHandle)
firstCell = A.LocalValues{1}
```

Assigning `A.Degree`, `A.LocalValues`, or `A.FunctionHandle` is not a supported update operation; construct a new `pdmat` instead.

4.4 Inspect And Elevate Inherited Bernstein Storage

Syntax

```
cellSubs = cells(A)
labels = lbls(A)
coefficients = coeffs(A, cellSubs)
n = ncell(A)
n = ncoeff(A)
n = npar(A)
vals = A.elevVals(degreeIncrement)
B = A.elevate(degreeIncrement)
```

Description

The storage queries use the shared **pdbase** cell and label order. **cells** returns one one-based physical-cell subscript per row; **lbls** returns the zero-based tensor labels $\mathbf{i} \in \prod_s \{0, \dots, m_s\}$; **coeffs** returns the selected flat coefficient leaf. **ncell**, **ncoeff**, and **npar** count physical cells, $\prod_s (m_s + 1)$ coefficients per cell, and parameter directions. **elevVals** accepts a scalar shorthand or direction-wise increment and returns an elevated nested coefficient tree without modifying **A**; **elevate** returns a new **pdmats** at the componentwise degree sum while preserving the represented known function and dynamic-class metadata. Mechanically, the backend binomially scales each tensor coefficient, performs an ℓ -dimensional discrete convolution with the fixed separable kernel $K_{\mathbf{M}-\mathbf{m}}[\mathbf{r}] = \prod_s \binom{M_s - m_s}{r_s}$, and divides by the target binomial weights. Section A.6 derives this exact representation change.

Examples and output

```
A = pdmat({[0 1]}, {1, 3}, Degree=1);
physicalCells = A.cells()
labels = A.lbls()
sourceCoefficients = cell2mat(A.coeffs(1))
counts = [A.ncell(), A.ncoeff(), A.npar()]
elevatedValues = A.elevVals(1);
elevatedLeaf = elevatedValues{1};
B = A.elevate(1);
elevatedTreeCoefficients = cell2mat(elevatedLeaf)
elevatedObjectCoefficients = cell2mat(B.coeffs(1))
degrees = [A.Degree, B.Degree]
```

```
physicalCells = 1
labels = 2×1 double
    0
    1
sourceCoefficients = 1×2 double
    1    3
counts = 1×3 double
    1    2    1
elevatedTreeCoefficients = 1×3 double
    1    2    3
elevatedObjectCoefficients = 1×3 double
    1    2    3
degrees = 1×2 double
    1    2
```

A direction-wise increment changes only the requested axes. This two-parameter example elevates the first axis while retaining the second.

```
A = pdmat({[0 1], [0 1]}, ...
    @(rho1, rho2) rho1 + 2*rho2, Degree=[1 1]);
B = A.elevate([1 0]);
sourceDegree = A.Degree
elevatedDegree = B.Degree
coefficientCounts = [A.ncoeff(), B.ncoeff()]
sameValue = [A.evaluate([0.25 0.75]), ...
    B.evaluate([0.25 0.75])]
```

```
sourceDegree = 1×2 double
    1    1
elevatedDegree = 1×2 double
    2    1
coefficientCounts = 1×2 double
    4    6
sameValue = 1×2 double
    1.7500    1.7500
```

Remarks

Invalid physical-cell selectors raise `pdbase:InvalidCellSubs`. Invalid increments raise `pdbase:InvalidDegreeIncrement`. Function-only data support exact `evaluate`, but their placeholder storage is not Bernstein coefficient evidence and cannot be used by `elevVals`, `elevate`, `algebra`, or `bernsteinTable`; the elevation paths raise `pdbase:MissingCoefficientEvidence`. A function constructed with an explicit verified `Degree` retains both its exact handle and Bernstein evidence.

4.5 evaluate

Syntax

```
val = evaluate(A, point)
val = A.evaluate(point)
```

Arguments

- **A** is a `pdmat` object.
- **point** is a finite real scalar for a one-parameter object, such as `0.25`, or a finite real row vector with one entry per parameter, such as `[0.25 12]`. Every entry must lie within the corresponding grid bounds.
- **val** is a numeric matrix with `size(A)`. Coefficient-backed objects use local Bernstein interpolation; function-backed objects call their stored function handle.

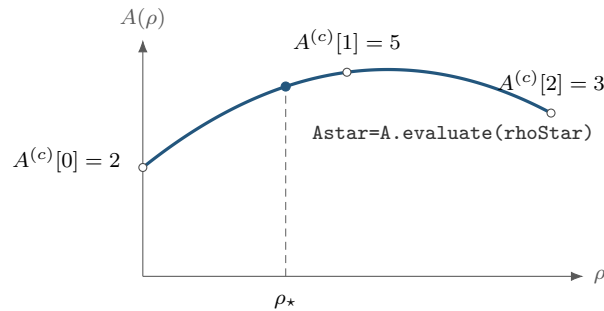
Remarks

A point with the wrong type, length, or finite-real shape raises `pdmat:InvalidPoint`; a point outside any grid bound raises `pdmat:PointOutOfBounds`. If a stored function fails or returns a nonfinite, complex, empty, or size-inconsistent matrix at the requested point, `evaluate` raises `pdmat:InvalidFunctionValue`.

For a degree-one scalar coefficient-backed object on $[\rho_1^{(1)}, \rho_1^{(2)}]$,

$$A(\rho) = (1 - \alpha)A^{(c)}[0] + \alpha A^{(c)}[1], \quad \alpha = \frac{\rho_1 - \rho_1^{(1)}}{\rho_1^{(2)} - \rho_1^{(1)}}.$$

The same local-coordinate rule is applied entrywise for matrix-valued data.



Evaluation follows the degree-two Bernstein curve with coefficients $[2, 5, 3]$; the dashed line locates the requested physical point.

Examples and output

Use the function-call form when the object is naturally one input among several MATLAB expressions.

```
A = pdmat({[0 1]}, {2, 4}, Degree=1);  
val = evaluate(A, 0.25)
```

```
val = 2.5000
```

At $\rho = 0.25$, the public local coordinate is $\alpha = 0.25$, so the interpolated value is $0.75 \cdot 2 + 0.25 \cdot 4 = 2.5$. The dot-call form is equivalent and can be convenient in command-window exploration.

```
A = pdmat({[0 1]}, {2, 4}, Degree=1);  
val = A.evaluate(0.75)
```

```
val = 3.5000
```

Function-backed objects route through the retained function handle, so non-polynomial expressions can still be evaluated exactly at a point.

```
F = pdmat({[0 pi]}, @(rho) sin(rho));  
val = F.evaluate(pi/2)
```

```
val = 1
```

4.6 Differentiate Known Data with rhodiff

Syntax

```
D = rhodiff(A)
D = rhodiff(A, rateBounds)
```

Arguments

`rhodiff` differentiates coefficient-backed `pdmat` data cell by cell and returns a `pdmat` with numeric rate-vertex rows. Explicit `rateBounds` supplies missing metadata; when `A.RateBounds` is already stored, the explicit bounds must match it exactly. Without either source of bounds, the call raises `pdmat:MissingRateBounds`. Along axis s , the local derivative coefficients are $\frac{m_s}{h_s^c} (A^{(c)}[\mathbf{i} + \mathbf{e}_s] - A^{(c)}[\mathbf{i}])$. For one parameter, degree $m > 0$ therefore returns degree $m - 1$, while degree zero remains degree zero. For more than one parameter, each nonzero-axis partial is elevated exactly from $\mathbf{m} - \mathbf{e}_s$ back to the common tensor degree $\mathbf{m} = \mathbf{A.Degree}$ before summation; a zero-degree axis contributes zero, and an all-zero tensor degree produces complete zero rate rows. The result is rate-dependent and generally discontinuous across cell faces. Sections [A.9](#) and [A.9.1](#) derive the reduction, re-elevation, and deterministic rate-row order. A function-only source raises `pdmat:FunctionOnlyDiff`; malformed or inconsistent bounds raise `pdmat:InvalidRateBounds` or `pdmat:RateBoundsMismatch`; data that already contain explicit rate rows, including a previous derivative, raise `pdmat:InvalidDiff`.

Examples and output

```
A = pdmat([0 1], {0, 2}, Degree=1, RateBounds=[-1 2]);
D = rhodiff(A);
disp("derivativeClass =")
disp(class(D))
disp("derivativeDegree =")
disp(D.Degree)
disp("derivativeRowValues =")
disp(cell2mat(D.coeffs(1)))
```

```
derivativeClass =
pdmat
derivativeDegree =
    0

derivativeRowValues =
    -2
     4
```

In several parameters, each partial is restored to the common tensor degree before the rate-weighted sum. Here the four rate rows are constant over all four local labels, but the stored degree remains $[1, 1]$.

```
A = pdmat({[0 1], [0 1]}, ...
    @(rho1, rho2) rho1 + 2*rho2, ...
    Degree=[1 1], RateBounds=[-1 1; -2 3]);
D = rhodiff(A);
derivativeDegree = D.Degree
derivativeRows = cell2mat(D.coeffs([1 1]))
```

```
derivativeDegree = 1×2 double
    1     1

derivativeRows = 4×4 double
    -5    -5    -5    -5
     5     5     5     5
    -3    -3    -3    -3
     7     7     7     7
```

4.7 Inspect Coefficients and Plot Values

4.7.1 bernsteinTable

Syntax

```
T = bernsteinTable(A)
T = bernsteinTable(A, cellSubs)
T = bernsteinTable(A, "oneLine")
T = bernsteinTable(A, cellSubs, "oneLine")
```

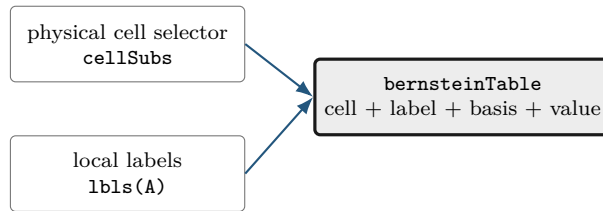
Arguments

- `cellSubs` selects one physical cell.
- `"oneLine"` returns one row per selected physical cell and rate vertex with a compact Bernstein expression.
- Without `"oneLine"`, an ordinary object uses seven metadata columns; an explicit-rate-row object additionally includes `RateVertexIndex` and `RateVertex`. Their deterministic vertex order is induced by applying `helper.combRows` to `num2cell(A.RateBounds,2).'`.

For one parameter, the `Basis` column reports

$$\binom{m}{j} (1 - \alpha)^{m-j} \alpha^j.$$

For multiple parameters, the basis is the tensor product of that expression in each local coordinate, using `alpha1`, `alpha2`, and so on. `LocalIndex` is the local Bernstein label. `CoeffSubscript` maps the local label back to the global coefficient subscript on the physical grid. `IsPhysicalNode` is true when that coefficient lies on a physical grid node.



The table reports stored coefficient evidence. It does not sample a function and does not create solver constraints.

Examples and output

```
A = pdmat({[0 0.2 1]}, {[0 1], [1 1], [1 2]}, Degree=1);
T = bernsteinTable(A, "oneLine");
disp(T)
T1 = bernsteinTable(A, 1, "oneLine");
disp(T1)
```

CellSubscript	Expression
{[1]}	"(1-alpha)*[0 1] + alpha*[1 1]"
{[2]}	"(1-alpha)*[1 1] + alpha*[1 2]"

CellSubscript	Expression
{[1]}	"(1-alpha)*[0 1] + alpha*[1 1]"

Use the full table when the metadata columns are the point of the example:

```
A = pdmat({[0 1]}, {1, 2, 3}, Degree=2);
T = bernsteinTable(A);
disp(T)
```

TermIndex	CellSubscript	CoeffSubscript	LocalIndex	Basis	IsPhysicalNode	Value
1	{[1]}	{[1]}	{[0]}	"(1-alpha)^2"	true	{[1]}
2	{[1]}	{[2]}	{[1]}	"2(1-alpha)alpha"	false	{[2]}
3	{[1]}	{[3]}	{[2]}	"alpha^2"	true	{[3]}

Remarks

A function-only object has no coefficient evidence and raises `pdmat:FunctionOnlyBernsteinTable`. Unknown text options, repeated selectors, or invalid physical-cell selectors raise `pdmat:InvalidBernsteinTableInput`.

For an explicit-rate-row object, `bernsteinTable(A,"oneLine")` returns one expression for every stored vertex rather than merging the alternatives:

```
A = pdmat([0 1], {[1, 3; 10, 14]}, ...
    Degree=1, RateBounds=[-1 2]);
T = bernsteinTable(A, "oneLine");
disp("vertexIndices =")
disp(T.RateVertexIndex)
disp("rateVertices =")
disp(vertcat(T.RateVertex{:}))
disp("expressionCount =")
disp(height(T))
```

```
vertexIndices =
     1
     2

rateVertices =
```

```

-1
 2

expressionCount =
 2

```

4.7.2 disp And display

Syntax

```

disp(A)
display(A)
A

```

Arguments

`disp(A)` prints a compact one-line summary. `display(A)` is the command-window display used when an expression is not terminated by a semicolon; it prints grid, degree, coefficient, and source metadata.

Examples and output

```

A = pdmat({[0 1]}, {1, 2}, Degree=1);
disp(A)
display(A)

```

```

pdmat [1 1] over 1-D grid, degree 1, source coefficient-backed, rate rows false
A =
  pdmat with 1-by-1 matrix values
  Parameters: 1
    rho_1: [0, 1], 2 nodes
  Degree: 1
  Physical cells: 1
  Coefficients per cell: 2
  Continuous: true
  Rate metadata: false
  Explicit rate rows: false
  Source: coefficient-backed
  Function handle: false

```

4.7.3 plot

Syntax

```
h = plot(A)
h = plot(A, dims)
h = plot(A, SamplesPerCell=n)
h = plot(A, dims, SamplesPerCell=n)
h = plot(A, ..., RateVertex=k)
h = plot(A, ..., Name, Value)
```

Arguments

- `dims` is one or two parameter indices. For one-parameter objects, the default is 1; otherwise the default is `[1 2]`, as in `plot(A, [1 2])`.
- `SamplesPerCell` is the package-owned sampling option. The default is 15; use `SamplesPerCell=4` for a coarse diagnostic plot.
- `RateVertex=k` selects one one-based stored rate row in the deterministic vertex order induced by applying `helper.combRows` to `num2cell(A.RateBounds,2).'`. Rate-row objects default to row one. Supplying this option for ordinary data or selecting beyond 2^ℓ raises `pdmatrix:InvalidRateVertex`.
- Remaining name-value pairs pass through to MATLAB graphics. One-dimensional plots pass them to `plot`; two-dimensional plots pass them to `surf`. Examples include `LineWidth`, `FaceAlpha`, and `EdgeColor`.
- The output `h` is the vector of graphics handles, one entry per matrix element.

Remarks

An invalid parameter index, a repeated dimension, or a request for more than two plotting dimensions raises `pdmatrix:InvalidPlotDimensions`. Malformed plotting options or a nonpositive integer `SamplesPerCell` raise `pdmatrix:InvalidPlotOptions`. Errors from MATLAB graphics options remain MATLAB graphics errors. Every unselected parameter is fixed at its lower grid bound, `A.GridInfo.Bounds(:,1)`. A plot of a three-parameter object is therefore a one- or two-dimensional slice, not a volume plot.

For explicit rate rows, select the vertex as part of the plotting call:

```
A = pdmat([0 1], {0, 2}, Degree=1, RateBounds=[-1 2]);
D = rhodiff(A);
plot(D, RateVertex=2, SamplesPerCell=20, LineWidth=2);
```

Examples and output

Each source listing below is the complete source used by `generate_plot_figures.m` for the adjacent result.

One-dimensional entries.

```
A = pdmat({[-1 1]}, @(rho) [rho, rho.^2]);
h = plot(A, SamplesPerCell=80, LineWidth=2);
set(h(1), "Color", [35 86 125]/255, "LineStyle", "-")
set(h(2), "Color", [0.25 0.25 0.25], "LineStyle", "--")
grid on
xlabel("rho")
ylabel("matrix entry")
title("One-dimensional pdmat entries")
legend(["A(1,1)", "A(1,2)"], "Location", "eastoutside")
```

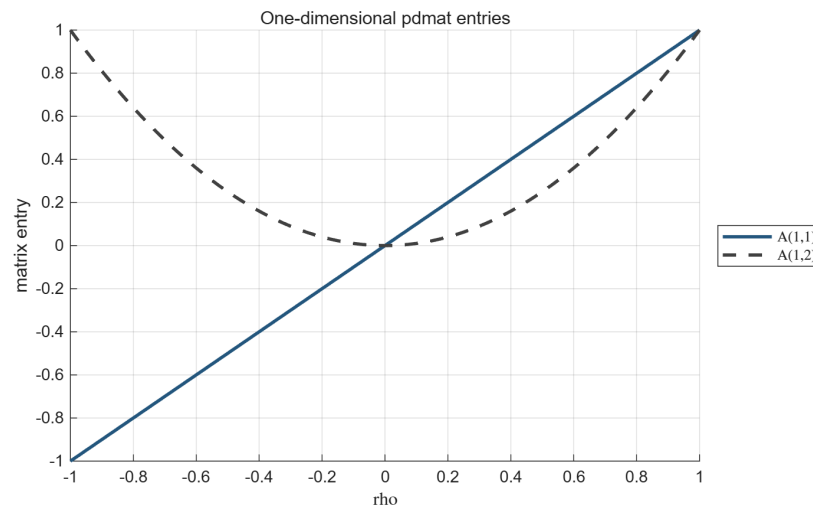


Figure 4.1: Result of the one-dimensional source above. The method returns one line handle per matrix entry.

Two-dimensional surface.

```
A = pdmat({[-1 1], [0 2]}, ...
    @(rho1, rho2) 1 + rho1.^2 + 0.5*rho2 + rho1.*rho2);
h = plot(A, [1 2], SamplesPerCell=45, ...
    EdgeColor="none", FaceAlpha=0.78);
set(h, "FaceColor", [35 86 125]/255)
grid on
view(35, 25)
xlabel("rho1", "Interpreter", "none")
ylabel("rho2", "Interpreter", "none")
zlabel("matrix entry")
title("Two-dimensional pdmat surface")
```

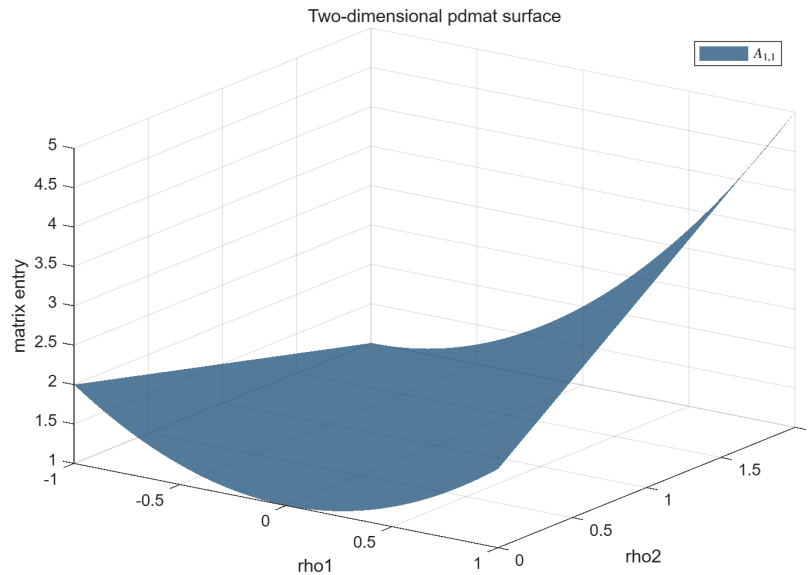


Figure 4.2: Result of the two-dimensional source above.

Two-dimensional matrix-valued surface.

```
A = pdmat({[-1 1], [0 2]}, @(rho1, rho2) [ ...
    1 + 0.8*rho1 + 0.25*rho2; ...
    1 - 0.8*rho1 + 0.25*rho2]);
h = plot(A, [1 2], SamplesPerCell=45, ...
    EdgeColor="none", FaceAlpha=0.62);
set(h(1), "FaceColor", [35 86 125]/255)
set(h(2), "FaceColor", [0.55 0.55 0.55])
grid on
view(35, 25)
xlabel("rho1", "Interpreter", "none")
ylabel("rho2", "Interpreter", "none")
zlabel("matrix entry")
title("Two entries of a 2-by-1 pdmat cross at rho1 = 0")
legend(h, ["A(1,1)", "A(2,1)"], ...
    "Location", "eastoutside")
```

The two entries are equal for every ρ_2 along $\rho_1 = 0$. Their ordering reverses on opposite sides: the first entry is lower when $\rho_1 < 0$ and higher when $\rho_1 > 0$. The crossing is therefore a property of the matrix function, not an artificial plotting offset.

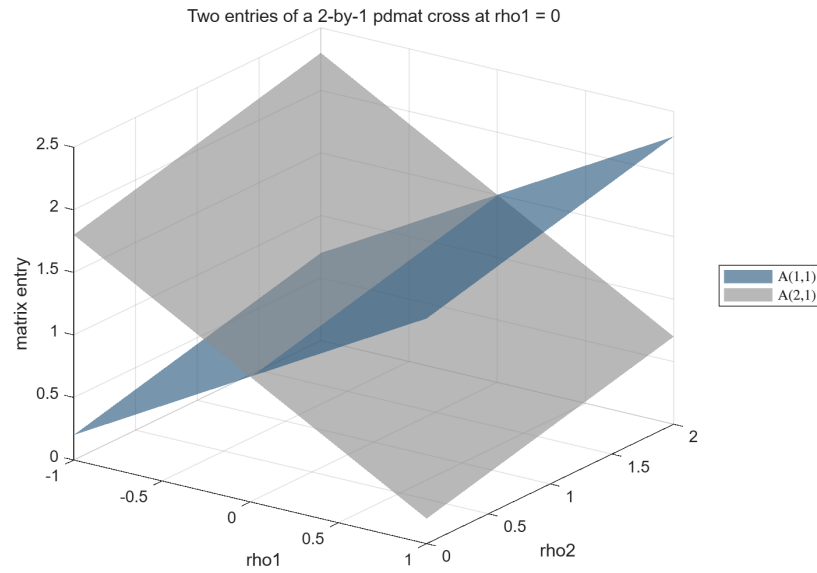


Figure 4.3: Two entry surfaces of a 2-by-1 matrix-valued function. The outside legend distinguishes the entries while their intersection remains visible along $\rho_1 = 0$.

4.8 Combine Known Matrices

A `pdmat` operation acts on complete matrix coefficients while the physical grid remains attached to the result. The blocks below deliberately show one operation family at a time: purpose, supported syntax, input, immediate output, and—when the result is spatial—a before/after picture.

4.8.1 Signs, Addition, And Subtraction

Use signs, sums, and differences to build known matrix expressions. For two parameter-dependent operands, the backend first checks matching outer bounds, forms the sorted union of the interior nodes on every axis, and exactly subdivides/re-expresses both coefficient trees on that common grid. It then elevates both operands componentwise to $\max(\mathbf{m}_A, \mathbf{m}_B)$ and combines matching coefficients. Numeric and affine constant operands are promoted on the active grid before the same coefficientwise operation. The corresponding MATLAB overload names are `plus`, `minus`, `uplus`, and `uminus`; sections [A.6](#) and [3.9.7](#) document the exact grid and degree mechanisms.

Syntax

```
S = A + B
S = A + M
S = M + A
D = A - B
D = A - M
D = M - A
P = +A
N = -A
```

Here `M` is a finite real scalar or size-compatible matrix. A scalar broadcasts to the matrix payload size. The first example makes the degree alignment visible.

Examples and output

```
A = pdmat({[0 1]}, {1, 2}, Degree=1);
B = pdmat({[0 1]}, {10, 20, 30}, Degree=2);
S = A + B;
sumDegree = S.Degree
sumCoefficients = cell2mat(S.coeffs(1))
```

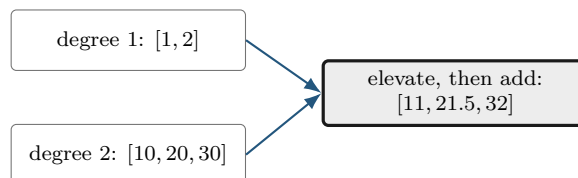
```
sumDegree = 2
sumCoefficients = 1×3 double
    11.0000    21.5000    32.0000
```

The middle coefficient of the elevated A is $(1 + 2)/2 = 1.5$, so the middle sum is $1.5 + 20 = 21.5$. Subtraction and unary signs act immediately on the same local coefficients.

```
D = 5 - A;
N = -A;
differenceCoefficients = cell2mat(D.coeffs(1))
negativeAtOne = N.evaluate(1)
positiveClass = class(+A)
```

```
differenceCoefficients = 1×2 double
     4     3
```

```
negativeAtOne = -2
positiveClass = 'pdmat'
```



When grids have the same outer bounds but different interior nodes, the union grid is used before addition or subtraction. The next example combines this refinement with unequal degrees; the two input objects remain unchanged.

```
A = pdmat({[0 1]}, @(rho) rho, Degree=1);
B = pdmat({[0 0.5 1]}, @(rho) 2*rho, Degree=2);
S = A + B;
mergedGrid = S.GridInfo.Vectors{1}
resultDegree = S.Degree
cell1Coefficients = cell2mat(S.coeffs(1))
cell2Coefficients = cell2mat(S.coeffs(2))
```

```
mergedGrid = 1×3 double
     0     0.5000     1.0000
```

```
resultDegree = 2
```

```
cell1Coefficients = 1×3 double
     0     0.7500     1.5000
```

```
cell2Coefficients = 1×3 double
    1.5000    2.2500    3.0000
```

4.8.2 Ordered Matrix Multiplication

Use `mtimes` for a matrix product. The inner payload dimensions must match. Two parameter-dependent factors produce the componentwise degree sum; a numeric factor is constant and does not raise any degree component.

Syntax

```
K = A * B
K = A * M
K = M * A
```

For one parameter, fix physical cell c and let $A = \sum_i B_i^m A^{(c)}[i]$ and $D = \sum_j B_j^n D^{(c)}[j]$. Then $K = AD$ has coefficients

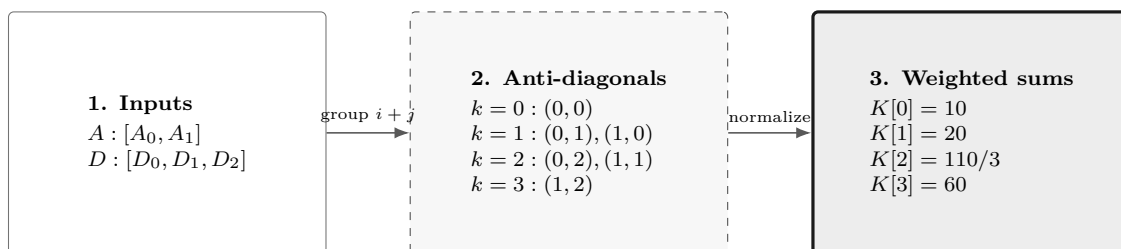
$$K^{(c)}[k] = \sum_{i+j=k} \frac{\binom{m}{i} \binom{n}{j}}{\binom{m+n}{k}} A^{(c)}[i] D^{(c)}[j].$$

The left/right matrix order is never exchanged. The formula is a weighted convolution of the local coefficients, not a simple sum of products. Its weights are combinatorial normalization ratios: multiplying $B_i^m B_j^n$ yields $\binom{m}{i} \binom{n}{j} / \binom{m+n}{i+j}$ times B_{i+j}^{m+n} . They are not Bernstein-basis evaluations at a selected local coordinate.

Examples and output

```
A = pdmat({[0 1]}, {1, 2}, Degree=1);
B = pdmat({[0 1]}, {10, 20, 30}, Degree=2);
K = A * B;
productDegree = K.Degree
productCoefficients = cell2mat(K.coeefs(1))
```

```
productDegree = 3
productCoefficients = 1×4 double
    10.0000    20.0000    36.6667    60.0000
```



Coefficient labels add under the weighted convolution; payload matrices multiply in the written order. The three panels make the convolution readable without crossing contribution arrows.

For tensor data, label addition and binomial weights are applied independently in each parameter direction. If $\mathbf{m}, \mathbf{n} \in \mathbb{N}^\ell$ are direction-wise basis degrees, the ℓ -dimensional ordered tensor convolution is

$$K^{(\mathbf{c})}[\mathbf{k}] = \sum_{\mathbf{i}+\mathbf{j}=\mathbf{k}} \left(\prod_{s=1}^{\ell} \frac{\binom{m_s}{i_s} \binom{n_s}{j_s}}{\binom{m_s+n_s}{k_s}} \right) A^{(\mathbf{c})}[\mathbf{i}] D^{(\mathbf{c})}[\mathbf{j}].$$

Two degree-one factors in two parameters produce degree two and $3^2 = 9$ coefficients per physical cell. Section A.8 derives the convolution by multiplying normalized tensor-product basis functions and grouping every ordered contribution with $\mathbf{k} = \mathbf{i} + \mathbf{j}$.

Remarks

Incompatible operands for `+`, `-`, and `*` raise `pdmat:InvalidAddition`, `pdmat:InvalidSubtraction`, or `pdmat:InvalidMultiplication`. Parameter counts or outer bounds that cannot be merged raise `pdmat:MixedGrid`. Coefficient algebra on a function-only object raises `pdmat:FunctionOnly` `Algebra`; construct the function with an explicit verified `Degree` when coefficient evidence is required.

4.9 Compare and Certify Known Data

Syntax

```
tf = A == B
C = A <= B
C = A >= B
C = A <= M
C = A >= M
```

Arguments

For two `pdmat` operands, `A==B` is a scalar logical coefficient-evidence comparison. Subtraction performs the compatibility checks, grid alignment, and degree elevation; the result is true exactly when the aligned coefficient difference is provably zero. It never creates a `pdlmi`. By contrast, `<=` and `>=` create a `pdlmi` from a compatible coefficient-backed known residual. A function-only object cannot supply finite coefficient evidence. Numeric `M` follows the ordinary finite-real size and scalar-broadcast rules.

Examples and output

```
A = pdmat([0 1], {1, 3}, Degree=1);
B = pdmat([0 1], {1, 2, 3}, Degree=2);
same = A == B
different = A == 2*A

yalmp('clear')
K = pdmat([0 1], {{1, 2; 3, 4}}, ...
    Degree=1, RateBounds=[-1 2]);
direct = K >= 0;
```

```

directCertificate = toYalmip(direct)
polya = direct.applyPolya(1);
polyaCertificate = toYalmip(polya)

```

```

same = logical
      1

different = logical
        0

directCertificate = logical
                1

polyaCertificate = logical
                    1

```

Remarks

Equality with a non-`pdmat` operand raises `pdmat:InvalidEquality`. Between two `pdmat` objects, subtraction owns compatibility checks and representation alignment, so incompatible operands propagate the corresponding subtraction, grid, continuity, or rate-bound error. A known `pdmat` equality is deliberately not a `pdlmi` residual. Direct and Pólya known-data exports reduce the complete family to one scalar logical certificate using the wrapper’s global inequality classification and inclusive absolute tolerance 10^{-10} . In semidefinite mode, every coefficient matrix is symmetrized as $(C + C^T)/2$ and its eigenvalues are tested against the oriented tolerance; scalar entries are not checked independently. Only element-wise mode tests every scalar matrix entry. A false result emits `pdlmi:InconclusiveCertificate` when exported and does not prove a continuous violation or indefiniteness. Putinar, SparseFullBox, and FullBox still return YALMIP constraints and may introduce auxiliary Gram variables even when the residual is known.

4.10 Reshape and Transform Known Matrices

These methods transform every local matrix coefficient in the same way. They are inherited from the centralized `pdbase` implementation, while the `pdmat` reconstruction hook preserves the `pdmat` class, known-data payloads, and metadata. They do not transpose, reshape, or concatenate the physical grid. Except for the documented `evaluator/display/shape` paths, structural methods require coefficient evidence.

4.10.1 Shape, Equality, And Display Protocols

Shape queries describe the matrix payload, not the grid. `isequal` compares object representations after compatible grid refinement and degree elevation; scalar logical operator equality is documented in section 4.9. The protocol methods support ordinary MATLAB indexing syntax and dot access; they are not a second storage API.

Syntax

```
sz = size(A)
d = size(A, dim)
n = height(A)
n = width(A)
n = length(A)
n = numel(A)
n = ndims(A)
tf = isequal(A, B, ...)
idx = end(A, k, n)
n = numArgumentsFromSubscript(A, S, ctx)
```

Examples and output

```
A = pdmat({[0 1]}, ...
    {[1 2 3; 4 5 6], [10 20 30; 40 50 60]}, Degree=1);
matrixSize = size(A)
rowCount = height(A)
columnCount = width(A)
longestDimension = length(A)
elementCount = numel(A)
dimensionCount = ndims(A)
equalsUnaryPlus = isequal(A,+A)
```

```
matrixSize = 1×2 double
           2     3

rowCount = 2
columnCount = 3
longestDimension = 3
elementCount = 6
dimensionCount = 2
equalsUnaryPlus =
    logical
         1
```

A function-only object can use shape queries and `isequal` compares its stored metadata and function handle. A coefficient-backed comparison that cannot form a common grid raises `pdmat:InvalidEquality`. The detailed multi-line `display` transcript is shown in section [4.7.2](#).

4.10.2 Transpose And Conjugate Transpose

Both transpose forms act on every coefficient. Because `pdmat` payloads are finite and real, conjugation does not change their values.

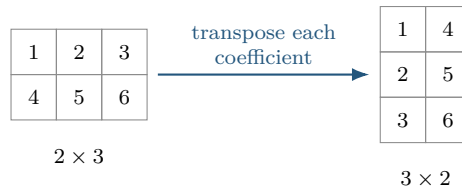
Syntax

```
T = transpose(A)
H = ctranpose(A)
T = A.'
H = A'
```

Examples and output

```
A = pdmat({[0 1]}, ...
          {[1 2 3; 4 5 6], [10 20 30; 40 50 60]}, Degree=1);
T = A.';
H = A';
transposeSize = size(T)
formsEqual = isequal(T,H)
transposeAtZero = T.evaluate(0)
```

```
transposeSize = 1×2 double
              3      2
formsEqual =
    logical
         1
transposeAtZero = 3×2 double
                1      4
                2      5
                3      6
```



4.10.3 Vectorization, Reshape, And Squeeze

Use these methods to change the matrix view while preserving column-major element order, the parameter grid, the degree, and every coefficient value. **reshape** accepts exactly two positive dimensions with at most one inferred [] dimension. **squeeze** retains the package's two-dimensional matrix convention.

Syntax

```
V = vec(A)
R = reshape(A, [m n])
R = reshape(A, m, n)
S = squeeze(A)
```

Examples and output

```
A = pdmat({[0 1]}, ...
          {[1 2 3; 4 5 6], [10 20 30; 40 50 60]}, Degree=1);
V = vec(A);
R = reshape(A, [3 2]);
S = squeeze(A);
vectorSize = size(V)
reshapeSize = size(R)
squeezeSize = size(S)
vectorAtZero = V.evaluate(0).'
```

```
vectorSize = 1×2 double
    6     1
reshapeSize = 1×2 double
    3     2
squeezeSize = 1×2 double
    2     3
vectorAtZero = 1×6 double
    1     4     2     5     3     6
```

Malformed or element-count-changing reshape requests raise `pdmat:InvalidReshape`.

4.10.4 Diagonals And Trace

`diag` extracts a selected diagonal from a matrix, or builds a square diagonal matrix from a vector. A finite integer offset `k` may not select an empty diagonal. `trace` returns a 1-by-1 `pdmat`.

Syntax

```
d = diag(A)
d = diag(A, k)
D = diag(v)
D = diag(v, k)
t = trace(A)
```

Examples and output

```
A = pdmat([[0 1]], [[1 2; 3 4], [10 20; 30 40]], Degree=1);
d = diag(A);
D = diag(d);
t = trace(A);
diagonalAtZero = d.evaluate(0)
rebuiltAtZero = D.evaluate(0)
traceAtZero = t.evaluate(0)
```

```
diagonalAtZero = 2×1 double
    1
    4
rebuiltAtZero = 2×2 double
    1    0
    0    4
traceAtZero = 5
```

Invalid offsets or empty selections raise `pdmat:InvalidDiag`.

4.10.5 Triangular Parts And Reductions

Triangular extraction preserves or zeros entries relative to a selected diagonal. Reductions act along matrix dimensions; "all" is supported by `sum` and `mean`.

Syntax

```
L = tril(A)
L = tril(A, k)
U = triu(A)
U = triu(A, k)
S = sum(A)
S = sum(A, dim)
S = sum(A, vecdim)
S = sum(A, "all")
M = mean(A)
M = mean(A, dim)
M = mean(A, vecdim)
M = mean(A, "all")
C = cumsum(A)
C = cumsum(A, dim)
C = cumsum(A, direction)
C = cumsum(A, dim, direction)
```

First inspect the two triangular results.

Examples and output

```
A = pdmat({[0 1]}, {[1 2; 3 4]}, [10 20; 30 40]), Degree=1);
L = tril(A);
U = triu(A);
lowerAtZero = L.evaluate(0)
upperAtZero = U.evaluate(0)
```

```
lowerAtZero = 2×2 double
    1    0
    3    4
upperAtZero = 2×2 double
    1    2
    0    4
```

Then apply reductions to the same coefficient matrix.

```
rowSum = sum(A, 2);
allMean = mean(A, "all");
running = cumsum(A, 2);
rowSumAtZero = rowSum.evaluate(0)
allMeanAtZero = allMean.evaluate(0)
runningAtZero = running.evaluate(0)
```

```
rowSumAtZero = 2×1 double
    3
    7
allMeanAtZero = 2.5000
runningAtZero = 2×2 double
    1    3
    3    7
```

`vecdim` is a nonempty vector of unique positive integer dimensions; dimensions above two are singleton no-ops. `direction` is "forward" or "reverse", case-insensitively. A direction supplied without a dimension uses the first nonsingleton matrix dimension, and `cumsum` along a dimension above two returns the same coefficient payloads. Invalid triangular offsets raise `pdmat:InvalidTriangularPart`. Invalid reduction calls raise `pdmat:InvalidSum`, `pdmat:InvalidMean`, or `pdmat:InvalidCumsum`; missing-value and native-output flags are not supported.

4.10.6 Reordering And Rotation

These methods reorder rows or columns inside every coefficient; they never reverse the scheduling grid.

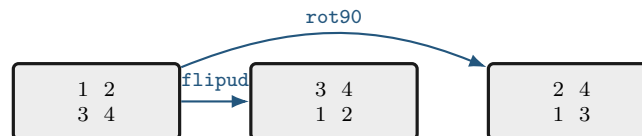
Syntax

```
B = flip(A)
B = flip(A, dim)
B = flipud(A)
B = fliplr(A)
B = rot90(A)
B = rot90(A, k)
```

Examples and output

```
A = pdmat({[0 1]}, {[1 2; 3 4], [10 20; 30 40]}, Degree=1);
F = flipud(A);
R = rot90(A);
flippedAtZero = F.evaluate(0)
rotatedAtZero = R.evaluate(0)
```

```
flippedAtZero = 2×2 double
    3     4
    1     2
rotatedAtZero = 2×2 double
    2     4
    1     3
```



Invalid dimensions or quarter-turn counts raise `pdmat:InvalidFlip` or `pdmat:InvalidRot90`.

4.10.7 Concatenation And Block Assembly

Concatenation joins payload matrices after compatible grid refinement and exact elevation to the componentwise maximum degree. Numeric blocks are promoted to constant known data. `cat` supports only dimensions 1 and 2.

Syntax

```
H = horzcat(A, B, ...)
V = vertcat(A, B, ...)
H = [A, B]
V = [A; B]
C = cat(dim, A, B, ...)
```

```
D = blkdiag(A, B, ...)
```

Examples and output

```
a = pdmat({[0 1]}, {1, 2}, Degree=1);
b = pdmat({[0 1]}, {10, 20}, Degree=1);
H = [a, b];
V = [a; b];
D = blkdiag(a, 5);
horizontalAtZero = H.evaluate(0)
verticalAtZero = V.evaluate(0)
blockDiagonalAtZero = D.evaluate(0)
```

```
horizontalAtZero = 1×2 double
    1    10
verticalAtZero = 2×1 double
    1
   10
blockDiagonalAtZero = 2×2 double
    1    0
    0    5
```

Incompatible blocks or syntax raise `pdmat:InvalidConcatenation`; unsupported dimensions raise `pdmat:UnsupportedCatDimension`; invalid block-diagonal inputs raise `pdmat:InvalidBlkdiag`. Function-only operands raise `pdmat:FunctionOnlyAlgebra`.

4.10.8 Repetition

`repmat` tiles the matrix payload. It accepts a positive scalar `m`, interpreted as `[m m]`, two positive counts, or a two-element count vector. Grid and Bernstein degree remain unchanged.

Syntax

```
R = repmat(A, m)
R = repmat(A, m, n)
R = repmat(A, [m n])
```

Examples and output

```
v = pdmat({[0 1]}, {[1 2], [3 4]}, Degree=1);
R = repmat(v, [2 2]);
repeatedSize = size(R)
repeatedDegree = R.Degree
repeatedAtZero = R.evaluate(0)
```

```
repeatedSize = 1×2 double
             2     4
repeatedDegree = 1
repeatedAtZero = 2×4 double
              1     2     1     2
              1     2     1     2
```

Invalid repetition shapes raise `pdmat:InvalidRepmat`.

4.11 Index and Assign pdmat Entries

Parenthesis indexing selects a matrix block from every coefficient. Assignment writes the selected block into every coefficient after the same common-grid refinement and componentwise maximum-degree elevation used by addition: the left object and parameter-dependent right-hand side are exactly re-expressed on the sorted union grid, both coefficient families are elevated, and then only the selected payload block is replaced at every aligned label. It does not select a physical cell; use `coeffs` for cell inspection. See sections [A.6](#) and [3.9.7](#).

Syntax

```
B = A(rows, cols)
B = A(1:end, end)
value = A.PropertyName
A(rows, cols) = numericValue
A(rows, cols) = B
```

Examples and output

```
A = pdmat({[0 1]}, ...
    {[1 2 3; 4 5 6], [10 20 30; 40 50 60]}, Degree=1);
lastCol = A(:, end);
A(:, 2) = 5;
selectedSize = size(lastCol)
assignedSize = size(A)
selectedAtZero = lastCol.evaluate(0)
assignedAtZero = A.evaluate(0)
```

```
selectedSize = 1×2 double
    2     1
assignedSize = 1×2 double
    2     3
selectedAtZero = 2×1 double
    3
    6
assignedAtZero = 2×3 double
    1     5     3
    4     5     6
```

Assignment also aligns a parameter-dependent right-hand side with a different grid and degree.

```
A = pdmat({[0 1]}, @(rho) [1 rho; 2 3*rho], Degree=1);
B = pdmat({[0 0.5 1]}, @(rho) [rho.^2; 1+rho], Degree=2);
A(:, 2) = B;
assignedGrid = A.GridInfo.Vectors{1}
assignedDegree = A.Degree
assignedValue = A.evaluate(0.25)
```

```
assignedGrid = 1×3 double
    0    0.5000    1.0000

assignedDegree = 2

assignedValue = 2×2 double
    1.0000    0.0625
    2.0000    1.2500
```

`subsref`, `subsasgn`, `end`, and `numArgumentsFromSubscript` implement this two-index MATLAB protocol. Invalid selectors raise `pdmat:InvalidSubscript`. Linear indexing, deletion, or non-parenthesis assignment raises `pdmat:UnsupportedAssignment`; an incompatible right-hand side raises `pdmat:InvalidAssignment`. Slicing and assignment require coefficient evidence and otherwise raise `pdmat:FunctionOnlyAlgebra`.

4.12 Boundaries and Related APIs

1. Function-only objects created without `Degree` do not have Bernstein coefficient evidence for `bernsteinTable`, coefficient algebra, differentiation, or inequalities.
2. Numeric operands must be finite real nonempty matrices with compatible sizes, or finite real scalars that can broadcast to the required size.
3. Ordinary one-row operands broadcast across explicit rate rows. Two explicit-rate-row operands must have exactly matching grids and nonempty rate bounds; they are not automatically refined. Multiplication supports explicit rate rows on at most one side and rejects rate-row by rate-row products.
4. Algebra, concatenation, indexing, and assignment preserve compatible rate metadata and stored row order. Mismatched rate boxes, grids, row counts, or rate-dependent product forms raise the owning operation's class-specific error.

Related APIs. `pdmatrix`, `evaluate`, `rhodiff`, `bernsteinTable`, `plot`, `pdlmi`, `pdbase`.

Chapter 5

pdvar: Decision Matrix Functions

`pdvar` creates continuous cell-local Bernstein decision expressions backed by YALMIP `sdpvar` coefficients. It participates in supported affine and known-data algebra. It does not choose solvers or call `optimize`. Comparison overloads create `pdlmi` objects, whose `toYalmip` method later converts them to YALMIP constraints. Shared matrix-shape, unary, structural, reduction, reordering, and repetition methods are implemented centrally by `pdbase` and remain complete public `pdvar` behavior; this chapter documents their decision-expression syntax, stable shape examples, validations, and rate-aware boundaries.

A degree- m `pdvar` represents a decision such as $P(\boldsymbol{\rho})$ or $y_k(\boldsymbol{\rho})$. For $m \geq 1$, neighboring cells reuse every decision handle on their complete shared face, which enforces value continuity without auxiliary equality constraints; for $m = 0$, the entire grid reuses one decision matrix. `rhodiff` applies cellwise forward differences scaled by $1/h_s^c$ and stores one row per vertex in $\mathcal{V}_{\mathcal{R}}$, while `value` recovers numeric coefficient payloads only after the caller has validated YALMIP's solver status. Products remain affine only when the other factor is known numeric or `pdmat` data.

5.1 API Map

Task	Entry
Construct decisions	<code>pdvar</code>
Inspect coefficients	<code>bernsteinTable</code> , <code>shared storage/evaluation/elevation</code>
Convert assigned coefficients	<code>value</code>
Differentiate rates	<code>rhodiff</code>
Affine/mixed algebra	<code>+</code> , <code>-</code> , <code>*</code> with known data
Matrix/index methods	<code>structural methods</code> , <code>indexing and assignment</code>
Build LMIs	<code><=</code> , <code>>=</code> , <code>==</code>

5.2 Construct pdvar Decisions

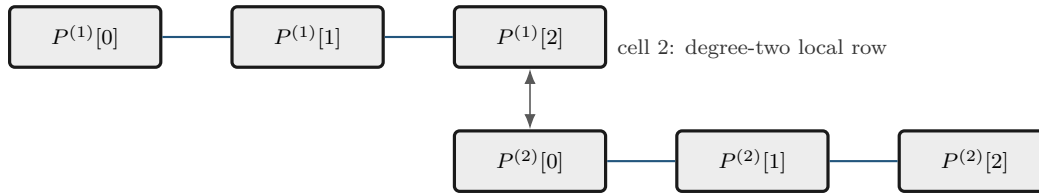
Syntax

```
P = pdvar(n, gridVectors)
P = pdvar(n, n, gridVectors)
P = pdvar(n, p, gridVectors)
P = pdvar(..., "full")
P = pdvar(..., "symmetric")
P = pdvar(..., Degree=0)
P = pdvar(..., Degree=1)
P = pdvar(..., Degree=degree)
P = pdvar(..., RateBounds=rb)
P = pdvar(..., ValidationMode=mode)
```

Arguments

- `pdvar(n, gridVectors)` creates an $n \times n$ square variable using scalar-size shorthand; for example, `pdvar(2, {[0 1]})`. Square variables default to symmetric YALMIP coefficients.
- `pdvar(n, n, gridVectors)` is the explicit square-size form; for example, `pdvar(2, 2, {[0 1]})`. It has the same default symmetric structure as the shorthand.
- `pdvar(n, p, gridVectors)` creates an $n \times p$ variable. Rectangular variables default to full YALMIP coefficients; for example, `pdvar(2, 3, {[0 1]})`.
- "full" and "symmetric" explicitly choose the YALMIP coefficient structure. "symmetric" requires a square size.
- `Degree=degree` accepts one finite nonnegative integer scalar shorthand or an ℓ -element vector and is stored as a 1-by- ℓ row $\mathbf{m} = (m_1, \dots, m_\ell)$. The omitted default is **1**. An explicitly supplied scalar on a multidimensional grid expands uniformly and emits `pdvar:ScalarDegreeExpansion` once; omitted defaults and one-dimensional forms remain warning-free. An all-zero vector creates one parameter-independent decision matrix shared across the grid. Otherwise neighboring cells share complete control faces; a zero component means constancy in that parameter direction.
- `RateBounds=rb` stores parameter-rate metadata for later `rhodiff(P)` calls. Use `RateBounds=[-1 1]` for one parameter; use a two-row matrix for two parameters, as shown in the rate-metadata example below.
- `ValidationMode=mode` accepts scalar text "fast" or "strict", case-insensitively, and defaults to "fast". It affects repeated validation of package-generated storage only; public options, grid, structure, affine safety, continuity, and source semantics remain global in both modes. The mode is not stored.

cell 1: degree-two local row



same YALMIP handle: $P^{(1)}[2] = P^{(2)}[0]$

For positive degree, neighboring physical cells reuse complete boundary-face YALMIP handles; no extra continuity LMI is generated.

Examples and output

The square shorthand creates a 2×2 continuous decision object. The default degree is one and the object carries no rate metadata.

```
yalmip('clear')
Ps = pdvar(2, {[0 1]});
P= Ps.coeffs([1]);
symmetricCoefficient = ishermitian(P{1})
variableClass = class(Ps)
matrixSize = size(Ps)
degree = Ps.Degree
rateDependent = Ps.HasRateDependence
```

```
symmetricCoefficient = 1
variableClass = 'pdvar'
matrixSize = 1×2 double
    2    2

degree = 1

rateDependent =
    logical
    0
```

Use two dimension arguments for a rectangular full variable.

```
yalmip('clear')
Pf = pdvar(2, 3, {[0 1]});
variableClass = class(Pf)
matrixSize = size(Pf)
degree = Pf.Degree
```

```
variableClass = 'pdvar'
matrixSize = 1×2 double
    2    3

degree = 1
```

The structure flag makes the coefficient structure explicit. This is useful when a square variable should use full, not symmetric, coefficients.

```
yalmp('clear')
Pfull = pdvar(2, 2, {[0 1]}, "full");
P = Pfull.coeffs([1]);
fullCoefficient = ishermitian(P{1})
variableClass = class(Pfull)
matrixSize = Pfull.MatrixSize
degree = Pfull.Degree
```

```
fullCoefficient = logical
0
```

```
variableClass = 'pdvar'
matrixSize = 1×2 double
2 2
```

```
degree = 1
```

Use Degree=0 for a parameter-independent decision variable stored with grid metadata.

```
yalmp('clear')
P0 = pdvar(1, [0 1 2], Degree=0);
variableClass = class(P0)
degree = P0.Degree
cellCount = P0.ncell()
```

```
variableClass = 'pdvar'
degree = 0
cellCount = 2
```

Direction-wise degree permits variation in selected parameters and exact constancy in the others.

```
yalmp('clear')
Pa = pdvar(1, {[0 1], [10 20]}, Degree=[2 0]);
degree = Pa.Degree
labels = Pa.lbls()
coefficientCount = Pa.ncoeff()
```

```
degree = 1×2 double
2 0
```

```
labels = 3×2 double
0 0
1 0
2 0
```

```
coefficientCount = 3
```

Rate bounds are metadata for later `rhodiff(P)` calls. `pdlmi` constrains rate vertices only after an expression such as `rhodiff` has stored rate-vertex rows.

```
yal mip('clear')
Pr = pdvar(1, {[0 1], [10 20]}, RateBounds=[-1 2; -3 4]);
variableClass = class(Pr)
rateDependent = Pr.HasRateDependence
rateBounds = Pr.RateBounds
```

```
variableClass = 'pdvar'
rateDependent =
    logical
     1

rateBounds = 2×2 double
    -1     2
    -3     4
```

Remarks

- Missing or misplaced dimensions and grids raise `pdvar:InvalidInput`; invalid sizes and degrees raise `pdvar:InvalidMatrixSize` or `pdvar:InvalidDegree`. Explicit multidimensional scalar degree expansion is accepted with warning `pdvar:ScalarDegreeExpansion`.
- Option failures use `pdvar:InvalidOptions`, `pdvar:UnknownOption`, or `pdvar:UnsupportedOption`; invalid modes raise `pdvar:InvalidValidationMode`.
- A symmetric nonsquare request raises `pdvar:InvalidStructure`; grid failures use `pdvar:InvalidGrid` or `pdvar:InvalidGridVector`.
- `RateBounds` must be a finite ℓ -by-2 matrix with each lower bound no greater than its upper bound; violations raise `pdbase:InvalidRateBounds`.

5.3 Inspect Object Properties

`pdvar` declares no additional class-owned properties. It exposes the read-only inherited `pdbase` properties listed in section 3.3; their private set access prevents user assignment. In particular, decision coefficients are inspected through `P.LocalValues`, while `P.Degree`, `P.IsContinuous`, and `P.RateBounds` describe their representation and rate metadata.

```
yal mip('clear')
P = pdvar(2, {[0 1]}, "symmetric", Degree=2, RateBounds=[-1 1]);
degree = P.Degree
isContinuous = P.IsContinuous
rateBounds = P.RateBounds
firstCellCoefficientCount = numel(P.coeffs(1))
```

```
degree = 2

isContinuous =
    logical
     1
```

```
rateBounds = 1×2 double
    -1      1
```

```
firstCellCoefficientCount = 3
```

Use public algebra to create a modified value; direct assignments such as `P.Degree = 1` are not supported.

5.4 Inspect, Evaluate, And Elevate Inherited Storage

Syntax

```
cellSubs = cells(P)
labels = lbls(P)
coefficients = coeffs(P, cellSubs)
n = ncell(P)
n = ncoeff(P)
n = npar(P)
X = evaluate(P, point)
X = P.evaluate(point)
vals = P.elevVals(degreeIncrement)
Q = P.elevate(degreeIncrement)
```

Description

The storage queries use the shared `pdbase` cell and label order and return the symbolic `sdpvar` coefficient payloads of the selected decision expression. `evaluate` reconstructs the affine symbolic matrix at one in-bounds parameter point; a derivative/rate-row object returns a 1-by- 2^ℓ cell array in stored rate-vertex order. `elevVals` returns only an elevated nested coefficient tree. `elevate` returns a new `pdvar` of the same dynamic class with the same represented expression, physical cells, YALMIP decision variables, metadata, and rate-row order at the higher degree; it does not create new decision freedom or modify `P`. The backend performs the same binomially normalized ℓ -dimensional convolution as known-data elevation, using the fixed separable kernel $K_{\mathbf{M}-\mathbf{m}}[\mathbf{r}] = \prod_s \binom{M_s - m_s}{r_s}$. Section A.6 derives this exact representation change.

Examples and output

```
yalmp('clear')
P = pdvar(2, {[0 1]}, "full", Degree=1);
physicalCells = P.cells()
labels = P.lbls()
coefficientTableSize = size(P.coeffs(1))
counts = [P.ncell(), P.ncoeff(), P.npar()]
X = P.evaluate(0.25);
Q = P.elevate(2);
sampleClass = class(X)
sampleSize = size(X)
elevatedClass = class(Q)
degrees = [P.Degree, Q.Degree]
elevatedCoefficientTableSize = size(Q.coeffs(1))
```

```
physicalCells = 1
labels = 2×1 double
    0
    1
coefficientTableSize = 1×2 double
    1    2
counts = 1×3 double
    1    2    1
sampleClass = 'sdpvar'
sampleSize = 1×2 double
    2    2
elevatedClass = 'pdvar'
degrees = 1×2 double
    1    3
elevatedCoefficientTableSize = 1×2 double
    1    4
```

Vector increments may elevate different axes by different amounts, including an axis on which the original decision is constant.

```
yalmp('clear')
P = pdvar(1, {[0 1], [0 1]}, Degree=[1 0]);
Q = P.elevate([1 2]);
sourceDegree = P.Degree
elevatedDegree = Q.Degree
coefficientCounts = [P.ncoeff(), Q.ncoeff()]
elevatedCoefficientTableSize = size(Q.coeffs([1 1]))
```

```
sourceDegree = 1×2 double
    1    0

elevatedDegree = 1×2 double
    2    2

coefficientCounts = 1×2 double
    2    9

elevatedCoefficientTableSize = 1×2 double
    1    9
```

Remarks

Invalid physical-cell selectors raise `pdbase:InvalidCellSubs`. Malformed points raise `pdvar:InvalidPoint`; out-of-bounds points raise `pdvar:PointOutOfBounds`. Invalid elevation increments raise `pdbase:InvalidDegreeIncrement`. Evaluation returns symbolic expressions until the user assigns or solves the underlying YALMIP variables; use `value` for the post-assignment numeric conversion documented in section 5.10.

5.5 rhodiff

Syntax

```
D = rhodiff(P)
D = rhodiff(P, rb)
```

Arguments

- `P` is a `pdvar` without existing rate-vertex rows.
- `rb` is a finite ℓ -by-2 numeric matrix of lower and upper parameter-rate bounds, with each lower bound no greater than its upper bound. It must equal `P.RateBounds` when the object already carries nonempty bounds.
- Without `rb`, `rhodiff(P)` requires nonempty `P.RateBounds`.
- `D` is a discontinuous `pdvar` with one coefficient row per rate vertex. In one parameter, degree $m > 0$ becomes $m - 1$, while degree zero remains zero. In two or more parameters, every nonzero-axis partial is elevated exactly from $\mathbf{m} - \mathbf{e}_s$ to the common tensor degree $\mathbf{m} = \mathbf{P.Degree}$ before summation; a zero-degree axis contributes zero. An all-zero tensor degree returns a complete zero row for every rate vertex.

For a coefficient row of direction-wise degree $\mathbf{m}$ on physical cell $\mathcal{H}_{\mathbf{c}}$, define $h_s^{\mathbf{c}} = \rho_s^{(c_s+1)} - \rho_s^{(c_s)} > 0$. Before rate substitution, differentiation along a nonzero-degree axis uses the forward difference

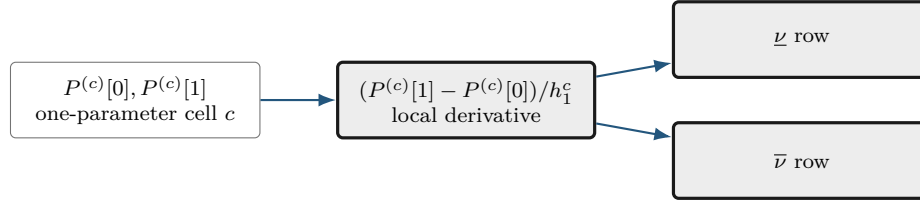
$$\left(\frac{\partial P^{(\mathbf{c})}}{\partial \rho_s} \right) [\mathbf{i}] = \frac{m_s}{h_s^{\mathbf{c}}} \left(P^{(\mathbf{c})}[\mathbf{i} + \mathbf{e}_s] - P^{(\mathbf{c})}[\mathbf{i}] \right), \quad i_s = 0, \dots, m_s - 1.$$

Thus the s -th partial has degree $\mathbf{m} - \mathbf{e}_s$. In one parameter this reduced degree is the public result degree. In two or more parameters, `rhodiff` elevates every nonzero partial back to the common degree $\mathbf{m}$ before adding the rate-weighted rows, so partials from different axes have compatible tensor labels. Section A.9 derives the reduction and exact re-elevation.

For a one-parameter degree-one coefficient pair $P^{(\mathbf{c})}[0], P^{(\mathbf{c})}[1]$, the rate-dependent derivative row reduces to

$$\dot{P}(\rho, \nu) = \nu \frac{P^{(\mathbf{c})}[1] - P^{(\mathbf{c})}[0]}{h_1^{\mathbf{c}}}.$$

This is the one-dimensional special case: the two rows correspond to the lower and upper allowed scheduling rates.



Differentiation changes local coefficients; rate substitution creates rows. Neither operation adds physical cells.

For ℓ parameters, nonuniform grids use a distinct step length in every direction and cell:

$$\dot{P}^{(c)}(\boldsymbol{\rho}, \boldsymbol{\nu}) = \sum_{s=1}^{\ell} \nu_s \frac{\partial P^{(c)}}{\partial \rho_s}(\boldsymbol{\rho}).$$

RateBounds fixes $\mathcal{R}$, so **rhodiff** evaluates this affine expression at the 2^ℓ vertices in $\mathcal{V}_{\mathcal{R}}$ and stores one row per vertex. Supply those bounds explicitly as **rhodiff**(P, **rb**), or initialize P with **RateBounds=rb**; if both are present they must match. Without bounds there is no finite vertex set to store, so **rhodiff**(P) raises **pdvar:MissingRateBounds**. Section A.9.1 gives the exact hypercube-local row and coefficient layout.

Remarks

Unsupported call arity or differentiating an expression that already contains rate-vertex rows raises **pdvar:InvalidDiff**. Explicit malformed rate bounds raise **pdvar:InvalidRateBounds**. Calling **rhodiff**(P) without stored bounds raises **pdvar:MissingRateBounds**; explicit bounds that disagree with P.**RateBounds** raise **pdvar:RateBoundsMismatch**.

Examples and output

When rate bounds live on the object, **rhodiff**(P) uses them directly.

```

yalmip('clear')
P = pdvar(2, {[0 1 2]}, "symmetric", RateBounds=[-1 1]);
D = rhodiff(P);
derivativeClass = class(D)
derivativeDegree = D.Degree
isContinuous = D.IsContinuous
coefficientTableSize = size(D.coeffs(1))
rateBounds = D.RateBounds

```

```

derivativeClass = 'pdvar'
derivativeDegree = 0
isContinuous =
    logical
    0
coefficientTableSize = 1×2 double
    2    1
rateBounds = 1×2 double
    -1    1

```

For a two-parameter object, the rate-vertex rows enumerate all lower/upper combinations from the two rows of `RateBounds`.

```
yal mip('clear')
Q = pdvar(1, {[0 1], [10 20]}, RateBounds=[-1 1; -2 2]);
E = rhodiff(Q);
% Rows 1:4 use [-1 -2], [-1 2], [1 -2], and [1 2], respectively.
derivativeClass = class(E)
derivativeDegree = E.Degree
coefficientTableSize = size(E.coeffs([1 1]))
rateVertexCount = coefficientTableSize(1)
coefficientsPerRateVertex = coefficientTableSize(2)
rateBounds = E.RateBounds
```

```
derivativeClass = 'pdvar'
derivativeDegree = 1
coefficientTableSize = 1×2 double
    4    4
rateVertexCount = 4
coefficientsPerRateVertex = 4
rateBounds = 2×2 double
    -1    1
    -2    2
```

A zero-degree axis contributes exactly zero but remains present in the common multivariate degree and in the rate-vertex enumeration.

```
yal mip('clear')
P = pdvar(1, {[0 1], [10 20]}, Degree=[2 0]);
D = rhodiff(P, [-1 1; -2 2]);
derivativeDegree = D.Degree
coefficientTableSize = size(D.coeffs([1 1]))
rateVertexCount = coefficientTableSize(1)
coefficientsPerRateVertex = coefficientTableSize(2)
```

```
derivativeDegree = 1×2 double
    2    0
coefficientTableSize = 1×2 double
    4    3
rateVertexCount = 4
coefficientsPerRateVertex = 3
```

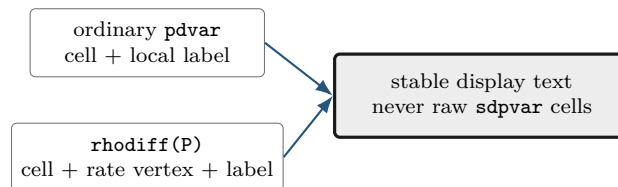
5.6 bernsteinTable

Syntax

```
T = bernsteinTable(P)
T = bernsteinTable(P, cellSubs)
T = bernsteinTable(P, "oneLine")
T = bernsteinTable(P, cellSubs, "oneLine")
```

Arguments

- P is a `pdvar` object.
- `cellSubs` selects one physical cell; omit it to list every cell.
- "oneLine" is the only supported text option. It returns one compact Bernstein expression per selected cell.
- T is a table. Ordinary rows contain the local label, basis, physical-node flag, and symbolic or numeric value. A rate-dependent derivative also includes `RateVertexIndex` and `RateVertex`.



Examples and output

```
yalmip('clear')
P = pdvar(1, {[0 1]});
T = bernsteinTable(P, "oneLine");
tableHeight = height(T)
tableWidth = width(T)
hasExpression = ismember("Expression", string(T.Properties.VariableNames))
```

```
tableHeight = 1
tableWidth = 2
hasExpression = logical
1
```

For `D = rhodiff(P)`, `bernsteinTable(D, cellSubs, "oneLine")` emits one row for every selected physical cell and rate-box vertex, rather than one row per local Bernstein coefficient. `RateVertexIndex` and `RateVertex` identify the lower/upper rate tuple; their tensor order is defined in Section A.9.1.

```
yal mip('clear')
P = pdvar(1, {[0 1], [0 1]}, RateBounds=[-1 1; -2 2]);
D = rhodiff(P);
T = bernsteinTable(D, [1 1], "oneLine");
vertexIndices = T.RateVertexIndex
rateVertices = vertcat(T.RateVertex{:})
expressionCount = height(T)
```

```
vertexIndices = 4×1 double
    1
    2
    3
    4

rateVertices = 4×2 double
   -1   -2
   -1    2
    1   -2
    1    2

expressionCount = 4
```

Remarks

`bernsteinTable` is an inspection method, not a solver call. Text options other than "oneLine", repeated selectors, invalid physical-cell selectors, or inconsistent rate-vertex rows raise `pdvar:InvalidBernsteinTableInput`.

5.7 Build Affine Algebra with Known Data

A `pdvar` result must remain affine in the underlying YALMIP decisions. The following blocks separate affine sums from products so the supported boundary is visible beside each example.

5.7.1 Signs, Addition, And Subtraction

Use this family to combine compatible decision expressions, coefficient-backed known data, finite real numeric data, and affine `sdpvar` matrices. For two parameter-dependent operands, alignment proceeds in a fixed order: verify matching outer bounds, form the sorted union of interior nodes on each axis, exactly subdivide/re-express both coefficient trees on that common grid, elevate both to the componentwise maximum degree, and finally add or subtract matching coefficients. Numeric and affine constant operands are promoted before the last two steps. The corresponding overload names are `plus`, `minus`, `uplus`, and `uminus`; see sections A.6 and 3.9.7.

Syntax

```
R = P + Q
R = P + A
R = A + P
R = P + M
R = M + P
R = P + X
R = X + P
R = P - Q
R = P - A
R = A - P
R = P - M
R = M - P
R = +P
R = -P
```

Here **A** is coefficient-backed **pdmatrix**, **M** is numeric, and **X** is affine **sdpvar**. A numeric scalar broadcasts to the payload size; an affine **sdpvar** is promoted as parameter-independent data.

Examples and output

```
yal mip('clear')
P = pdvar(2, {[0 1]}, "full");
X = sdpvar(2, 2, 'full');
S = P + X;
D = eye(2) - P;
N = -P;
sumClass = class(S)
sumSize = size(S)
sumDegree = S.Degree
sumContainsDecision = S.ContainsDecision
differenceClass = class(D)
negativeClass = class(N)
```

```
sumClass = 'pdvar'
sumSize = 1×2 double
         2     2
sumDegree = 1
sumContainsDecision =
    logical
         1
differenceClass = 'pdvar'
negativeClass = 'pdvar'
```

A nonlinear symbolic operand such as `eta*eta` is rejected. Rate-vertex rows from `rhodiff` may be added to compatible ordinary expressions; the ordinary rows are broadcast over the active rate vertices.

The next example exercises both stages of parameter-dependent alignment: a coarse degree-one decision is refined to the known operand's two-cell grid and elevated to degree two before coefficientwise addition.

```
yal mip('clear')
P = pdvar(1, {[0 1]}, Degree=1);
A = pdmat({[0 0.5 1]}, @(rho) rho.^2, Degree=2);
S = P + A;
resultClass = class(S)
mergedGrid = S.GridInfo.Vectors{1}
resultDegree = S.Degree
storageCounts = [S.ncell(), S.ncoeff()]
```

```
resultClass = 'pdvar'

mergedGrid = 1×3 double
            0    0.5000    1.0000

resultDegree = 2

storageCounts = 1×2 double
                2     3
```

5.7.2 Multiplication By Known Or Numeric Data

The `mtimes` overload supports multiplication only when decision dependence occurs on at most one side. A coefficient-backed `pdmat` or finite real numeric matrix may multiply a `pdvar` from either side. A scalar `pdvar` may scale a known matrix. Two decision-bearing factors, including $P*Q$ and $P*\eta$ for a decision `sdpvar` `eta`, are nonlinear and rejected.

Syntax

```
R = A * P
R = P * A
R = M * P
R = P * M
```

The first example shows numeric left and right matrix products without exposing incidental YALMIP variable identifiers.

Examples and output

```
yal mip('clear')
P = pdvar(2, 1, {[0 1]}, "full");
L = [1 2] * P;
R = P * [4 5];
S = 3 * P;
leftSize = size(L)
rightSize = size(R)
scaledSize = size(S)
degrees = [L.Degree R.Degree S.Degree]
```

```
leftSize = 1×2 double
    1     1
rightSize = 1×2 double
    2     2
scaledSize = 1×2 double
    2     1
degrees = 1×3 double
    1     1     1
```

A known parameter-dependent factor adds its Bernstein degree componentwise through ordered coefficient convolution. On one parameter cell, a quadratic decision and affine known factor produce

$$(PA)[k] = \sum_{i+j=k} \frac{\binom{2}{i}\binom{1}{j}}{\binom{3}{k}} P[i]A[j], \quad k = 0, 1, 2, 3.$$

The same backend rule is documented at `bernProd` in section 3.9.5 and derived step by step in section A.8. In ℓ parameters, the ordered products are grouped by $\mathbf{k} = \mathbf{i} + \mathbf{j}$ and weighted by the product of the one-dimensional binomial normalization ratios; the result degree is the componentwise sum.

```
yal mip('clear')
P = pdvar(2, {[0 1]}, "symmetric", Degree=2);
A = pdmat([0 1], {eye(2), 2*eye(2)}, Degree=1);
```

```

L = A * P;
R = P * A;
knownLeftClass = class(L)
knownLeftDegree = L.Degree
knownRightClass = class(R)
knownRightDegree = R.Degree
coefficientsPerCell = numel(L.coeffs(1))

```

```

knownLeftClass = 'pdvar'
knownLeftDegree = 3
knownRightClass = 'pdvar'
knownRightDegree = 3
coefficientsPerCell = 4

```

Direction-wise degrees add independently. A known factor that varies only in the second parameter raises only the second component of the result degree.

```

yal mip('clear')
P = pdvar(1, {[0 1], [0 1]}, Degree=[2 0]);
A = pdmat({[0 1], [0 1]}, ...
    @(rho1, rho2) 1 + rho2, Degree=[0 1]);
L = A * P;
R = P * A;
leftDegree = L.Degree
rightDegree = R.Degree
coefficientsPerCell = L.ncoeff()

```

```

leftDegree = 1×2 double
    2     1

```

```

rightDegree = 1×2 double
    2     1

```

```

coefficientsPerCell = 6

```

A derivative object may be multiplied by numeric data or a matching-grid known object while preserving every rate row. Products with rate dependence on both sides, products of a derivative with an ordinary decision expression, and products involving metadata-only rate dependence are rejected.

Two decision-bearing factors are also rejected: their product would be quadratic in YALMIP decision coefficients. The supported boundary is numeric or known `pdmat` data multiplying one affine `pdvar` expression on either side, with the written matrix order preserved.

Remarks

Incompatible affine operands raise `pdvar:InvalidAddition` or `pdvar:InvalidSubtraction`. Bad dimensions, two decision-bearing factors, nonlinear symbolic data, or unsupported rate combinations raise `pdvar:InvalidMultiplication`. Incompatible parameter bounds raise `pdvar:MixedGrid`; a function-only `pdmat` operand raises `pdvar:FunctionOnlyAlgebra`.

5.8 Reshape and Transform Decision Matrices

Every method below is inherited from the centralized `pdbase` implementation and transforms each local affine YALMIP payload while the `pdvar` reconstruction hook preserves the physical grid, degree, class, decision metadata, and compatible rate rows. The examples report stable classes, dimensions, and metadata rather than YALMIP's internal variable numbers.

5.8.1 Shape, Equality, And MATLAB Protocols

Shape queries describe the matrix payload. `isequal` is an exact identity check: grid, degree, matrix size, metadata, and local symbolic payloads must already match; it does not merge or elevate operands.

Syntax

```
sz = size(P)
d = size(P, dim)
n = height(P)
n = width(P)
n = length(P)
n = numel(P)
n = ndims(P)
tf = isequal(P, Q, ...)
idx = end(P, k, n)
n = numArgumentsFromSubscript(P, S, ctx)
```

Examples and output

```
yalmip('clear')
P = pdvar(2, 3, {[0 1]}, "full");
matrixSize = size(P)
rowCount = height(P)
columnCount = width(P)
longestDimension = length(P)
elementCount = numel(P)
dimensionCount = ndims(P)
equalsUnaryPlus = isequal(P,+P)
```

```
matrixSize = 1×2 double
           2     3
```

```
rowCount = 2
columnCount = 3
longestDimension = 3
elementCount = 6
dimensionCount = 2
equalsUnaryPlus =
    logical
         1
```

`end`, `subsref`, `subsasgn`, and `numArgumentsFromSubscript` support the two-index and dot-access forms shown in [section 5.9](#); they do not expose physical-cell storage.

5.8.2 Transpose And Conjugate Transpose

Both forms transpose every local affine payload without breaking the YALMIP coefficient graph.

Syntax

```
T = transpose(P)
H = ctranspose(P)
T = P.'
H = P'
```

Examples and output

```
yal mip('clear')
P = pdvar(2, 3, {[0 1]}, "full");
T = P.';
H = P';
transposeSize = size(T)
conjugateTransposeSize = size(H)
transposeClass = class(T)
conjugateTransposeClass = class(H)
```

```
transposeSize = 1×2 double
    3     2
conjugateTransposeSize = 1×2 double
    3     2
transposeClass = 'pdvar'
conjugateTransposeClass = 'pdvar'
```



5.8.3 Vectorization, Reshape, And Squeeze

`vec` uses MATLAB column-major order. `reshape` accepts two positive dimensions with at most one inferred `[]` dimension and preserves the element count. `squeeze` retains the two-dimensional package convention.

Syntax

```
V = vec(P)
R = reshape(P, [m n])
R = reshape(P, m, n)
S = squeeze(P)
```

Examples and output

```
yal mip('clear')
P = pdvar(2, 3, {[0 1]}, "full");
V = vec(P);
R = reshape(P, [3 2]);
S = squeeze(P);
vectorSize = size(V)
reshapeSize = size(R)
squeezeSize = size(S)
```

```
vectorSize = 1×2 double
    6     1
reshapeSize = 1×2 double
    3     2
squeezeSize = 1×2 double
    2     3
```

Malformed or size-changing reshape requests raise `pdvar:InvalidReshape`.

5.8.4 Diagonals And Trace

`diag` extracts a diagonal or constructs a diagonal matrix from a vector; the selected offset may not be empty. `trace` returns a scalar decision expression.

Syntax

```
d = diag(P)
d = diag(P, k)
D = diag(v)
D = diag(v, k)
t = trace(P)
```

Examples and output

```
yalmip('clear')
P = pdvar(3, {[0 1]}, "full");
d = diag(P);
D = diag(d);
t = trace(P);
diagonalSize = size(d)
rebuiltSize = size(D)
traceSize = size(t)
```

```
diagonalSize = 1×2 double
    3     1
rebuiltSize = 1×2 double
    3     3
traceSize = 1×2 double
    1     1
```

Invalid offsets or empty selections raise `pdvar:InvalidDiag`.

5.8.5 Triangular Parts And Reductions

Triangular methods zero the complementary part of every coefficient. Reductions act on matrix dimensions; "all" is available for `sum` and `mean`.

Syntax

```
L = tril(P)
L = tril(P, k)
U = triu(P)
U = triu(P, k)
S = sum(P)
S = sum(P, dim)
S = sum(P, vecdim)
S = sum(P, "all")
M = mean(P)
M = mean(P, dim)
M = mean(P, vecdim)
M = mean(P, "all")
C = cumsum(P)
C = cumsum(P, dim)
C = cumsum(P, direction)
C = cumsum(P, dim, direction)
```

First form the two triangular views.

Examples and output

```
yalmip('clear')
P = pdvar(3, {[0 1]}, "full");
L = tril(P);
U = triu(P);
lowerSize = size(L)
upperSize = size(U)
degrees = [L.Degree U.Degree]
```

```
lowerSize = 1×2 double
    3     3
upperSize = 1×2 double
    3     3
degrees = 1×2 double
    1     1
```

Then reduce a rectangular expression.

```
yalmip('clear')
P = pdvar(2, 3, {[0 1]}, "full");
rowSum = sum(P, 2);
allMean = mean(P, "all");
running = cumsum(P, 2);
rowSumSize = size(rowSum)
allMeanSize = size(allMean)
runningSize = size(running)
```

```
rowSumSize = 1×2 double
    2     1
allMeanSize = 1×2 double
    1     1
runningSize = 1×2 double
    2     3
```

`vecdim` is a nonempty vector of unique positive integer dimensions; dimensions above two are singleton no-ops. `direction` is "forward" or "reverse", case-insensitively. A direction supplied without a dimension uses the first nonsingleton matrix dimension, and `cumsum` along a dimension above two returns the same symbolic payloads. Invalid triangular offsets raise `pdvar:InvalidTriangularPart`. Invalid reduction calls raise `pdvar:InvalidSum`, `pdvar:InvalidMean`, or `pdvar:InvalidCumsum`; missing-value and native-output flags are not supported.

5.8.6 Reordering And Rotation

These methods rearrange matrix entries inside every symbolic coefficient; they do not reverse the parameter grid.

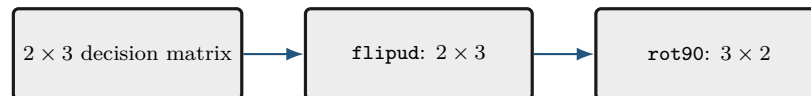
Syntax

```
Q = flip(P)
Q = flip(P, dim)
Q = flipud(P)
Q = fliplr(P)
Q = rot90(P)
Q = rot90(P, k)
```

Examples and output

```
yalmp('clear')
P = pdvar(2, 3, {[0 1]}, "full");
F = flipud(P);
R = rot90(P);
flippedSize = size(F)
rotatedSize = size(R)
flippedClass = class(F)
rotatedClass = class(R)
```

```
flippedSize = 1×2 double
             2     3
rotatedSize = 1×2 double
             3     2
flippedClass = 'pdvar'
rotatedClass = 'pdvar'
```



Invalid calls raise `pdvar:InvalidFlip` or `pdvar:InvalidRot90`.

5.8.7 Concatenation And Block Assembly

Assembly accepts compatible `pdvar`, coefficient-backed `pdmat`, affine `sdpvar`, and finite real numeric blocks. It aligns grids and elevates all blocks exactly to the componentwise maximum degree while preserving an affine expression. `cat` supports only matrix dimensions 1 and 2.

Syntax

```
H = horzcat(P, Q, ...)
V = vertcat(P, Q, ...)
H = [P, Q]
V = [P; Q]
C = cat(dim, P, Q, ...)
D = blkdiag(P, Q, ...)
```

Examples and output

```
yalmp('clear')
P = pdvar(2, {[0 1]}, "full");
H = [P, P];
V = [P; P];
D = blkdiag(P, eye(2));
horizontalSize = size(H)
verticalSize = size(V)
blockDiagonalSize = size(D)
```

```
horizontalSize = 1×2 double
                2     4
verticalSize = 1×2 double
              4     2
blockDiagonalSize = 1×2 double
                   4     4
```

Incompatible blocks or rate metadata raise `pdvar:InvalidConcatenation`; unsupported dimensions raise

`pdvar:UnsupportedCatDimension`; invalid block-diagonal inputs raise `pdvar:InvalidBlkdiag`. Incompatible bounds raise `pdvar:MixedGrid`; function-only known blocks raise `pdvar:FunctionOnlyAlgebra`.

5.8.8 Repetition

`repmat` tiles the matrix payload while retaining the parameter metadata. It accepts a positive scalar, two positive counts, or a two-element count vector.

Syntax

```
Q = repmat(P, m)
Q = repmat(P, m, n)
Q = repmat(P, [m n])
```

Examples and output

```
yalmip('clear')
P = pdvar(1, 2, {[0 1]}, "full");
Q = repmat(P, [2 2]);
repeatedSize = size(Q)
repeatedDegree = Q.Degree
containsDecision = Q.ContainsDecision
```

```
repeatedSize = 1×2 double
              2      4
repeatedDegree = 1
containsDecision =
    logical
         1
```

Invalid repetition shapes raise `pdvar:InvalidRepmat`.

5.9 Index and Assign `pdvar` Entries

Parenthesis indexing selects a matrix block from every symbolic coefficient. Assignment writes a compatible numeric, `pdvar`, coefficient-backed `pdmat`, or affine `sdpvar` block into that position after the same common-grid refinement and componentwise maximum-degree elevation used by affine addition. The left object and parameter-dependent right-hand side are exactly re-expressed on the sorted union grid before the selected payload block is replaced at every aligned label. It does not select one physical cell. See sections [A.6](#) and [3.9.7](#).

Syntax

```
Q = P(rows, cols)
Q = P(1:end, end)
value = P.PropertyName
P(rows, cols) = numericValue
P(rows, cols) = Q
P(rows, cols) = A
P(rows, cols) = X
```

Examples and output

```
yal mip('clear')
P = pdvar(2, 3, {[0 1]}, "full");
lastCol = P(:, end);
P(:, 1) = 0;
selectedSize = size(lastCol)
assignedSize = size(P)
assignedClass = class(P)
assignedDegree = P.Degree
containsDecision = P.ContainsDecision
```

```
selectedSize = 1×2 double
             2     1
assignedSize = 1×2 double
             2     3
assignedClass = 'pdvar'
assignedDegree = 1
containsDecision =
    logical
         1
```

Assignment may therefore increase both the grid resolution and the degree of the complete decision expression while retaining its affine decision content.

```
yal mip('clear')
P = pdvar(2, 2, {[0 1]}, "full", Degree=1);
A = pdmat([0 0.5 1]), ...
    @(rho) [rho.^2; 1+rho], Degree=2);
P(:, 2) = A;
assignedGrid = P.GridInfo.Vectors{1}
assignedDegree = P.Degree
assignedClass = class(P)
storageCounts = [P.ncell(), P.ncoeff()]
containsDecision = P.ContainsDecision
```

```
assignedGrid = 1×3 double
             0    0.5000    1.0000

assignedDegree = 2
assignedClass = 'pdvar'

storageCounts = 1×2 double
             2     3

containsDecision =
    logical
         1
```

Invalid selectors raise `pdvar:InvalidSubscript`. Linear indexing, deletion, or non-parenthesis assignment raises `pdvar:UnsupportedAssignment`; incompatible or nonlinear right-hand sides raise `pdvar:InvalidAssignment`.

5.10 value

Syntax

```
A = value(P)
rows = value(rhodiff(P))
```

`value` converts assigned `pdvar` coefficients into known, coefficient-backed `pdmat` data. An ordinary expression returns one `pdmat`, preserving its grid, matrix size, degree, local coefficient order, and continuity metadata. A rate-vertex expression created by `rhodiff` returns a $1\text{-by-}2^\ell$ cell array of `pdmat` objects in the deterministic lower/upper vertex order induced by applying `helper.combRows` to `num2cell(RateBounds,2)'`. The returned `pdmat` objects do not carry rate metadata. Every symbolic coefficient must have an assigned finite real YALMIP value; otherwise the method raises `pdvar:UnassignedValue`.

Examples and output

```
yalmip('clear')
P = pdvar(1, [0 1], Degree=2);
c = P.coeffs(1);
assign([c{:}], [1 2 4]);
A = value(P);
assignedCoefficients = cell2mat(A.coeffs(1))
```

```
assignedCoefficients = 1×3 double
    1     2     4
```

For rate rows, `value` returns one known object per rate vertex rather than collapsing the rows.

```
yalmip('clear')
P = pdvar(1, [0 1], Degree=1);
c = P.coeffs(1);
assign([c{:}], [1 3]);
D = rhodiff(P, [-1 1]);
rows = value(D);
rowCount = numel(rows)
lowerRowClass = class(rows{1})
upperRowClass = class(rows{2})
lowerRateCoefficients = cell2mat(rows{1}.coeffs(1))
upperRateCoefficients = cell2mat(rows{2}.coeffs(1))
```

```
rowCount = 2
lowerRowClass = 'pdmat'
upperRowClass = 'pdmat'
lowerRateCoefficients = -2
upperRateCoefficients = 2
```

5.11 Comparisons To `pdlmi`

The `le`, `ge`, and `eq` overloads implement `<=`, `>=`, and `==`. Inequalities select either semidefinite or entry-wise meaning once for the complete original residual; equality is always direct coefficient equality.

Syntax

```
C = P <= 0
C = P >= 0
C = lhs <= rhs
C = lhs >= rhs
C = lhs == rhs
```

Arguments

- `rhs` may be compatible numeric, coefficient-backed `pdmatrix`, affine `sdpvar`, or `pdvar` data. Ordinary operands use the same grid refinement, componentwise maximum-degree alignment, and promotion as subtraction.
- For `<=` and `>=`, every coefficient in every physical cell and rate row is scanned before assembly. The complete relation is semidefinite only if every original coefficient is square Hermitian. Numeric Hermitian testing uses the inclusive rule $\|A - A^T\|_\infty \leq 10^{-10}$; symbolic coefficients use YALMIP's `ishermitian`. If any coefficient fails, the complete inequality is entry-wise and warning `pdlmi:ElementwiseInequality` is issued once. Modes are never mixed coefficient by coefficient or entry by entry.
- `==` accepts rectangular and nonsymmetric residuals without that warning. It equates corresponding coefficient entries directly. Ordinary rows compare with ordinary rows; derivative rows compare only with compatible derivative rows.
- `C` is a `pdlmi` wrapper. A comparison assembles constraints but does not solve an optimization problem.

Examples and output

Comparison creates a `pdlmi` wrapper with one stored constraint for each physical cell and local Bernstein coefficient.

```
yalmip('clear')
P = pdvar(2, {[0 0.5 1]}, "symmetric");
Cneg = P <= 0;
negativeWrapperClass = class(Cneg)
negativeConstraintCount = numel(Cneg.Constraints)
Cpos = P >= 0;
positiveWrapperClass = class(Cpos)
positiveConstraintCount = numel(Cpos.Constraints)
```

```

negativeWrapperClass = 'pdlmi'
negativeConstraintCount = 4
positiveWrapperClass = 'pdlmi'
positiveConstraintCount = 4

```

A rectangular residual selects entry-wise inequality globally. Equality is entry-wise by definition and is warning-free.

```

yalmp('clear')
V = pdvar(3, 2, {[0 1]}, "full");
warnId = "pdlmi:ElementwiseInequality";
warnState = warning("off", warnId);
restoreWarning = onCleanup(@() warning(warnState));
Centry = V >= 0;
Ceq = V == sdpvar(3, 2, 'full');
entrywiseConstraintGroups = numel(Centry.Constraints)
equalityConstraintGroups = numel(Ceq.Constraints)
identity = V == V;
identityConstraints = identity.toYalmip()
clear restoreWarning

```

```

entrywiseConstraintGroups = 2
equalityConstraintGroups = 2

```

```

identityConstraints =

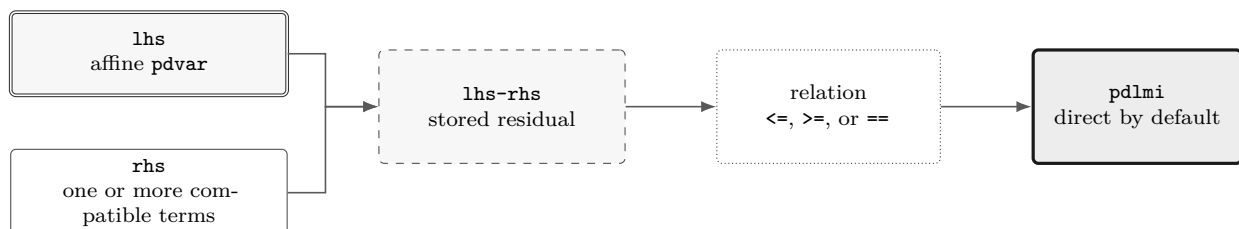
```

```

[]

```

Each group above corresponds to one local coefficient and contains six scalar entries in MATLAB column-major order. Under Pólya, Putinar, SparseFullBox, or FullBox, each entry of an entry-wise inequality receives an independent scalar certificate. Equality is direct-only: constructor certificate options and `applyPolya`, `applyPutinar`, `applySparseFullBoxPreorder`, or `applyFullBoxPreorder` raise `pdlmi:UnsupportedEqualityCertificate` before validating an increment, width, or order. Mixed ordinary/derivative row kinds raise `pdvar:InvalidEqualityRows`. Use overloaded `==` to build a modeling equality; use `isequal` only for MATLAB structural comparison.



5.12 Boundaries and Related APIs

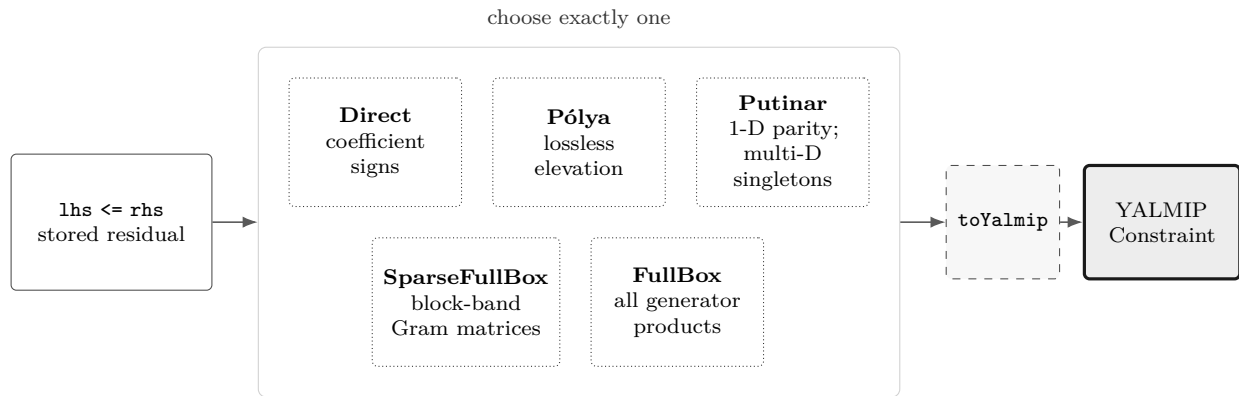
1. Constructor-created `pdvar` objects support every finite nonnegative scalar or direction-wise integer degree. Each cell stores $\prod_s (m_s + 1)$ Bernstein control matrices; a zero-degree axis is constant in that direction.
2. Products may contain decision variables on at most one side; nonlinear decision products such as `P * Q` are outside this slice.
3. Function-only `pdmatrix` operands cannot enter `pdvar` algebra unless they were constructed with explicit Bernstein evidence.
4. `rhodiff` of an existing rate-vertex derivative expression is not part of the current public workflow.

Related APIs. `pdvar`, `rhodiff`, `bernsteinTable`, `mtimes`, `pdlmi`, `pdmatrix`.

Chapter 6

pdlmi: Finite Certificate Assembly

`pdlmi` stores finite certificates assembled from `pdvar` expressions or coefficient-backed `pdmatrix` inequalities. Square Hermitian inequality families use semidefinite constraints; other inequalities use entry-wise scalar constraints; `pdvar` equality uses direct coefficient equations. Direct assembly is the default, while cell-local Pólya elevation, a dimension-aware Putinar selector, `FullBox` with symmetric block-band Gram matrices (`SparseFullBox`), and the dense full-box preordering (`FullBox`) are explicit inequality alternatives. It is the package boundary between Bernstein objects and ordinary YALMIP solver calls.



The five certificate paths are alternatives. Selecting Pólya, Putinar, SparseFullBox, or FullBox rebuilding replaces, rather than compounds, the previous selection.

6.1 API Map

Task	Entry
Construct wrapper	<code>pdlmi</code> , <code><=</code> / <code>>=</code> / <code>==</code>
Count constraints	<code>direct coefficient constraints</code>
Select Pólya elevation	<code>UsePolya</code> , <code>PolyaDegree</code> , and <code>applyPolya</code>
Select Putinar certificate	<code>UsePutinar</code> , <code>PutinarOrder</code> , and <code>applyPutinar</code>
Select SparseFullBox	<code>UseSparseFullBoxPreorder</code> , <code>SparseFullBoxOrder</code> , <code>BandWidth</code> , and <code>applySparseFullBoxPreorder</code>
Select FullBox	<code>UseFullBoxPreorder</code> , <code>FullBoxOrder</code> , and <code>applyFullBoxPreorder</code>
Export to YALMIP	<code>toYalmip</code>
Solve	<code>solver-facing use</code> , <code>Examples</code>

6.2 Inspect Certificate State

All `pdlmi` properties have private set access and are readable with dot notation. `Constraints` contains the assembled YALMIP entries; `Residual` and `Relation` retain the original comparison for rebuilding. The remaining properties record the selected certificate state.

Property	Meaning
<code>Constraints</code>	Stored finite YALMIP constraint entries.
<code>Residual</code> , <code>Relation</code>	Original <code>pdvar</code> residual and one of " <code><=</code> ", " <code>>=</code> ", or " <code>==</code> ".
<code>UsePolya</code> , <code>PolyaDegree</code>	Pólya selector and nonnegative degree increment.
<code>UsePutinar</code> , <code>PutinarOrder</code>	Putinar selector and absolute Gram order.
<code>UseSparseFullBoxPreorder</code> , <code>SparseFullBoxOrder</code> , <code>BandWidth</code>	<code>SparseFullBox</code> selector, absolute Gram order, and bandwidth w of the symmetric block-band Gram matrices $Q_J^{(w)}$.
<code>UseFullBoxPreorder</code> , <code>FullBoxOrder</code>	Full-box selector and absolute Gram order.

For example, `C.Relation`, `C.Residual.Degree`, and `C.UseSparseFullBoxPreorder` inspect a wrapper. Apply methods return a rebuilt value; assigning these properties is unsupported. Selected Pólya and Gram parameters are stored as 1-by- ℓ rows. An intermediate `SparseFullBox` result stores all three sparse properties. A width-one request canonicalizes to actual Direct state, where those fields are `false`, zero, and zero; a request with `BandWidth >= max(SparseFullBoxOrder + 1)` canonicalizes to actual FullBox state, where only `UseFullBoxPreorder`/`FullBoxOrder` record the selection.

6.3 Construct pdlmi Comparisons

Syntax

```
C = pdlmi(expr, relation)
C = pdlmi(expr, "==")
C = pdlmi(expr, relation, "UsePolya")
C = pdlmi(expr, relation, UsePolya=true, PolyaDegree=d)
C = pdlmi(expr, relation, "UsePolya", "PolyaDegree", d)
C = pdlmi(expr, relation, "UsePutinar")
C = pdlmi(expr, relation, UsePutinar=true, PutinarOrder=r)
C = pdlmi(expr, relation, PutinarOrder=r)
C = pdlmi(expr, relation, "UseSparseFullBoxPreorder")
C = pdlmi(expr, relation, UseSparseFullBoxPreorder=true)
C = pdlmi(expr, relation, BandWidth=w)
C = pdlmi(expr, relation, SparseFullBoxOrder=r)
C = pdlmi(expr, relation, UseSparseFullBoxPreorder=true, ...
    BandWidth=w, SparseFullBoxOrder=r)
C = pdlmi(expr, relation, "UseFullBoxPreorder")
C = pdlmi(expr, relation, UseFullBoxPreorder=true)
C = pdlmi(expr, relation, FullBoxOrder=r)
C = pdlmi(expr, relation, UseFullBoxPreorder=true, FullBoxOrder=r)
C = pdlmi(expr, relation, UseFullBoxPreorder=false, FullBoxOrder=r)
C = pdlmi(expr, relation, ValidationMode=mode)
C = lhs <= rhs
C = lhs >= rhs
C = lhs == rhs
```

Arguments

- **expr** is a **pdvar** residual, or a coefficient-backed **pdmat** residual for "**<=**" or "**>=**"; **relation** is exactly "**<=**", "**>=**", or "**==**". Inequalities are classified globally from the original coefficient family. **pdvar** equality is entry-wise direct coefficient assembly and accepts rectangular or nonsymmetric residuals. **pdmat** equality is instead the scalar logical comparison **A==B** and never creates a **pdlmi**.
- **UsePolya** is a logical scalar option with default **false**. Its bare form, "**UsePolya**", and **UsePolya=true** without a degree select increment one.
- **PolyaDegree=d** accepts one finite nonnegative integer scalar shorthand or an ℓ -element vector and is stored as a 1-by- ℓ row. It elevates the residual componentwise by **d** before assembling constraints. Supplying it without **UsePolya** enables Pólya and issues **pdlmi:ImplicitUsePolya**; **UsePolya=false** with any positive component raises **pdlmi:ConflictingPolyaOptions**.
- **UsePutinar** is a logical scalar option with default **false**. Its bare form and the explicit value **true** select the dimension-appropriate Putinar certificate at the smallest admissible order.
- **PutinarOrder=r** accepts one finite nonnegative integer scalar shorthand or an ℓ -element vector and is stored as a 1-by- ℓ absolute Bernstein Gram order, not a degree increment. For assembled residual degree M in one parameter, its default and minimum are $\lfloor M/2 \rfloor$. For tensor degree **M**

in two or more parameters, the default and componentwise minimum are $\lceil \mathbf{M}/2 \rceil$. Supplying the order without `UsePutinar` enables Putinar assembly. Pairing `UsePutinar=false` with any explicit order, including zero, raises `pdlmi:ConflictingPutinarOptions`.

- `UseSparseFullBoxPreorder` is a logical scalar option with default `false`. Its bare form and the explicit value `true` select `SparseFullBox` with default `BandWidth=2`; `BandWidth` or `SparseFullBoxOrder` alone also enables this family. Explicit `UseSparseFullBoxPreorder=false` conflicts with either sparse parameter.
- `BandWidth=w` remains one finite positive integer bandwidth for the symmetric block-band Gram matrices $Q_J^{(w)}$. `SparseFullBoxOrder=r` accepts scalar shorthand or an ℓ -element vector and is stored as a 1-by- ℓ absolute Gram order. Its default and minimum are $\lfloor M/2 \rfloor$ in one parameter and componentwise $\lceil \mathbf{M}/2 \rceil$ otherwise, where $\mathbf{M} = \mathbf{C}.\text{Residual.Degree}$. Width one canonicalizes to actual Direct state after order validation; width at least $\max_s(r_s + 1)$ canonicalizes to actual `FullBox` state; intermediate widths select `SparseFullBox` state.
- `UseFullBoxPreorder` is a logical scalar option with default `false`. Its bare form and the explicit value `true` select full-box assembly at the smallest admissible order.
- `FullBoxOrder=r` accepts scalar shorthand or an ℓ -element vector and is stored as a 1-by- ℓ absolute order, not a degree increment. Supplying it without `UseFullBoxPreorder` enables full-box assembly. When `FullBox` is enabled, the minimum is $\lfloor M/2 \rfloor$ for a one-parameter residual of degree M , and componentwise $\lceil \mathbf{M}/2 \rceil$ for two or more parameters. Explicitly pairing `UseFullBoxPreorder=false` with `FullBoxOrder=r` suppresses that automatic selection: `C` remains direct and records the normalized row only as inspectable metadata; the order is not applied to assembly.
- Pólya, Putinar, `SparseFullBox`, and `FullBox` selection are mutually exclusive and apply only to inequalities. Any recognized certificate option on an equality raises `pdlmi:UnsupportedEqualityCertificate` before width or order validation. Enabling more than one inequality family raises `pdlmi:ConflictingRelaxations`.
- `ValidationMode=mode` accepts scalar text "fast" or "strict", case-insensitively, and defaults to "fast". It is transient and not stored. Every `apply...` call accepts its own trailing `ValidationMode`; a constructor selection is not inherited by later calls.
- `C` is direct coefficient-wise assembly unless one opt-in inequality mode is selected. Comparisons such as `lhs >= rhs` are the ordinary user-facing entry point and accept compatible numeric, `pdmat`, affine `sdpvar`, and `pdvar` right-hand sides.

The inspectable properties `Constraints`, `Residual`, and `Relation` expose the assembled finite constraints and the original comparison data used for rebuilding. `UsePolya/PolyaDegree`, `UsePutinar/PutinarOrder`, `UseSparseFullBoxPreorder/SparseFullBoxOrder/BandWidth`, and `UseFullBoxPreorder/FullBoxOrder` report the selected mode and parameters. Their set access is private. Comparison-created direct objects have every selector false and every numeric certificate property zero. The full-box explicit-false combination described above is the exception: it retains `FullBoxOrder=r` while `UseFullBoxPreorder` remains false and the stored constraints remain direct. Putinar and `SparseFullBox` deliberately have no corresponding exception.

Examples and output

The direct constructor is equivalent to the comparison path when the relation and options are explicit.

```
yal mip('clear')
P = pdvar(2, {[0 1]}, "symmetric");
C = pdlmi(P, ">=", UsePolya=false, PolyaDegree=0);
wrapperClass = class(C)
constraintCount = numel(C.Constraints)
usePolya = C.UsePolya
polyaDegree = C.PolyaDegree
usePutinar = C.UsePutinar
putinarOrder = C.PutinarOrder
useFullBox = C.UseFullBoxPreorder
fullBoxOrder = C.FullBoxOrder
useSparseFullBox = C.UseSparseFullBoxPreorder
sparseFullBoxOrder = C.SparseFullBoxOrder
bandWidth = C.BandWidth
```

```
wrapperClass = 'pdlmi'
constraintCount = 2
usePolya =
    logical
    0
polyaDegree = 0
usePutinar =
    logical
    0
putinarOrder = 0
useFullBox =
    logical
    0
fullBoxOrder = 0
useSparseFullBox =
    logical
    0
sparseFullBoxOrder = 0
bandWidth = 0
```

Remarks

1. A residual outside the supported `pdvar`/`pdmat` forms raises `pdlmi:InvalidExpression`; a function-only `pdmat` raises `pdlmi:MissingCoefficientEvidence`; `pdlmi(A,"==")` for `pdmat` raises `pdlmi:UnsupportedPdmatEquality`; an unsupported relation raises `pdlmi:InvalidRelation`.
2. Malformed, unknown, or duplicate options raise `pdlmi:InvalidOptions`, `pdlmi:UnknownOption`, or `pdlmi:DuplicateOption`; invalid validation modes raise `pdlmi:InvalidValidationMode`; nonlogical selectors use their corresponding `InvalidUse...` identifier.
3. Malformed Pólya increments, Putinar orders, sparse orders, widths, and full-box orders raise, respectively, `pdlmi:InvalidPolyaDegree`, `pdlmi:InvalidPutinarOrder`, `pdlmi:`

`InvalidSparseFullBoxOrder`, `pdlmi:InvalidBandWidth`, and `pdlmi:InvalidFullBoxOrder`. An integer below an active Gram-family minimum raises `pdlmi:PutinarOrderTooLow`, `pdlmi:SparseFullBoxOrderTooLow`, or `pdlmi:FullBoxOrderTooLow`.

4. Explicit false plus a Putinar order raises `pdlmi:ConflictingPutinarOptions`; explicit false plus a sparse order or width raises `pdlmi:ConflictingSparseFullBoxOptions`; explicit false plus a positive Pólya increment raises `pdlmi:ConflictingPolyaOptions`. The full-box explicit-false form remains direct and retains the supplied order as metadata.
5. An inequality with any nonsquare or non-Hermitian original coefficient is assembled entry-wise and emits `pdlmi:ElementwiseInequality`; it is not rejected. Numeric Hermitian testing accepts equality at the inclusive 10^{-10} infinity-norm tolerance.
6. Equality is direct-only. Every `apply...` method raises `pdlmi:UnsupportedEqualityCertificate` before validating its supplied increment, width, or order.

6.4 Assemble Direct Coefficient Constraints

For a `pdvar` residual, including equality behavior, each physical cell, local Bernstein coefficient, and rate-vertex row if present creates one stored YALMIP constraint entry. For an ordinary expression with N_c physical cells and N_b Bernstein coefficients per cell, the number of entries is $N_c N_b$; with N_v rate vertices it is $N_c N_b N_v$. A semidefinite-mode entry is one matrix constraint. An entry-wise-mode entry is the column-major vector of scalar matrix entries compared independently. An equality entry is the corresponding direct coefficient equation and may be rectangular. For a coefficient-backed known `pdmatrix` inequality, Direct and Pólya assembly instead scan the complete numeric coefficient family and store exactly one logical cell containing the global certificate; `toYalmip` exports that state as one scalar logical.

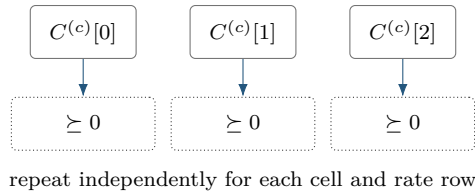
The idea is the Bernstein convex-hull property. After orienting the cell residual $\mathcal{F}^{(c)}$ as the sign-normalized positive target $S^{(c)}$, suppress the already-stacked rate-row index and write

$$S^{(c)}(\alpha) = \sum_{\mathbf{i}} B_{\mathbf{i}}^{\mathbf{M}}(\alpha) C^{(c)}[\mathbf{i}],$$

where $\mathbf{M}$ is the assembled residual tensor degree and $\mathbf{i} \in \prod_s \{0, \dots, M_s\}$. The semidefinite direct certificate requires $C^{(c)}[\mathbf{i}] \succeq 0$ for every physical cell $\mathbf{c}$ and local label $\mathbf{i}$, repeated independently for every stored rate row. This is sufficient, not necessary.

For an entry-wise inequality, replace each matrix coefficient by $\mathcal{C}(:)$ and impose the oriented scalar sign on every component. The global classification is made before Pólya elevation or Gram assembly, so every coefficient and every certificate in one wrapper uses the same mode. Putinar, `SparseFullBox`, and `FullBox` create one independent scalar Gram certificate for each column-major entry. Direct equality simply imposes $\mathcal{C}(:) == 0$; it does not use a positivity certificate or an implicit tolerance. Section B.4 proves the convex-hull implication and separates this sufficient coefficient test from point sampling.

The diagram specializes to one parameter, degree two, and one ordinary (non-rate) row; therefore c is a scalar one-based cell index and $i \in \{0, 1, 2\}$.



Syntax

```
C = pdlmi(expr, relation)
C = lhs <= rhs
C = lhs >= rhs
C = lhs == rhs
```

The constructor and comparison forms above create direct coefficient constraints when no inequality certificate is selected. For a `pdvar` residual, each stored coefficient is constrained in the oriented positive-semidefinite or entry-wise mode; equality constrains each coefficient entry directly to zero. Its stored-constraint count is therefore the number of physical cells times the number of local Bernstein coefficients, with an additional factor for active rate vertices. As stated above, a coefficient-backed known `pdmatrix` inequality instead stores exactly one logical cell containing the global numeric certificate.

Examples and output

```
yal mip('clear')
P = pdvar(1, {[0 1]}, Degree=2);
C = P >= 0;

constraintCount = numel(C.Constraints)
usePolya = C.UsePolya
usePutinar = C.UsePutinar
useSparseFullBox = C.UseSparseFullBoxPreorder
useFullBox = C.UseFullBoxPreorder
```

```
constraintCount = 3
usePolya =
    logical
    0
usePutinar =
    logical
    0
useSparseFullBox =
    logical
    0
useFullBox =
    logical
    0
```

Here the degree-two, one-parameter residual has three local Bernstein coefficients, so direct assembly stores three coefficient constraints. The wrapper remains direct because no certificate selector was enabled; a call such as `C.applyPolya(1)` returns a new wrapper with an elevated coefficient family and leaves `C` unchanged.

6.5 Select Pólya Assembly with applyPolya

With `UsePolya=true`, `pdlmi` first elevates the stored residual componentwise by `d = PolyDegree`. For a `pdvar` residual it then constrains every elevated local coefficient and stored rate row: if the assembled residual degree is `M`, the count is $N_c \prod_s (M_s + d_s + 1)$ for ordinary expressions and $N_c N_v \prod_s (M_s + d_s + 1)$ for rate-row expressions. A coefficient-backed known `pdmat` residual instead stores exactly one logical cell containing the global numeric certificate, including after Pólya elevation. An all-zero increment is valid and has the same coefficient count as `Direct`. For an entry-wise `pdvar` inequality, elevation preserves the global mode and the oriented sign is applied independently to every column-major scalar entry of each elevated coefficient. Equality wrappers reject this method.

Syntax

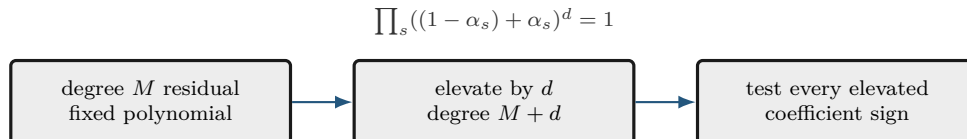
```
Cpolya = C.applyPolya()
Cpolya = C.applyPolya(d)
Cpolya = C.applyPolya(..., ValidationMode=mode)
```

`applyPolya` is a value-class method: it returns a new `pdlmi`, leaves `C` unchanged, and rebuilds from `C.Residual`. Selecting a new increment therefore replaces rather than compounds a previous selection. The no-argument form uses one in every direction; an invalid scalar or vector increment raises `pdlmi:InvalidPolyaDegree`.

Examples and output

```
yalmip('clear')
P = pdvar(1, {[0 1]}, Degree=1);
direct = P >= 0;
polya = direct.applyPolya(2);
directDegree = direct.PolyaDegree
usePolya = polya.UsePolya
polyaDegree = polya.PolyaDegree
constraintCount = numel(polya.Constraints)
```

```
directDegree = 0
usePolya =
    logical
     1
polyaDegree = 2
constraintCount = 4
```



Pólya degree is a representation increment that may differ by direction; it is not a new decision degree, a grid refinement, or a Gram order.

The mathematical background, its interpretation as lossless tensor degree elevation, and the limits of fixed-increment coefficient positivity are developed in section B.5.

6.6 Select Putinar Assembly with `applyPutinar`

For an entry-wise inequality, the constructions below are applied independently to each scalar matrix entry in MATLAB column-major order. Gram variables are not shared between entries. Equality wrappers reject Putinar selection.

The Putinar selector is dimension-aware. With one parameter it deliberately uses the same exact even/odd Markov–Lukács assembly as `applyFullBoxPreorder`. Let $Z_r(\alpha) = \mathbf{b}_r(\alpha) \otimes I_n$, where $\mathbf{b}_r = [B_0^r, \dots, B_r^r]^\top$. For selected absolute order r , the even form matches at degree $2r$,

$$S^{(c)} = Z_r^\top Q_0 Z_r + \alpha(1 - \alpha) Z_{r-1}^\top Q_1 Z_{r-1},$$

where the second block is omitted at $r = 0$. The odd form matches at degree $2r + 1$,

$$S^{(c)} = (1 - \alpha) Z_r^\top Q_L Z_r + \alpha Z_r^\top Q_U Z_r.$$

All present Gram matrices are positive semidefinite. The one-parameter default and minimum are $r = \lfloor M/2 \rfloor$ for assembled residual degree M .

For two or more parameters, the selector instead uses the fixed-order, box-specific singleton-generator Bernstein–Gram quadratic module. After normalizing the relation so that $S^{(c)}(\alpha)$ is positive semidefinite, it represents

$$S^{(c)}(\alpha) = S_0(\alpha) + \sum_{s=1}^{\ell} \alpha_s (1 - \alpha_s) S_s(\alpha).$$

Only singleton box generators appear in this multivariate branch: products such as $\alpha_1(1 - \alpha_1)\alpha_2(1 - \alpha_2)$ belong to the full-box preordering, not this module.

Syntax

```
Cputinar = C.applyPutinar()
Cputinar = C.applyPutinar(r)
Cputinar = C.applyPutinar(..., ValidationMode=mode)
```

Here $\mathbf{r}$ is scalar shorthand or a direction-wise absolute Bernstein Gram order $\mathbf{r}$. The no-argument form uses the dimension-dependent minima above; an explicit order must satisfy the same applicable componentwise bound. In the multivariate branch, the unweighted block S_0 uses tensor-product basis degree $\mathbf{r}$, while S_s uses degree $\mathbf{r} - \mathbf{e}_s$. A nominal block with a negative component is omitted. The target is degree-elevated without changing its represented polynomial, and the multivariate Gram sum is matched exactly at tensor degree $2\mathbf{r}$.

Examples and output

```
yalmp('clear')
P = pdvar(1, {[0 1]}, Degree=2);
direct = P >= 0;
putinar = direct.applyPutinar();
directUsesPutinar = direct.UsePutinar
directOrder = direct.PutinarOrder
directConstraintCount = numel(direct.Constraints)
selectedUsesPutinar = putinar.UsePutinar
selectedOrder = putinar.PutinarOrder
selectedConstraintCount = numel(putinar.Constraints)
```

```
directUsesPutinar =
    logical
     0
directOrder = 0
directConstraintCount = 3
selectedUsesPutinar =
    logical
     1
selectedOrder = 1
selectedConstraintCount = 5
```

For this one-parameter scalar degree-two cell, order one uses the exact even Markov–Lukács form: an unweighted Gram block of dimension two and one generator-weighted block of dimension one, followed by three exact coefficient identities. The result is identical to `direct.applyFullBoxPreorder()`. Each physical cell and each active rate-vertex row receives independent copies of these PSD blocks and identities.

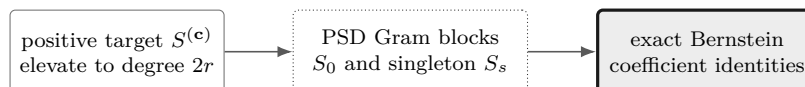
Remarks

For the multivariate backend, the smallest useful worked case is $\ell = 2$. Let $g_s = \alpha_s(1 - \alpha_s)$. In the uniform specialization $\mathbf{r} = (r, r)$, Putinar uses exactly the singleton-generator module

$$S^{(\mathbf{c})} = S_0 + g_1 S_1 + g_2 S_2,$$

where S_0 has Bernstein basis degree (r, r) , S_1 has degree $(r - 1, r)$, and S_2 has degree $(r, r - 1)$. Each S_j is a PSD Gram form. MATLAB elevates the sign-normalized target and all present terms to tensor degree $(2r, 2r)$, then creates one exact coefficient match per local Bernstein label. These PSD blocks and equalities are assembled independently for every physical cell, active rate row, and scalar matrix entry; they are never shared across those axes.

The diagram below specializes to the even one-parameter example above.



The method changes the finite certificate, not the stored residual, decision degree, or physical grid.

`applyPutinar` is a value-class method: it returns a new `pdlmi`, leaves `C` unchanged, and rebuilds from the original `Residual` and `Relation`. Reapplication replaces any earlier Pólya, Putinar,

SparseFullBox, or FullBox selection. It adds no implicit margin and makes no solver call. Sections B.6 and B.7 derive the one-dimensional parity form and the multivariate singleton-generator module used by this selector.

Remarks

1. Malformed scalar or vector orders raise `pdlmi:InvalidPutinarOrder`; an order below $\lfloor M/2 \rfloor$ in one parameter, or below componentwise $\lceil \mathbf{M}/2 \rceil$ in two or more parameters, raises `pdlmi:PutinarOrderTooLow`.
2. A malformed selector raises `pdlmi:InvalidUsePutinar`; explicit false plus any supplied order raises `pdlmi:ConflictingPutinarOptions`.
3. Simultaneous relaxation families raise `pdlmi:ConflictingRelaxations`; expression, relation, and option validation in section 6.3 still applies. Global inequality classification is unchanged: the family is semidefinite only when every original coefficient across all cells and rate rows is square Hermitian. Otherwise the complete family uses independent entry-wise scalar Putinar certificates in MATLAB column-major order.

6.7 Select SparseFullBox with applySparseFullBoxPreorder

SparseFullBox is FullBox with each dense Gram matrix Q_J replaced by a symmetric block-band Gram matrix $Q_J^{(w)}$ of scalar bandwidth w . It uses the same one-parameter parity terms, multivariate box-generator masks, relation orientation, and exact target-coefficient identities as FullBox. Only the admissible entries of the Gram matrices change: the cell residual $\mathcal{F}^{(c)}$, physical grid, decision degree $\mathbf{m}$, assembled residual degree $\mathbf{M}$, generator weights, and coefficient identities remain unchanged.

Syntax

```
Csp = C.applySparseFullBoxPreorder()
Csp = C.applySparseFullBoxPreorder(w)
Csp = C.applySparseFullBoxPreorder(w, r)
Csp = C.applySparseFullBoxPreorder(..., ValidationMode=mode)
```

Arguments

$\mathbf{w}$ is one finite positive integer bandwidth for $Q_J^{(w)}$ and defaults to two. $\mathbf{r}$ is scalar shorthand or an optional direction-wise absolute SparseFullBox Gram order. It defaults to $\lfloor M/2 \rfloor$ for a one-parameter residual of degree M , or componentwise $\lceil \mathbf{M}/2 \rceil$ for two or more parameters. Every explicit order is validated against that same componentwise minimum before endpoint normalization.

Description

For selected absolute order $\mathbf{r}$, use the paper notation χ_J for the indicator of J and $\mathbf{a}_J = \mathbf{r} - \chi_J$.

SparseFullBox uses

$$S^{(c)}(\alpha) = \sum_{J \subseteq \{1, \dots, \ell\}} \left(\prod_{s \in J} \alpha_s (1 - \alpha_s) \right) (\mathbf{b}_{\mathbf{a}_J}(\alpha) \otimes I_n)^\top Q_J^{(w)} (\mathbf{b}_{\mathbf{a}_J}(\alpha) \otimes I_n), \quad Q_J^{(w)} \succeq 0,$$

where every $Q_J^{(w)}$ is symmetric block-band with bandwidth w , and terms for which $\mathbf{a}_J$ has a negative component are omitted. The one-parameter parity-specific FullBox formulas use the same replacement $Q_J \mapsto Q_J^{(w)}$. Exact coefficient matching is unchanged. In one parameter, $w = 1$ is block diagonal and the implementation canonicalizes it to Direct, $w = 2$ is block tridiagonal, and $w = 3$ is block pentadiagonal. Increasing w gives a nested sufficient hierarchy, but feasibility need not improve strictly on a particular model. Section B.9 explains how the retained tensor windows determine the banded Gram map and why the two endpoint requests rebuild canonical Direct or FullBox state.

Examples and output

```
yalmp('clear')
P = pdvar(1, {[0 1]}, Degree=4);
direct = P >= 0;
sparse = direct.applySparseFullBoxPreorder(2, 2);
directEndpoint = direct.applySparseFullBoxPreorder(1, 2);
denseEndpoint = direct.applySparseFullBoxPreorder(3, 2);
sparseState = [sparse.UseSparseFullBoxPreorder, ...
    sparse.SparseFullBoxOrder, sparse.BandWidth, ...
    sparse.UseFullBoxPreorder, numel(sparse.Constraints)]
directEndpointState = [directEndpoint.UseSparseFullBoxPreorder, ...
    directEndpoint.UseFullBoxPreorder, directEndpoint.BandWidth, ...
    numel(directEndpoint.Constraints)]
denseEndpointState = [denseEndpoint.UseSparseFullBoxPreorder, ...
    denseEndpoint.UseFullBoxPreorder, denseEndpoint.FullBoxOrder, ...
    numel(denseEndpoint.Constraints)]
```

```
sparseState = 1×5 double
    1     2     2     0     8
directEndpointState = 1×4 double
    0     0     0     5
denseEndpointState = 1×4 double
    0     1     2     7
```

The intermediate result retains SparseFullBox state with $r = 2$ and $w = 2$; in one parameter its Gram matrices are block tridiagonal, and all five target coefficients are matched exactly. With one returns actual Direct state with no sparse metadata or Gram variables. Since $w = 3 = r + 1$, the dense endpoint returns actual FullBox state with two dense PSD blocks followed by the same five identities. These are canonical public states, not merely equivalent constraint lists.

Remarks

1. `applySparseFullBoxPreorder` is a value-class method: it returns a new wrapper, leaves the source unchanged, and rebuilds from **Residual** and **Relation**, replacing any earlier Pólya, Putinar, SparseFullBox, or FullBox selection. It adds no positivity margin and makes no solver call.

2. Every physical cell and active rate row owns independent Gram matrices $Q_J^{(w)}$ and coefficient identities. Semidefinite mode builds matrix Gram terms; entry-wise mode builds an independent scalar certificate for every matrix entry in MATLAB column-major order.
3. Width one canonicalizes to actual Direct state. A width satisfying $w \geq \max(r + 1)$ canonicalizes to actual FullBox state. For $1 < w < \max_s(r_s + 1)$, the result records `UseSparseFullBoxPreorder=true`, `SparseFullBoxOrder=r`, and `BandWidth=w`.
4. Invalid widths raise `pdlmi:InvalidBandWidth`. Malformed orders raise `pdlmi:InvalidSparseFullBoxOrder`; orders below the dimension-dependent minimum raise `pdlmi:SparseFullBoxOrderTooLow`. More than two positional inputs raise `pdlmi:InvalidApplyOptions`.
5. Constructor selector conflicts use `pdlmi:ConflictingSparseFullBoxOptions` or `pdlmi:ConflictingRelaxations`. Equality raises `pdlmi:UnsupportedEqualityCertificate` before width or order validation.

6.8 Select FullBox with `applyFullBoxPreorder`

The mathematical construction is a full-box preordering in the unit-box local coordinates $\alpha = \phi_c^{-1}(\rho) \in [0, 1]^\ell$; the software state is `FullBox`. It is a box-specific Bernstein–Gram representation, not a general Putinar certificate, general-domain SOS interface, or full-space SOS representation. It adds no implicit positivity margin. For an entry-wise inequality, every column-major scalar entry receives an independent copy of the selected Gram construction. Equality wrappers reject `FullBox` selection.

Syntax

```
Cbox = C.applyFullBoxPreorder()
Cbox = C.applyFullBoxPreorder(r)
Cbox = C.applyFullBoxPreorder(..., ValidationMode=mode)
```

The no-argument form selects the smallest admissible absolute order. The explicit form accepts scalar shorthand or a direction-wise `r`; it does not add `r` to the residual degree. Both forms are value-class selectors: they return a new `pdlmi`, leave `C` unchanged, and rebuild from `C.Residual`. Reapplication therefore replaces an earlier `FullBox` order or a Pólya, Putinar, or `SparseFullBox` selection rather than compounding it.

The idea is to represent the oriented positive target as a sum of PSD Gram forms multiplied by functions that are nonnegative on the local box. For selected direction-wise absolute order `r`, let χ_J be the indicator of J , set $\mathbf{a}_J = \mathbf{r} - \chi_J$, and define $Z_{\mathbf{a}_J} = \mathbf{b}_{\mathbf{a}_J} \otimes I_n$:

$$S^{(c)}(\alpha) = \sum_{J \subseteq \{1, \dots, \ell\}} \left(\prod_{s \in J} \alpha_s (1 - \alpha_s) \right) Z_{\mathbf{a}_J}(\alpha)^\top Q_J Z_{\mathbf{a}_J}(\alpha), \quad Q_J \succeq 0.$$

Terms for which $\mathbf{a}_J$ has a negative component are omitted. In one parameter the implementation uses the even/odd Markov–Lukács forms, with minimum absolute order $\lfloor M/2 \rfloor$ for assembled residual degree M . For two or more parameters it uses every subset product of the ℓ box generators

and componentwise minimum order $\lceil \mathbf{M}/2 \rceil$. The selected order fixes the dense Bernstein Gram bases; it is not added to $\mathbf{M}$.

Examples and output

```
yal mip('clear')
P = pdvar(1, {[0 1]}, Degree=2);
direct = P >= 0;
box = direct.applyFullBoxPreorder();
directConstraintCount = numel(direct.Constraints)
useFullBox = box.UseFullBoxPreorder
fullBoxOrder = box.FullBoxOrder
fullBoxConstraintCount = numel(box.Constraints)
```

```
directConstraintCount = 3
useFullBox =
    logical
         1
fullBoxOrder = 1
fullBoxConstraintCount = 5
```

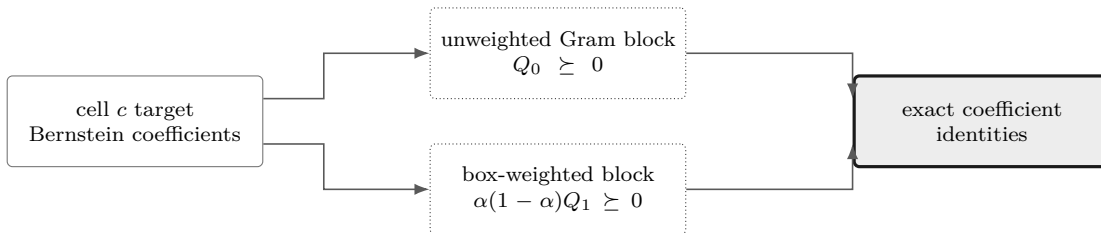
At this order the scalar degree-two cell has two PSD Gram constraints followed by three exact Bernstein coefficient identities.

Remarks

For $\ell = 2$, the FullBox state differs from Putinar by retaining the product generator. With $g_s = \alpha_s(1 - \alpha_s)$, the uniform specialization $\mathbf{r} = (r, r)$ has the exact backend form

$$S^{(c)} = S_0 + g_1 S_1 + g_2 S_2 + g_1 g_2 S_{12}.$$

The Bernstein basis degrees are respectively (r, r) , $(r-1, r)$, $(r, r-1)$, and $(r-1, r-1)$; each present term is a PSD Gram block. All four terms are elevated and coefficient-matched at tensor degree $(2r, 2r)$. Thus the additional S_{12} block is precisely the two-generator distinction from the singleton Putinar module, not a change to the residual, grid, or decision degree. The implementation builds the PSD blocks and exact equalities independently for each cell, rate row, and scalar matrix entry.



Sections B.6 and B.8 derive the one-dimensional parity forms, explain the multivariate subset masks, and distinguish this implemented box preordering from the multivariate singleton-generator Putinar branch, full-space SOS, and general polynomial parsers. Assembly is independent for every physical cell and every active rate-vertex row; Gram variables are not shared across either boundary. For $\geq$, the method represents the residual. For $\leq$, it negates the target before forming the positive-semidefinite representation. Within each cell and row, the stored constraints are ordered as all PSD

Gram blocks followed by exact equalities matching every Bernstein coefficient. Consequently, the number of stored constraints is not the direct coefficient count.

Remarks

1. Malformed orders raise `pdlmi:InvalidFullBoxOrder`; an admissible integer below the minimum raises `pdlmi:FullBoxOrderTooLow`. The constructor rejects malformed selectors with `pdlmi:InvalidUseFullBoxPreorder`.
2. Simultaneous Pólya, Putinar, SparseFullBox, or FullBox selection raises `pdlmi:ConflictingRelaxations`; expression, relation, and option validation in section 6.3 still applies.
3. Global inequality classification is unchanged: the family is semidefinite only when every original coefficient across all cells and rate rows is square Hermitian. Otherwise the complete family uses independent entry-wise scalar FullBox certificates in MATLAB column-major order.

6.9 Export with toYalmip

Syntax

```
F = toYalmip(C)
F = C.toYalmip()
```

Arguments

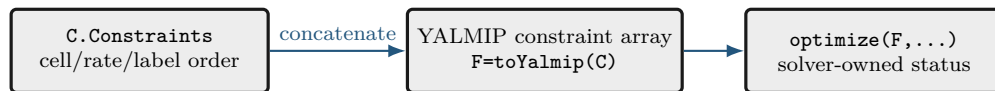
For a `pdvar` residual, `toYalmip` concatenates the stored constraint cells into a YALMIP constraint array for Direct, Pólya, Putinar, SparseFullBox, and FullBox objects. For a coefficient-backed known `pdmat` residual, Direct and Pólya instead return the one stored scalar logical certificate. Its numeric check uses the wrapper’s global inequality classification and inclusive absolute tolerance 10^{-10} : semidefinite mode symmetrizes each coefficient matrix as $(C + C^T)/2$ and tests its eigenvalues, whereas only element-wise mode tests individual scalar entries. A false result emits warning `pdlmi:InconclusiveCertificate`: failure of this sufficient coefficient certificate does not prove a violation or indefiniteness of the continuous function. Known-data Putinar, SparseFullBox, and FullBox objects still export YALMIP constraints and may contain auxiliary Gram variables. The method has no package-specific options and does not add a solver wrapper, objective, margin, or diagnostic layer.

Examples and output

The function-call and dot-call forms return equivalent YALMIP constraint arrays.

```
yalmip('clear')
P = pdvar(2, {[0 1]}, "symmetric");
C = P >= 0;
wrapperClass = class(C)
storedConstraintCount = numel(C.Constraints)
F1 = toYalmip(C);
F2 = C.toYalmip();
firstIsConstraint = isa(F1, "lmi") || isa(F1, "constraint")
firstConstraintCount = length(F1)
secondIsConstraint = isa(F2, "lmi") || isa(F2, "constraint")
secondConstraintCount = length(F2)
```

```
wrapperClass = 'pdlmi'
storedConstraintCount = 2
firstIsConstraint =
    logical
     1
firstConstraintCount = 2
secondIsConstraint =
    logical
     1
secondConstraintCount = 2
```



After this boundary, the package is no longer assembling Bernstein data; YALMIP owns solver choice, diagnostics, and optimization status.

Examples and output

Known coefficient evidence can be certified without introducing a solver model on the Direct and Pólya paths.

```
yalmip('clear')
A = pdmat([0 1], {[1, 2; 3, 4]}, ...
    Degree=1, RateBounds=[-1 2]);
direct = A >= 0;
directCertificate = toYalmip(direct)
polya = direct.applyPolya(1);
polyaCertificate = toYalmip(polya)
```

```
directCertificate = logical
 1
polyaCertificate = logical
 1
```

6.10 Boundaries and Related APIs

`pdvar`, `rhodiff`, `applyPolya`, `applyPutinar`, `applySparseFullBoxPreorder`, `applyFullBoxPreorder`, `toYalmip`, `optimize`.

6.10.1 Solve with YALMIP

`pdlmi` does not own solver settings or wrap `optimize`. After `toYalmip`, use ordinary YALMIP and validate the returned status before retaining an objective or recovered coefficient data:

```
opts = sdpsettings("solver", solverName, "verbose", 0);
sol = optimize([C1.toYalmip, C2.toYalmip], objective, opts);
assert(sol.problem == 0, sol.info)
objectiveValue = value(objective);
```

A result with `problem ~= 0` is not an optimized certificate value. In particular, numerical, unknown, aborted, memory, and space-exhaustion statuses are inconclusive about both the original continuous problem and the value of γ .

Chapter 7

Shared Helper Utilities

The `+helper` namespace contains exactly seven implementation utilities whose ownership genuinely crosses class boundaries: `cellGet`, `chk`, `combRows`, `isZero`, `mkGrid`, `mkNest`, and `normalizeDegree`. They are developer-facing building blocks, not an alternative modeling API: ordinary users should enter through `pdbase`, `pdmatrix`, `pdvar`, and `pdlmi`. The object-aware `bernTbl`, `mapVals`, and `matSubs` mechanisms are protected `pdbase` backend utilities and are documented in section 3.9.6, not as `helper.*` functions.

7.1 `helper.cellGet`

Purpose. Read one flat coefficient leaf from a nested physical-cell tree.

Syntax

```
leaf = helper.cellGet(vals, subs)
```

Description

`vals` is addressed as `vals{i1}{i2}...{i_ell}` and `subs` supplies one index per level. The returned `leaf` is the selected flat coefficient cell or rate-row table; no copy, reshaping, or multi-index cell interpretation is introduced.

Examples and output

```
vals = {{{10, 20}}, {{30, 40}}};  
leaf = cell2mat(helper.cellGet(vals, [2 1]))
```

```
leaf = 1×2 double  
      30      40
```

Remarks

The helper deliberately relies on ordinary MATLAB cell indexing. Callers validate the subscript length and bounds before traversal.

7.2 helper.chk

Purpose. Apply shared validation predicates while retaining a caller-owned error identifier and message.

Syntax

```
val = helper.chk(val, errId, msg, tags...)
val = helper.chk(val, errId, msg, tags..., Name, Value)
```

Description

Predicate tags are `numeric`, `real`, `cell`, `struct`, `nonempty`, `scalar`, `vector`, `matrix`, `finite`, `integer`, `positive`, `nonnegative`, `increasing`, and `rowbounds`. Named tests are `Size`, `Numel`, `MinNumel`, `Min`, and `Max`. Successful validation returns `val` unchanged.

Examples and output

```
validated = helper.chk([1 2], "demo:BadValue", "bad value", ...
    "numeric", "real", "finite", "Size", [1 2])
```

```
validated = 1×2 double
           1     2
```

Remarks

A failed data predicate raises `errId`. An unknown predicate, or a named test without a value, raises `helper:InvalidValidatorCall`; these errors signal a validator-programming defect rather than bad caller data.

7.3 helper.combRows

Purpose. Enumerate Cartesian combinations in the ordering shared by physical cells, local labels, and rate vertices.

Syntax

```
rows = helper.combRows(vecs)
```

Description

`vecs` is a cell array containing one vector per tensor axis. `rows` contains every Cartesian combination, with earlier dimensions varying more slowly.

Examples and output

```
rows = helper.combRows({0:1, 10:11})
```

```
rows = 4×2 double
    0    10
    0    11
    1    10
    1    11
```

Remarks

This ordering is part of the storage contract. Reordering the rows would misalign tensor labels, physical-cell traversal, and derivative rate vertices.

7.4 helper.isZero

Purpose. Prove the distinct zero notions used by known-data and affine algebra fast paths.

Syntax

```
tf = helper.isZero(val, "num")
tf = helper.isZero(val, "add", matrixSize)
tf = helper.isZero(vals, "vals")
tf = helper.isZero(obj, "obj")
```

Description

"num" requires a nonempty finite real numeric zero; "add" additionally enforces scalar expansion or the supplied matrix shape; "vals" recursively checks nested, symbolic, and rate-structured payloads; and "obj" accepts coefficient-backed `pdmat` or `pdvar` evidence.

Examples and output

```
numericZero = helper.isZero(zeros(2), "add", [2 2])
A = pdmat([0 1], {0, 0}, Degree=1);
objectZero = helper.isZero(A, "obj")
```

```
numericZero =
    logical
    1
objectZero =
    logical
    1
```

Remarks

Function-only `pdmatrix` placeholders are not coefficient evidence and therefore do not prove object zero. Invalid modes raise `helper:InvalidZeroMode`; the wrong mode-specific arity raises `helper:InvalidZeroCall`.

7.5 `helper.mkGrid`

Purpose. Validate tensor-grid vectors and construct the shared `GridInfo` record.

Syntax

```
info = helper.mkGrid(grid)
info = helper.mkGrid(grid, owner)
```

Description

`grid` is a nonempty cell array of finite, real, strictly increasing vectors with at least two nodes each. `owner` defaults to `"pdbase"` and prefixes validation identifiers. The result contains `Vectors`, Cartesian-product `Points`, per-axis `Bounds`, and `NumNodes`.

Examples and output

```
info = helper.mkGrid({[0 1], [10 20]}, "demo");
bounds = info.Bounds
points = info.Points
```

```
bounds = 2x2 double
    0     1
   10    20
```

```
points = 4x2 double
    0    10
    0    20
    1    10
    1    20
```

Remarks

Malformed containers use `owner + ":InvalidGrid"`; malformed individual axes use `owner + ":InvalidGridVector"`. Cell counts are derived by consumers and are not duplicated in `GridInfo`.

7.6 helper.mkNest

Purpose. Allocate recursive physical-cell storage and construct each leaf from its complete cell subscript.

Syntax

```
vals = helper.mkNest(nCell, mkLeaf)
vals = helper.mkNest(nCell, mkLeaf, prefix)
```

Description

`nCell` gives the positive cell count per parameter direction and `mkLeaf` receives one completed one-based subscript row. `prefix` is recursion state used internally; ordinary callers omit it. The result is addressed as `vals{i1}{i2}...{i_ell}`.

Examples and output

```
vals = helper.mkNest([2 1], @(subs) {sum(subs)});
secondLeaf = cell2mat(helper.cellGet(vals, [2 1]))
```

```
secondLeaf = 3
```

Remarks

`mkLeaf` owns the leaf payload. The helper supplies deterministic physical-cell subscripts but does not infer coefficient count, matrix size, or rate semantics.

7.7 helper.normalizeDegree

Purpose. Validate scalar shorthand or direction-wise Bernstein degrees while preserving the caller's error namespace.

Syntax

```
degree = helper.normalizeDegree(value, nPar, errId, label)
```

Description

`value` must be one finite nonnegative integer scalar or an `nPar`-element vector. A scalar expands uniformly, a column vector is reshaped, and every accepted result is a 1-by-`nPar` double row. `errId` is raised for empty, nonnumeric, complex, nonfinite, noninteger, negative, or incorrectly sized input; `label` supplies the public option name in the error message and defaults to "Degree".

Examples and output

```
degree = helper.normalizeDegree([0; 2; 1], 3, ...  
    "demo:InvalidDegree", "Degree")
```

```
degree = 1×3 double  
    0     2     1
```

Remarks

The helper performs normalization only. Constructors and certificate selectors own user-facing warnings, minimum-order checks, and their class-specific error identifiers.

Chapter 8

Worked Solver Examples

8.1 Deterministic Putinar, SparseFullBox, And FullBox Selection

This example selects default Putinar and FullBox certificates, an explicit FullBox order, and the three canonical SparseFullBox outcomes without solving an optimization problem. The reported values depend only on the public assembly contract, not on incidental YALMIP decision-variable identifiers.

```
yal mip('clear')
Pa = pdvar(1, {[0 1], [10 20]}, Degree=[2 0]);
directA = Pa >= 0;
polyaA = directA.applyPolya([0 1]);
decisionDegree = Pa.Degree
polyaIncrement = polyaA.PolyaDegree
elevatedDegree = polyaA.Residual.Degree + polyaA.PolyaDegree
constraintCount = numel(polyaA.Constraints)
```

```
decisionDegree = 1×2 double
    2     0

polyaIncrement = 1×2 double
    0     1

elevatedDegree = 1×2 double
    2     1

constraintCount = 6
```

This anisotropic example has three original labels and six labels after direction-wise elevation: $(2 + 1)(0 + 1) = 3$ and $(2 + 1)(1 + 1) = 6$. The zero second component of `Pa.Degree` makes the decision constant in the second parameter; `PolyaDegree=[0 1]` changes only its certificate representation.

```
yal mip('clear')
P = pdvar(1, {[0 1]}, Degree=2);
direct = P >= 0;
putinarDefault = direct.applyPutinar();
boxDefault = direct.applyFullBoxPreorder();
boxHigher = boxDefault.applyFullBoxPreorder(2);
polya = direct.applyPolya(1);
boxFromPolya = polya.applyFullBoxPreorder();
putinarFromBox = boxHigher.applyPutinar();
Fbox = toYalmip(boxDefault);

directState = [direct.UseFullBoxPreorder, direct.FullBoxOrder, ...
    numel(direct.Constraints)]
defaultState = [boxDefault.UseFullBoxPreorder, boxDefault.FullBoxOrder, ...
    numel(boxDefault.Constraints)]
putinarState = [putinarDefault.UsePutinar, putinarDefault.PutinarOrder, ...
    numel(putinarDefault.Constraints)]
higherState = [boxHigher.FullBoxOrder, numel(boxHigher.Constraints)]
replacementState = [boxFromPolya.UsePolya, ...
    boxFromPolya.UseFullBoxPreorder, boxFromPolya.FullBoxOrder]
putinarReplacementState = [putinarFromBox.UsePutinar, ...
    putinarFromBox.UseFullBoxPreorder, putinarFromBox.PutinarOrder]
exportedConstraintCount = length(Fbox)
```

```

directState = 1×3 double
    0    0    3
defaultState = 1×3 double
    1    1    5
putinarState = 1×3 double
    1    1    5
higherState = 1×2 double
    2    7
replacementState = 1×3 double
    0    1    1
putinarReplacementState = 1×3 double
    1    0    1
exportedConstraintCount = 5

```

For this one-dimensional degree-two scalar residual, order one uses two PSD Gram blocks followed by three exact coefficient identities. Order two matches a degree-four target and stores two PSD blocks followed by five identities. The original `direct` object remains unchanged, and the selection made from `polya` replaces Pólya rather than elevating its already assembled constraints. Likewise, selecting Putinar from `boxHigher` rebuilds from the original residual and clears the full-box selection. The next degree-four residual makes width two an intermediate `SparseFullBox` certificate, while widths one and three expose its canonical endpoints.

```

P4 = pdvar(1, {[0 1]}, Degree=4);
direct4 = P4 >= 0;
sparse4 = direct4.applySparseFullBoxPreorder(2, 2);
direct4Endpoint = sparse4.applySparseFullBoxPreorder(1, 2);
full4Endpoint = sparse4.applySparseFullBoxPreorder(3, 2);
sparse4State = [sparse4.UseSparseFullBoxPreorder, ...
    sparse4.SparseFullBoxOrder, sparse4.BandWidth, ...
    numel(sparse4.Constraints)]
direct4State = [direct4Endpoint.UseSparseFullBoxPreorder, ...
    direct4Endpoint.UseFullBoxPreorder, ...
    numel(direct4Endpoint.Constraints)]
full4State = [full4Endpoint.UseSparseFullBoxPreorder, ...
    full4Endpoint.UseFullBoxPreorder, full4Endpoint.FullBoxOrder, ...
    numel(full4Endpoint.Constraints)]

```

```

sparse4State = 1×4 double
    1    2    2    8
direct4State = 1×3 double
    0    0    5
full4State = 1×4 double
    0    1    2    7

```

The intermediate wrapper stores three width-two PSD blocks followed by five identities. Width one rebuilds actual `Direct` state with five coefficient constraints. Width three satisfies $b = r + 1$ and rebuilds actual `FullBox` state with two dense PSD blocks followed by the five identities. Every apply call is value-semantic and starts from the stored original residual, so the source wrapper and any earlier selection remain unchanged.

8.2 Solve for a Parameter-Dependent Lyapunov Matrix

The first solver smoke test in `+tests/+pdlmi/test_solver_smoke.m` contrasts a constant Lyapunov matrix with a degree-one, parameter-dependent one. It solves the same decay and positivity constraints in both cases. The degree-one solution is then materialized as known `pdmatrix` data and plotted. `pdmatrix` is a decision-expression class and does not itself provide `plot`; applying `value` to its local YALMIP payloads after a successful solve supplies the numeric Bernstein coefficients for `pdmatrix`.

```
yalmp('clear')
grid = {[0 1]};
A = pdmatrix(grid, @(rho) (1 - rho) * [-1 -1; 1 -1] + ...
    rho * [-1 -10; 0.1 -1], Degree=1);
solver = 'lmilab';
if exist('mosekopt', 'file') ~= 0
    solver = 'mosek';
end
opts = sdpsettings('solver', solver, 'verbose', 0);

Pc = pdmatrix(2, grid, Degree=0); CconstantDecay = Pc * A + A' * Pc <= -1e-10 * eye(2);
CconstantPositive = Pc >= eye(2); solConstant = optimize([CconstantDecay.toYalmip, ...
    CconstantPositive.toYalmip], [], opts);

Pd = pdmatrix(2, grid); CdependentDecay = Pd * A + A' * Pd <= -1e-10 * eye(2);
CdependentPositive = Pd >= eye(2); solDependent = optimize([CdependentDecay.toYalmip, ...
    CdependentPositive.toYalmip], [], opts); assert(solDependent.problem == 0,
    solDependent.info)

Pplot = pdmatrix(grid, ...
    {cellfun(@value, Pd.LocalValues{1}, UniformOutput=false)}, ...
    Degree=Pd.Degree);
figure(Name="Parameter-dependent Lyapunov matrix")
plot(Pplot, SamplesPerCell=40, LineWidth=1.5)
title("Solved parameter-dependent Lyapunov matrix")
```

The figure contains one curve for each entry of the solved 2-by-2 matrix $P(\rho)$, sampled on the parameter interval. It is a diagnostic visualization of the feasible solution, not a certificate beyond the coefficient-wise constraints supplied to the solver. The constant-case result is retained to exercise the comparison in the smoke test; only a MOSEK solve is used there to certify its intended infeasibility distinction.

8.3 Solve a Rate-Dependent Block LMI

This example is adapted from `+tests/+pdlmi/test_solver_smoke.m`. The goal is to find a parameter-dependent $P(\rho)$ and scalar γ such that, on every Bernstein coefficient and rate vertex,

$$\begin{bmatrix} \dot{P} + PA + A^\top P & PB & C^\top \\ B^\top P & -\gamma I & D^\top \\ C & D & -\gamma I \end{bmatrix} \preceq 0, \quad P(\rho) \succeq 0.$$

The derivative term $\dot{P}$ is built with `rhodiff(P, [-1 1])`. The comparison operators create `pdlmi` wrappers, and `toYalmip` hands the coefficient-wise constraints to YALMIP. The second listing assigns the solver- and tolerance-dependent numeric value of γ to `gammaValue`. For this model, `numel(E1.Constraints)` returns 6, and `numel(E2.Constraints)` returns 2.

```

yalmip('clear')
grid = linspace(0, 1, 2);
A = pdmat(grid, @(rho) [-1, 0.5; -1, -2] + ...
    rho * [-1.3, -20; 2, -10], Degree=1);
B = pdmat(grid, @(rho) [1, -4; -1, -1] + ...
    rho * [2.2, 0.5; -6, -5], Degree=1);
Cmat = eye(2);
Dmat = zeros(2);

P = pdvar(2, grid);
diffP = rhodiff(P, [-1 1]);
gamma = pdvar(1, grid, Degree=0);
E1 = [diffP + P * A + A' * P, P * B, Cmat';
    B' * P, -gamma * eye(2), Dmat';
    Cmat, Dmat, -gamma * eye(2)] <= 0;
E2 = P >= 0;
decayConstraintCount = numel(E1.Constraints)
positivityConstraintCount = numel(E2.Constraints)

```

```

decayConstraintCount = 6
positivityConstraintCount = 2

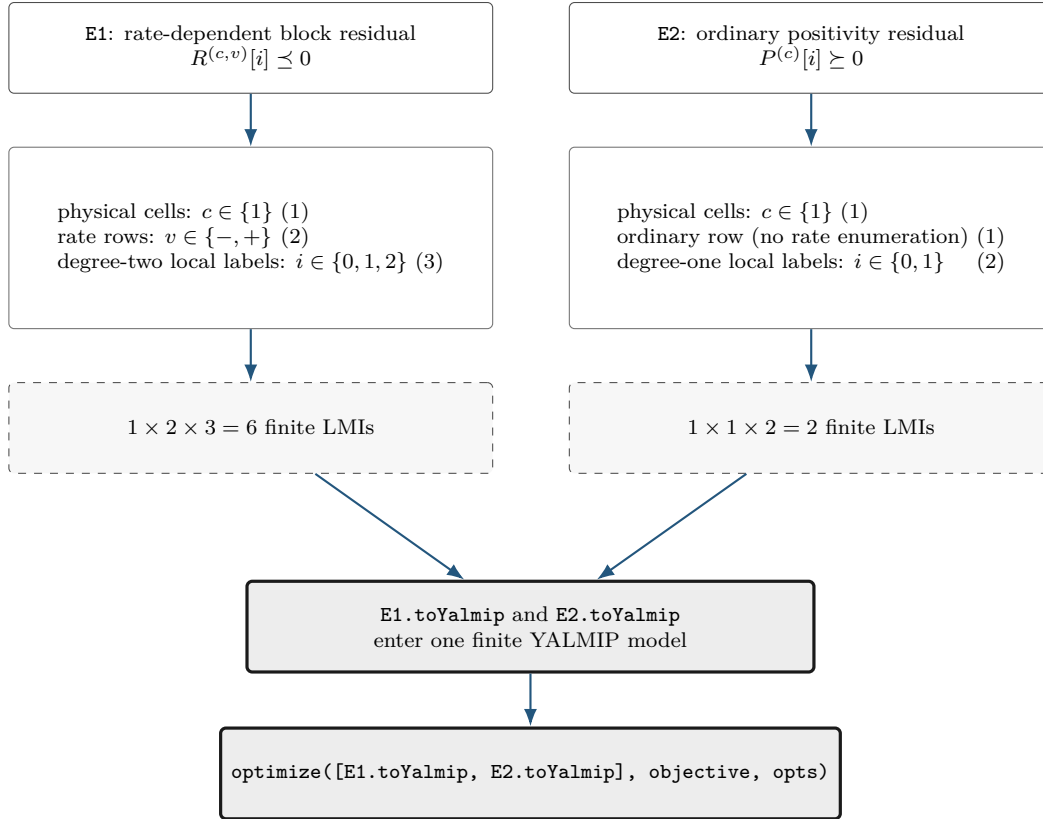
```

```

objective = gamma.LocalValues{1}{1};
solver = 'lmilab';
if exist('mosekopt', 'file') ~= 0
    solver = 'mosek';
end
opts = sdpsettings('solver', solver, 'verbose', 0);
sol = optimize([E1.toYalmip, E2.toYalmip], objective, opts);
assert(sol.problem == 0, sol.info)
gammaValue = value(objective);
assert(isfinite(gammaValue))
gammaValue

```

The constraint counts above follow directly from the three independent assembly axes: physical-cell identity, rate-row identity, and local Bernstein label. In this tested example the two wrappers use the same physical grid but different rate-row and degree structures.



The numbers (6) and (2) are certificate-assembly counts: they state how many finite semidefinite constraints represent E1 and E2. They are not evidence that the optimization succeeds. Feasibility is checked separately by `sol.problem`, and the finite objective is checked after the shared `optimize` call.

Appendix A

Bernstein Representation and Exact Operations

This appendix develops the representation used by the package from the one-dimensional degree-one case to cell-local tensor polynomials. The reader needs only scalar polynomials, matrices, and elementary calculus. Every conversion, elevation, product, derivative, and subdivision below is exact; none is a sampling approximation. Farouki gives a broader historical and numerical account [3].

Each main construction follows the same reading path: *Basic idea*, *Formula*, *Worked example*, and *Visual conclusion*. The extra staging is intentional: first identify what the operation means, then read its indices, and only afterward connect it to package storage.

A.1 Convex Combinations Before Polynomials

Basic idea. A convex combination of numbers or matrices has the form

$$X = \sum_{i=0}^m w_i X_i, \quad w_i \geq 0, \quad \sum_{i=0}^m w_i = 1.$$

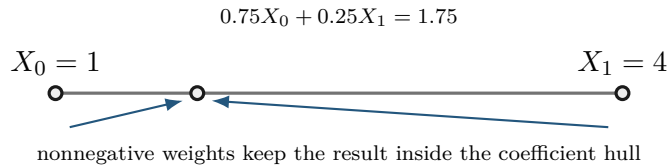
For scalars, X lies between the smallest and largest X_i . For symmetric matrices, if every $X_i \succeq 0$, then $X \succeq 0$, because

$$u^\top X u = \sum_i w_i u^\top X_i u \geq 0 \quad \text{for every } u.$$

Formula. Bernstein bases are useful because their basis values are precisely such weights at every point of a cell.

Worked example. Take $X_0 = 1$, $X_1 = 4$, and weights 0.75 and 0.25. Their combination is 1.75, which remains between the two inputs.

Visual conclusion.



The implication is one-way. A polynomial may be positive everywhere while one Bernstein coefficient is negative; Appendix B uses that fact to motivate stronger certificates.

A.2 Degree-One Interpolation On One Physical Cell

Basic idea. Let $\mathcal{H}_c = [\rho_1^{(c)}, \rho_1^{(c+1)}]$, with width $h_1^c = \rho_1^{(c+1)} - \rho_1^{(c)} > 0$. Its forward map and inverse coordinate are

$$\phi_c(\alpha) = \rho_1^{(c)} + h_1^c \alpha, \quad \alpha = \phi_c^{-1}(\rho_1) = \frac{\rho_1 - \rho_1^{(c)}}{h_1^c} \in [0, 1].$$

Formula. The degree-one Bernstein basis is

$$B_0^1(\alpha) = 1 - \alpha, \quad B_1^1(\alpha) = \alpha.$$

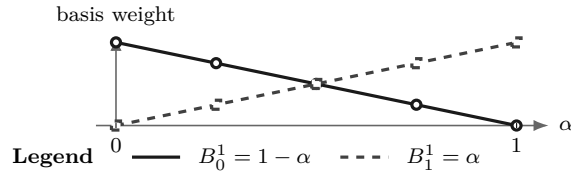
Therefore

$$C^{(c)}(\alpha) = (1 - \alpha)C^{(c)}[0] + \alpha C^{(c)}[1].$$

At the lower physical face, $\alpha = 0$ and the value is $C^{(c)}[0]$; at the upper face, $\alpha = 1$ and the value is $C^{(c)}[1]$. Between them the matrix follows the straight segment joining those two coefficient matrices.

Worked example. For example, $C^{(c)}[0] = 2$ and $C^{(c)}[1] = 4$ give $C^{(c)}(0.25) = 0.75(2) + 0.25(4) = 2.5$, exactly as in section 4.5.

Visual conclusion.



A.3 Higher Degree And The Partition Of Unity

Basic idea. Higher degree adds interior shape controls while keeping the same physical cell. It does not add physical grid nodes.

Formula. For degree m , define

$$B_i^m(\alpha) = \binom{m}{i} (1 - \alpha)^{m-i} \alpha^i, \quad i = 0, \dots, m.$$

Each basis function is nonnegative on $[0, 1]$. The binomial theorem gives

$$\sum_{i=0}^m B_i^m(\alpha) = ((1 - \alpha) + \alpha)^m = 1.$$

Thus

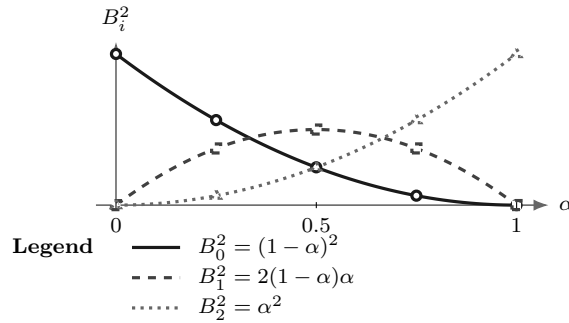
$$C^{(c)}(\alpha) = \sum_{i=0}^m B_i^m(\alpha) C^{(c)}[i]$$

is a convex combination of its local coefficient matrices.

Worked example. For degree two,

$$B_0^2 = (1 - \alpha)^2, \quad B_1^2 = 2(1 - \alpha)\alpha, \quad B_2^2 = \alpha^2.$$

Visual conclusion.



At every α , the three plotted heights are nonnegative and sum to one.

The label i identifies a basis position inside the selected physical cell. It is not a physical grid-node subscript.

A.4 Worked Conversion: $1 + 2\alpha + 3\alpha^2$

Consider

$$p(\alpha) = 1 + 2\alpha + 3\alpha^2.$$

Seek degree-two Bernstein coefficients b_0, b_1, b_2 :

$$p = b_0(1 - \alpha)^2 + 2b_1(1 - \alpha)\alpha + b_2\alpha^2.$$

Matching powers of α gives

$$b_0 = 1, \quad b_1 = 2, \quad b_2 = 6.$$

Indeed,

$$1B_0^2 + 2B_1^2 + 6B_2^2 = (1 - 2\alpha + \alpha^2) + (4\alpha - 4\alpha^2) + 6\alpha^2 = 1 + 2\alpha + 3\alpha^2.$$

The three values 1, 2, 6 are coefficients of basis functions, not samples at three equally spaced physical points. Only the two endpoint coefficients are endpoint values. At $\alpha = 1/2$,

$$p(1/2) = \frac{1}{4}(1) + \frac{1}{2}(2) + \frac{1}{4}(6) = 2.75.$$

```
p = pmat({[0 1]}, {1, 2, 6}, Degree=2);
valueAtHalf = p.evaluate(0.5)
T = bernsteinTable(p, "oneLine");
disp(T)
```

```
valueAtHalf = 2.7500
```

```
CellSubscript
```

```
Expression
```

```
{[1]}
```

```
"(1-alpha)^2*1 + 2(1-alpha)alpha*2 + alpha^2*6"
```

A.5 Exact Power–Bernstein Conversion

More generally, use brackets for algebraic coefficient labels as well:

$$p(\alpha) = \sum_{j=0}^n a[j] \alpha^j = \sum_{k=0}^n b[k] B_k^n(\alpha).$$

Power coefficients convert to Bernstein coefficients by

$$b[k] = \sum_{j=0}^k \frac{\binom{k}{j}}{\binom{n}{j}} a[j], \quad k = 0, \dots, n.$$

The inverse map is

$$a[j] = \binom{n}{j} \sum_{i=0}^j (-1)^{j-i} \binom{j}{i} b[i], \quad j = 0, \dots, n.$$

These are changes of basis inside the same degree- n polynomial space. For matrix polynomials they act entrywise. For tensor polynomials they act along one coordinate axis at a time.

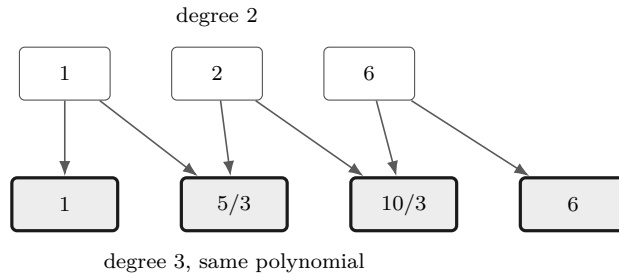
A.6 Lossless Degree Elevation

In one parameter, a polynomial on physical cell c with degree m has a Bernstein representation of every higher degree. One elevation step gives

$$\widehat{C}^{(c)}[0] = C^{(c)}[0], \quad \widehat{C}^{(c)}[k] = \frac{k}{m+1} C^{(c)}[k-1] + \left(1 - \frac{k}{m+1}\right) C^{(c)}[k], \quad \widehat{C}^{(c)}[m+1] = C^{(c)}[m].$$

For the worked coefficients $[1, 2, 6]$, elevation from degree two to degree three produces

$$\left[1, \frac{5}{3}, \frac{10}{3}, 6\right].$$



Direct elevation from m to $M \geq m$ is

$$\widehat{C}^{(c)}[k] = \sum_{i=\max(0, k-M+m)}^{\min(m, k)} C^{(c)}[i] \frac{\binom{m}{i} \binom{M-m}{k-i}}{\binom{M}{k}}.$$

This formula is a binomially normalized discrete convolution. Define

$$U_m^{(c)}[i] = \binom{m}{i} C^{(c)}[i], \quad E_{M-m}[j] = \binom{M-m}{j}, \quad \widehat{U}_M^{(c)}[k] = \binom{M}{k} \widehat{C}^{(c)}[k].$$

Then

$$\widehat{U}_M^{(c)} = U_m^{(c)} * E_{M-m}, \quad \widehat{C}^{(c)}[k] = \frac{(U_m^{(c)} * E_{M-m})[k]}{\binom{M}{k}},$$

where $*$ is ordinary one-dimensional discrete convolution and out-of-range entries are zero. Thus elevation first binomially scales the coefficient row, convolves it with the fixed binomial kernel of degree $M - m$, and divides by the degree- M binomial weight. The new coefficients are fixed convex combinations of the old ones. Elevation therefore changes representation but adds no decision freedom. Constructing a new higher-degree **pdvar** does enlarge the decision parameterization.

For a tensor coefficient array of degree $\mathbf{m} = (m_1, \dots, m_\ell)$, elevate independently to any component-wise target $\mathbf{M} \geq \mathbf{m}$. With

$$U_{\mathbf{m}}^{(\mathbf{c})}[\mathbf{i}] = \left(\prod_{s=1}^{\ell} \binom{m_s}{i_s} \right) C^{(\mathbf{c})}[\mathbf{i}], \quad E_{\mathbf{M}-\mathbf{m}}[\mathbf{j}] = \prod_{s=1}^{\ell} \binom{M_s - m_s}{j_s},$$

the elevated array satisfies

$$\left(\prod_{s=1}^{\ell} \binom{M_s}{k_s} \right) \widehat{C}^{(\mathbf{c})}[\mathbf{k}] = (U_{\mathbf{m}}^{(\mathbf{c})} *_\ell E_{\mathbf{M}-\mathbf{m}})[\mathbf{k}],$$

where $*_\ell$ denotes ℓ -dimensional discrete convolution. This tensor convolution is deterministic and preserves the represented polynomial exactly. Public **elevate** and **elevVals** accept a scalar uniform shorthand or a direction-wise increment vector.

A.7 Physical Tensor Grids And Local Tensor Labels

Suppose direction s has the axis grid

$$\mathcal{G}_s = \{\rho_s^{(1)}, \dots, \rho_s^{(k_s)}\}, \quad \rho_s^{(1)} < \rho_s^{(2)} < \dots < \rho_s^{(k_s)}.$$

The counts k_s may differ. Thus $\mathcal{G} = \mathcal{G}_1 \times \dots \times \mathcal{G}_\ell$ has $\prod_s k_s$ physical nodes. A physical hypercube is identified separately by the one-based multi-index $\mathbf{c} = (c_1, \dots, c_\ell)$, with $\mathcal{H}_{\mathbf{c}} = \prod_s [\rho_s^{(c_s)}, \rho_s^{(c_s+1)}]$. Define its directional width and forward affine map by

$$\begin{aligned} h_s^{\mathbf{c}} &= \rho_s^{(c_s+1)} - \rho_s^{(c_s)}, \\ \phi_{\mathbf{c}} : [0, 1]^\ell &\longrightarrow \mathcal{H}_{\mathbf{c}}, \\ \phi_{\mathbf{c},s}(\alpha_s) &= \rho_s^{(c_s)} + h_s^{\mathbf{c}} \alpha_s, \\ \boldsymbol{\alpha} &= \phi_{\mathbf{c}}^{-1}(\boldsymbol{\rho}), \quad s = 1, \dots, \ell. \end{aligned}$$

For any cellwise quantity C , write $C^{(\mathbf{c})} = C \circ \phi_{\mathbf{c}}$ for its pullback to the unit box. For direction-wise degree $\mathbf{m} = (m_1, \dots, m_\ell)$,

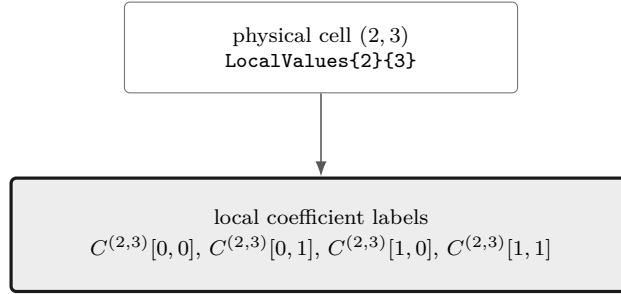
$$C^{(\mathbf{c})}(\boldsymbol{\alpha}) = \sum_{\mathbf{i} \in \prod_s \{0, \dots, m_s\}} B_{\mathbf{i}}^{\mathbf{m}}(\boldsymbol{\alpha}) C^{(\mathbf{c})}[\mathbf{i}], \quad B_{\mathbf{i}}^{\mathbf{m}} = \prod_{s=1}^{\ell} B_{i_s}^{m_s}(\alpha_s).$$

There are $\prod_s (k_s - 1)$ physical hypercubes and $\prod_s (m_s + 1)$ zero-based local Bernstein labels per hypercube. A zero component $m_s = 0$ leaves only label $i_s = 0$, so the polynomial is constant in direction s . Physical node and hypercube indices remain one-based. The familiar $(m + 1)^\ell$ count is the uniform-degree specialization $\mathbf{m} = m\mathbf{1}$.

```
obj = pdbase({[0 1 2], [10 20 30 40]}, [1 1], 1);
physicalCellCount = obj.ncell()
coefficientsPerCell = obj.ncoeff()
localLabels = obj.lbls()
```

```
physicalCellCount = 6
coefficientsPerCell = 4
localLabels = 4×2 double
    0    0
    0    1
    1    0
    1    1
```

The nested tree `LocalValues{c1}{c2}...{cell}` follows physical-cell coordinates. Only after reaching a leaf does the flat coefficient cell follow the `lbls` order. Earlier dimensions vary more slowly in both `cells` and `lbls`.



Select the physical-cell leaf first; only then read the flat local coefficient labels in `lbls` order.

A.8 Products As Ordered Weighted Convolution

Basic idea. Multiplying two Bernstein polynomials creates every ordered pair of input coefficients. Pairs with the same label sum contribute to the same output. For matrix payloads, “ordered” matters: the left factor stays on the left.

Formula. First consider one parameter and fix the one-based physical cell c . Let

$$P = \sum_{i=0}^m B_i^m P^{(c)}[i], \quad Q = \sum_{j=0}^n B_j^n Q^{(c)}[j].$$

The basis-product identity

$$B_i^m B_j^n = \frac{\binom{m}{i} \binom{n}{j}}{\binom{m+n}{i+j}} B_{i+j}^{m+n}$$

gives

$$(PQ)^{(c)}[k] = \sum_{i+j=k} P^{(c)}[i] Q^{(c)}[j] \frac{\binom{m}{i} \binom{n}{j}}{\binom{m+n}{k}}.$$

This is an ordered matrix convolution. In general $P^{(c)}[i] Q^{(c)}[j] \neq Q^{(c)}[j] P^{(c)}[i]$.

Equivalently, define the binomially scaled rows

$$\tilde{P}^{(c)}[i] = \binom{m}{i} P^{(c)}[i], \quad \tilde{Q}^{(c)}[j] = \binom{n}{j} Q^{(c)}[j], \quad \tilde{R}^{(c)}[k] = \binom{m+n}{k} (PQ)^{(c)}[k].$$

Then

$$\tilde{R}^{(c)} = \tilde{P}^{(c)} * \tilde{Q}^{(c)}.$$

The convolution remains ordered for matrix payloads: each summand is $\tilde{P}^{(c)}[i]\tilde{Q}^{(c)}[j]$, never the reversed product.

Worked example. Take quadratic coefficients $P = [1, 2, 6]$ and affine coefficients $Q = [2, 4]$. The anti-diagonals give

$$\begin{aligned} R[0] &= 1(2) = 2, \\ R[1] &= (1(4) + 2(2))/3 = 4, \\ R[2] &= (2(4) + 6(2))/3 = 28/3, \\ R[3] &= 6(4) = 24. \end{aligned}$$

```
P = pdmat({[0 1]}, {1, 2, 6}, Degree=2);
Q = pdmat({[0 1]}, {2, 4}, Degree=1);
R = P * Q;
productDegree = R.Degree
productCoefficients = cell2mat(R.coeffs(1))
```

```
productDegree = 3
productCoefficients = 1x4 double
    2.0000    4.0000    9.3333   24.0000
```

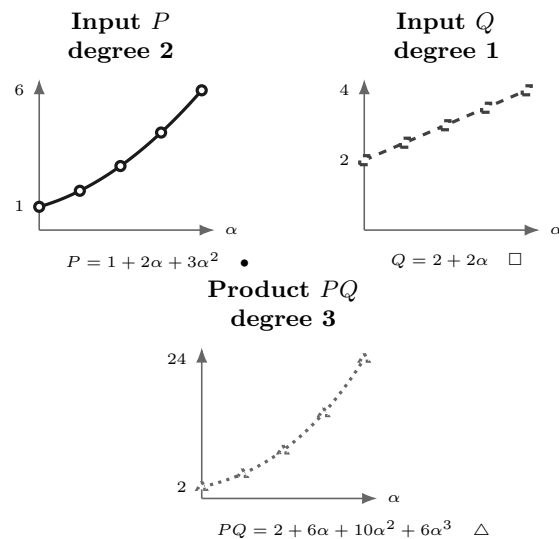
The product degree in the transcript is three: degrees add, so $2 + 1 = 3$. The four outputs are coefficients in that cubic Bernstein basis, not values sampled at four points.

The same calculation can be checked as a function identity. On $\alpha \in [0, 1]$,

$$P(\alpha) = 1 + 2\alpha + 3\alpha^2, \quad Q(\alpha) = 2 + 2\alpha,$$

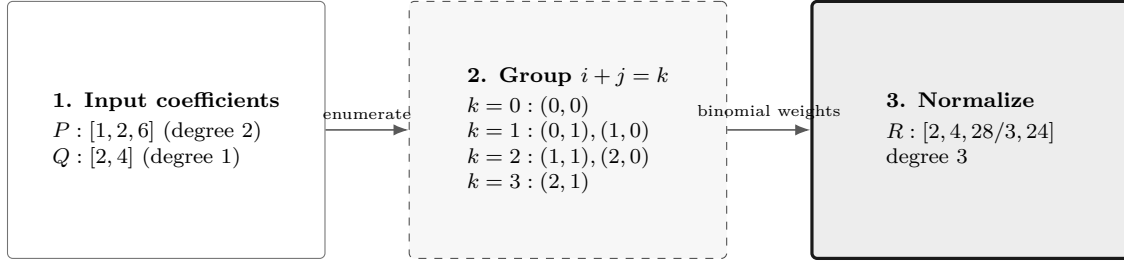
and

$$P(\alpha)Q(\alpha) = 2 + 6\alpha + 10\alpha^2 + 6\alpha^3.$$



Direct formula labels sit outside the plot regions. Solid/circle, dashed/square, and dotted/triangle treatments distinguish the three functions without relying on color.

Visual conclusion.



Only the panel-to-panel arrows encode flow. Contributions are listed inside their anti-diagonal group, so no edges cross labels or equations.

This is the algorithm behind protected `bernProd` (section 3.9.5) and public `pdmat`/`pdvar` multiplication (sections 4.8.2 and 5.7).

For tensor degrees $\mathbf{m}$ and $\mathbf{n}$, labels add componentwise and the product has degree $\mathbf{m} + \mathbf{n}$:

$$(PQ)^{(c)}[\mathbf{k}] = \sum_{\mathbf{i}+\mathbf{j}=\mathbf{k}} P^{(c)}[\mathbf{i}]Q^{(c)}[\mathbf{j}] \prod_{s=1}^{\ell} \frac{\binom{m_s}{i_s} \binom{n_s}{j_s}}{\binom{m_s+n_s}{k_s}}.$$

In binomially scaled form, the tensor rule is the ℓ -dimensional discrete convolution

$$\left(\prod_{s=1}^{\ell} \binom{m_s+n_s}{k_s} \right) (PQ)^{(c)}[\mathbf{k}] = \left(\left\{ \left(\prod_{s=1}^{\ell} \binom{m_s}{i_s} \right) P^{(c)}[\mathbf{i}] \right\}_{\mathbf{i}} *_{\ell} \left\{ \left(\prod_{s=1}^{\ell} \binom{n_s}{j_s} \right) Q^{(c)}[\mathbf{j}] \right\}_{\mathbf{j}} \right) [\mathbf{k}].$$

Labels add componentwise; physical-cell identity does not change, and every matrix product retains the written left-to-right factor order.

A.9 Differentiation And Rate Vertices

Basic idea. Differentiate inside each physical cell. Adjacent control differences produce the derivative coefficients; multiplying by a bounded rate then creates one stored row per rate-box vertex. Value continuity of P does not force the cell-local derivatives to match across a face.

Formula. For one parameter, on physical cell c of width h_1^c ,

$$\frac{\partial P}{\partial \rho} = \frac{m}{h_1^c} \sum_{i=0}^{m-1} (C^{(c)}[i+1] - C^{(c)}[i]) B_i^{m-1}(\alpha).$$

Worked example. For the worked polynomial $[1, 2, 6]$ on $[0, 2]$, $m = 2$ and $h = 2$, so the derivative coefficients are

$$\left[\frac{2}{2}(2-1), \frac{2}{2}(6-2) \right] = [1, 4].$$

This represents $dp/d\rho = 1 + 3\alpha$.

In ℓ dimensions with tensor degree $\mathbf{m}$, differentiation with respect to ρ_s replaces each coefficient by

$$\frac{m_s}{h_s^c} (C^{(c)}[\mathbf{i} + \mathbf{e}_s] - C^{(c)}[\mathbf{i}]),$$

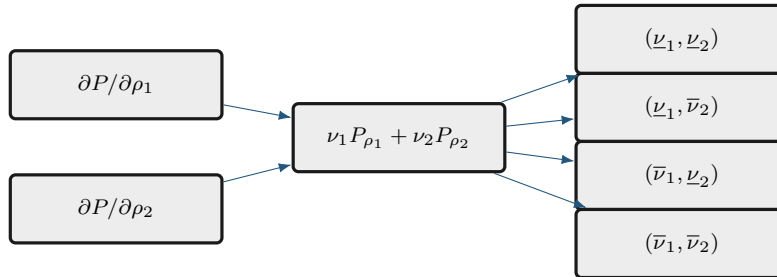
when $m_s > 0$, reduces only that component to $m_s - 1$, and leaves the other label components unchanged. A zero-degree axis has identically zero partial derivative. For two or more parameters, the implementation elevates every nonzero partial exactly from $\mathbf{m} - \mathbf{e}_s$ back to the common tensor degree $\mathbf{m}$ before adding them; in one parameter it retains the established reduced-degree result. If every component is zero, `rhodiff` stores a complete zero coefficient row for every rate vertex.

Finally,

$$\dot{P} = \sum_{s=1}^{\ell} \nu_s \frac{\partial P}{\partial \rho_s}, \quad \boldsymbol{\nu} = \dot{\boldsymbol{\rho}}.$$

This matrix expression is affine in $\boldsymbol{\nu}$. Every value inside the rate box $\mathcal{R}$ is therefore a convex combination of the 2^ℓ vertex matrices in $\mathcal{V}_{\mathcal{R}}$. Because the positive- and negative-semidefinite cones are convex, the requested semidefinite sign holds on all of $\mathcal{R}$ whenever it holds at every vertex. `rhodiff` stores one row for each vertex.

Visual conclusion.



A.9.1 Rate-Vertex Row Storage

Let $\text{RateBounds}(s, :) = [\underline{\nu}_s \ \bar{\nu}_s]$. The rate box $\mathcal{R} = \prod_s [\underline{\nu}_s, \bar{\nu}_s]$ has the finite vertex set

$$\mathcal{V}_{\mathcal{R}} = \prod_{s=1}^{\ell} \{\underline{\nu}_s, \bar{\nu}_s\}, \quad |\mathcal{V}_{\mathcal{R}}| = 2^\ell.$$

For a derivative object $D = \text{rhodiff}(P, \text{RateBounds})$ and one physical hypercube $\mathbf{c} = (c_1, \dots, c_\ell)$, the leaf $\mathbf{D}.\text{LocalValues}\{c_1\} \cdots \{c_\ell\}$ is a 2^ℓ -by- $\prod_s (\mathbf{D}.\text{Degree}(s) + 1)$ cell array. Its rows enumerate vertices $\boldsymbol{\nu}^{(q)} \in \mathcal{V}_{\mathcal{R}}$; its columns enumerate local Bernstein labels in `combRows` order. Each entry already contains $\sum_s \nu_s^{(q)} \partial P^{(\mathbf{c})} / \partial \rho_s$. The rows add rate alternatives inside one existing hypercube; they do not add physical cells.

For $\ell = 2$, the lower/upper tensor order is

row	ν_1	ν_2
1	$\underline{\nu}_1$	$\underline{\nu}_2$
2	$\underline{\nu}_1$	$\bar{\nu}_2$
3	$\bar{\nu}_1$	$\underline{\nu}_2$
4	$\bar{\nu}_1$	$\bar{\nu}_2$

The last parameter direction changes fastest. This is the same tensor ordering used by the package's coefficient and cell traversal, so `pdlmi` can constrain every rate row without reconstructing the rate box.

A.10 de Casteljau Evaluation And Exact Subdivision

de Casteljau recursion evaluates a Bernstein polynomial using only affine combinations. For one parameter, on a fixed physical cell c , begin with $C^{(c,0)}[i] = C^{(c)}[i]$ and define

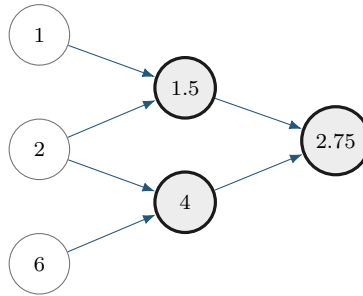
$$C^{(c,r)}[i] = (1 - \tau)C^{(c,r-1)}[i] + \tau C^{(c,r-1)}[i + 1], \quad r = 1, \dots, m.$$

The value at $\alpha = \tau$ is $C^{(c,m)}[0]$.

For $[1, 2, 6]$ and $\tau = 1/2$, the recursion is

$$[1, 2, 6] \longrightarrow [1.5, 4] \longrightarrow [2.75].$$

The left edge of this triangle gives exact coefficients $[1, 1.5, 2.75]$ on the left subinterval; the right edge gives $[2.75, 4, 6]$ on the right.



One recursion triangle supplies both the evaluation and the two child-cell coefficient lists.

Tensor evaluation or subdivision applies the same recursion along one parameter direction at a time. Subdivision changes the physical-cell partition and can tighten coefficient hulls while preserving the polynomial. The package uses exact coefficient transformations internally where documented, but it does not currently expose a public de Casteljau, subdivision, or adaptive-refinement API.

A.11 What Changes, And What Does Not

The following distinctions prevent several common modeling mistakes.

Operation	Changes	Preserves
Basis conversion	coefficient coordinates	polynomial, degree, physical cell
Degree elevation	Bernstein degree and coefficient list	polynomial, physical grid, decision freedom
Grid subdivision	physical-cell partition and local coefficients	polynomial on the original domain
Higher-degree <code>pdvar</code>	number of decision coefficients	grid and matrix size
Multiplication	polynomial degree and coefficients	physical-cell identity
Differentiation	degree and coefficient values	physical-cell partition before rate rows

These representation facts support the finite certificates in the next appendix. They do not by themselves select a solver or prove that one certificate is necessary.

A.12 Further Reading

For a concise orientation before the primary paper by Farouki [3], see the [Bernstein polynomial overview](#) and the [de Casteljau overview](#). They are background references for the basis and recursion; the package's cell-local storage and certificate behavior are defined by this manual and the source implementation.

Appendix B

Polynomial-Matrix Certificates on Hypercubes

This appendix starts with positive semidefinite matrices, builds SOS and Gram representations, and then explains the five certificate paths implemented by `pdlmi`: Direct, Pólya, Putinar, SparseFullBox, and FullBox. Background constructions are included so a new reader can recognize the mathematical landscape; only explicitly marked methods are package APIs.

As in the preceding appendix, each major certificate is revealed as *Basic idea* $\rightarrow$ *Formula* $\rightarrow$ *Worked example* $\rightarrow$ *Visual conclusion*. Keep two questions separate while reading: what polynomial matrix is represented, and what finite cone is being used to certify its sign?

Fix one physical cell $\mathcal{H}_c$ and one stored rate row. Write the theoretical cell residual as $\mathcal{F}^{(c)} \prec 0$ and define the sign-normalized positive target $S^{(c)} = -\mathcal{F}^{(c)}$. The theoretical problem is

$$S^{(c)}(\boldsymbol{\alpha}) \succ 0 \quad \text{for every } \boldsymbol{\alpha} \in [0, 1]^\ell.$$

The package's Direct and Gram constraints prove non-strict inequalities. A strict model must state its own margin, for example $S^{(c)} - \varepsilon I \succeq 0$. The package adds no hidden margin.

B.1 Positive Semidefinite Matrices And LMIs

Basic idea. A real symmetric matrix Q is positive semidefinite when

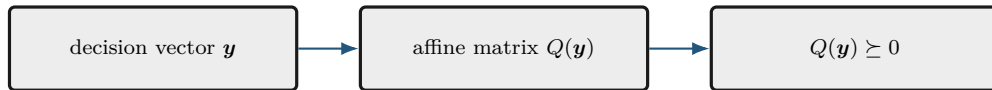
$$u^\top Q u \geq 0 \quad \text{for every vector } u.$$

Formula. Let $\mathbf{y} = (y_1, \dots, y_N)$ collect the scalar decision coordinates. An LMI is an affine matrix expression constrained to the PSD cone:

$$Q(\mathbf{y}) = Q_0 + \sum_{k=1}^N y_k Q_k \succeq 0.$$

The unknowns enter affinely, so this is a semidefinite program. A polynomial matrix sign condition appears infinite because it asks for one PSD matrix at every point. A certificate replaces those pointwise conditions by finitely many PSD matrices and affine coefficient identities.

Visual conclusion.



B.2 SOS And Gram Matrices

Basic idea. A polynomial in $\xi = (\xi_1, \dots, \xi_\ell)$ is a finite sum

$$p(\xi) = \sum_{\eta \in \mathbb{N}^\ell} p_\eta \xi^\eta, \quad \xi^\eta = \prod_{s=1}^{\ell} \xi_s^{\eta_s},$$

with only finitely many nonzero coefficients. Its total degree is $\deg p = \max\{|\eta| : p_\eta \neq 0\}$, where $|\eta| = \sum_s \eta_s$. A scalar polynomial is a sum of squares (SOS) if $p = q_1^2 + \dots + q_t^2$, which immediately implies $p \geq 0$.

Formula. Suppose $\deg p \leq 2d$. Collect every monomial of total degree at most d into

$$v_d(\xi) = (\xi^\beta)_{|\beta| \leq d} \in \mathbb{R}^{q_d}, \quad q_d = \binom{\ell + d}{d}.$$

The scalar Gram form is

$$p(\xi) = v_d(\xi)^\top Q v_d(\xi), \quad Q \in \mathbb{S}^{q_d}, \quad Q \succeq 0.$$

A factorization $Q = L^\top L$ yields $p = \|Lv_d\|_2^2$, so the Gram form is SOS. Expanding it gives the coefficient identities

$$p_\eta = \sum_{\substack{|\beta| \leq d, |\gamma| \leq d \\ \beta + \gamma = \eta}} Q_{\beta, \gamma} \quad \text{for every } \eta.$$

These identities are affine in Q . A Gram matrix generally is not unique because the same monomial ξ^η can arise from several pairs (β, γ) ; the certificate searches for any PSD Q satisfying all coefficient identities.

For a symmetric n -by- n polynomial matrix, lift the monomial basis to

$$Z_d(\xi) = v_d(\xi) \otimes I_n \in \mathbb{R}^{q_d n \times n}, \quad S^{(c)}(\xi) = Z_d(\xi)^\top Q Z_d(\xi), \quad Q \in \mathbb{S}^{q_d n}, \quad Q \succeq 0.$$

Then $u^\top S^{(c)}(\xi) u = (Z_d(\xi) u)^\top Q (Z_d(\xi) u) \geq 0$ for every u , so $S^{(c)}(\xi) \succeq 0$. Matrix-SOS relaxations are developed in [11].

The package performs the analogous matching in tensor-product Bernstein coordinates. For a tensor degree $\mathbf{a} = (a_1, \dots, a_\ell) \in \mathbb{N}^\ell$, define

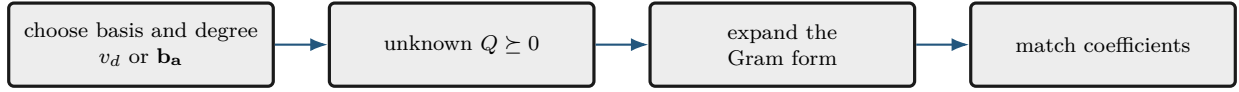
$$\mathbf{b}_\mathbf{a}(\alpha) = \bigotimes_{s=1}^{\ell} \begin{bmatrix} B_0^{a_s}(\alpha_s) & \dots & B_{a_s}^{a_s}(\alpha_s) \end{bmatrix}^\top, \quad Z_\mathbf{a}(\alpha) = \mathbf{b}_\mathbf{a}(\alpha) \otimes I_n.$$

The vector $\mathbf{b}_\mathbf{a}$ has $\prod_s (a_s + 1)$ entries. Therefore $Z_\mathbf{a}$ has $\prod_s (a_s + 1)$ block rows of size n -by- n , equivalently $n \prod_s (a_s + 1)$ scalar rows, and Q has scalar dimension $n \prod_s (a_s + 1)$. The unweighted form $Z_\mathbf{a}^\top Q Z_\mathbf{a}$ has coordinatewise degree $2\mathbf{a}$. Multiplication by

$$g_J(\alpha) = \prod_{s \in J} \alpha_s (1 - \alpha_s)$$

adds $2\chi_J$ to that coordinatewise degree. Consequently, a box-weighted term matched at direction-wise degree $2\mathbf{r}$ uses $\mathbf{a} = \mathbf{r} - \chi_J$. The uniform choice $\mathbf{r} = r\mathbf{1}$ is a specialization. In the odd one-dimensional interval form, the endpoint weights α and $1 - \alpha$ instead add degree one. Expanding each weighted Bernstein Gram form and equating it to the target Bernstein array produces affine coefficient identities, just as in the monomial example.

Visual conclusion.



B.3 Full-Space SOS

An unweighted monomial Gram form proves $S^{(\mathbf{c})}(\boldsymbol{\xi}) \succeq 0$ on all of $\mathbb{R}^\ell$. That can be much stronger than positivity only on a box. For example, $\alpha(1 - \alpha)$ is nonnegative on $[0, 1]$ but negative outside it. The package does not expose a general full-space SOS parser; this construction is background for understanding weighted certificates.

B.4 Direct Bernstein Coefficients

Basic idea. If every local coefficient matrix has the requested sign, every convex combination of those coefficients has the same sign. This yields a small, transparent sufficient certificate.

Formula. After any rate rows have been stacked, write the residual and target coefficients on physical cell $\mathbf{c}$ as $\mathcal{F}^{(\mathbf{c})} = \sum_{\mathbf{i}} B_{\mathbf{i}}^{\mathbf{M}} \mathcal{C}^{(\mathbf{c})}[\mathbf{i}]$ and $C^{(\mathbf{c})}[\mathbf{i}] = -\mathcal{C}^{(\mathbf{c})}[\mathbf{i}]$, where $\mathbf{i} \in \prod_s \{0, \dots, M_s\}$. Then

$$S^{(\mathbf{c})}(\boldsymbol{\alpha}) = \sum_{\mathbf{i}} B_{\mathbf{i}}^{\mathbf{M}}(\boldsymbol{\alpha}) C^{(\mathbf{c})}[\mathbf{i}].$$

Because the basis weights are nonnegative and sum to one,

$$C^{(\mathbf{c})}[\mathbf{i}] \succeq 0 \ \forall \mathbf{i} \implies S^{(\mathbf{c})}(\boldsymbol{\alpha}) \succeq 0.$$

This is the package default. It introduces no Gram variables and creates one constraint per physical cell, rate row, and local label. The converse is false.

Worked example. Consider

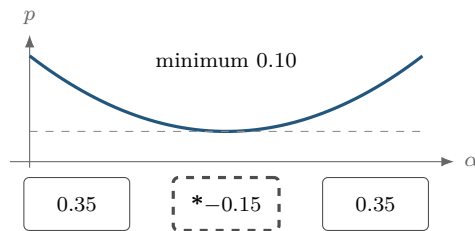
$$p(\alpha) = \alpha^2 - \alpha + 0.35 = (\alpha - 0.5)^2 + 0.10.$$

It is strictly positive, with minimum 0.10, but its degree-two Bernstein coefficients are

$$[0.35, -0.15, 0.35].$$

The negative middle coefficient defeats the direct certificate without making the polynomial negative.

Visual conclusion.



One negative Bernstein coefficient does not imply a negative curve.

The same polynomial has the PSD Bernstein Gram representation

$$\mathbf{b}_1(\alpha) = \begin{bmatrix} 1 - \alpha \\ \alpha \end{bmatrix}, \quad Q = \begin{bmatrix} 0.35 & -0.15 \\ -0.15 & 0.35 \end{bmatrix} \succeq 0, \quad p = \mathbf{b}_1^\top Q \mathbf{b}_1.$$

The eigenvalues of Q are 0.20 and 0.50. A complete Gram certificate can therefore succeed when coefficient-wise signs fail.

B.5 Pólya As Lossless Elevation

Basic idea. Pólya selection does not change the residual polynomial. It rewrites that same polynomial at a higher Bernstein degree and then retries the direct coefficient test.

Formula. Classical Pólya theory concerns homogeneous polynomials strictly positive on a simplex [14]. For one box coordinate,

$$\lambda_0 = 1 - \alpha, \quad \lambda_1 = \alpha, \quad \lambda_0 + \lambda_1 = 1.$$

For ℓ coordinates, the implemented multiplier is

$$\prod_{s=1}^{\ell} (\lambda_{s,0} + \lambda_{s,1})^d = 1.$$

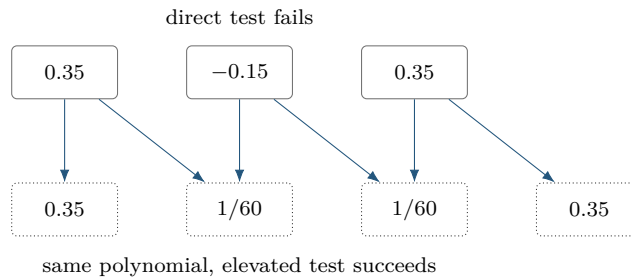
Regrouping is exactly Bernstein degree elevation: the represented residual does not change.

Worked example. For the worked coefficients,

$$[0.35, -0.15, 0.35] \longrightarrow \left[0.35, \frac{1}{60}, \frac{1}{60}, 0.35 \right]$$

after one elevation. Every new coefficient is positive, so the elevated coefficient test succeeds.

Visual conclusion.



`applyPolya(d)` accepts scalar uniform shorthand or a direction-wise increment $\mathbf{d}$, elevates from $\mathbf{M}$ to $\mathbf{M} + \mathbf{d}$, and then applies coefficient constraints to every cell and rate row. It adds neither Gram variables nor decision freedom. A fixed-level failure is inconclusive; the package makes no tensor-box completeness claim.

B.6 Exact Interval Markov–Lukács Forms

Basic idea. On one interval, endpoint factors can encode the domain exactly. Even and odd target degrees require different weighted Gram patterns, so parity is a structural choice rather than a cosmetic one.

Formula. In one variable, interval nonnegativity has exact parity-specific descriptions [8, 9]. If $\deg p \leq 2r$,

$$p \geq 0 \text{ on } [0, 1] \iff p = s_0 + \alpha(1 - \alpha)s_1,$$

where s_0, s_1 are SOS of degrees at most $2r$ and $2r - 2$. If $\deg p \leq 2r + 1$,

$$p \geq 0 \text{ on } [0, 1] \iff p = (1 - \alpha)s_L + \alpha s_U,$$

where s_L, s_U are SOS of degree at most $2r$.

For polynomial matrices, with

$$Z_d(\alpha) = \begin{bmatrix} B_0^d I_n & \cdots & B_d^d I_n \end{bmatrix}^\top,$$

the even and odd forms are

$$S^{(e)} = Z_r^\top Q_0 Z_r + \alpha(1 - \alpha)Z_{r-1}^\top Q_1 Z_{r-1}, \quad Q_0, Q_1 \succeq 0,$$

and

$$S^{(o)} = (1 - \alpha)Z_r^\top Q_L Z_r + \alpha Z_r^\top Q_U Z_r, \quad Q_L, Q_U \succeq 0.$$

The second even block is absent at $r = 0$. See [10, 11] for polynomial-matrix context.

Worked example. A cubic residual has degree $2r + 1$ with $r = 1$. It therefore uses two degree-one Gram bases: one multiplied by $1 - \alpha$ and one by α . This is the generic one-dimensional odd-degree full-box pattern.

Visual conclusion.

even $2r$ unweighted Gram + $\alpha(1 - \alpha)$ -weighted Gram	odd $2r + 1$ $(1 - \alpha)$ lower-face Gram + α upper-face Gram
--	---

Partition Q into n -by- n blocks Q_{ij} . The Bernstein coefficient at label k of $\alpha^a(1 - \alpha)^b Z_d^\top Q Z_d$, expressed at total degree $2d + a + b$, is

$$\mathcal{G}_k^{d,a,b}(Q) = \frac{1}{\binom{2d+a+b}{k}} \sum_{\substack{0 \leq i,j \leq d \\ i+j=k-a}} \binom{d}{i} \binom{d}{j} Q_{ij}.$$

Equating this affine map to each target coefficient gives exact linear identities; $Q \succeq 0$ supplies the LMIs. For one parameter, `applyFullBoxPreorder` implements these exact parity forms, while `applySparseFullBoxPreorder` uses the same parity terms and identities with symmetric block-band Gram matrices $Q_j^{(w)}$ at intermediate bandwidths. Their minimum absolute order is $\lfloor M/2 \rfloor$, where M is the assembled residual degree.

B.7 Putinar Modules And Schmüdgen Preorderings

For

$$K = \{\xi : g_1(\xi) \geq 0, \dots, g_q(\xi) \geq 0\},$$

a Putinar quadratic module has the form

$$\sigma_0 + \sum_{j=1}^q \sigma_j g_j, \quad \sigma_j \text{ SOS.}$$

Under the Archimedean hypothesis, every polynomial strictly positive on K has such a representation at some degree [12]. Compactness alone is not that theorem statement. The package implements one box-specific, fixed-order selector, not a general-domain or general-generator parser.

For one parameter, `applyPutinar(r)` dispatches to the same exact even/odd Markov–Lukács forms as `applyFullBoxPreorder(r)`. Its absolute default and minimum are $r = \lfloor M/2 \rfloor$. Even targets use the unweighted and $\alpha(1 - \alpha)$ -weighted Gram blocks and match at degree $2r$; odd targets use the two endpoint-weighted Gram blocks and match at degree $2r + 1$.

For $\ell \geq 2$, use the local box generators

$$g_s(\boldsymbol{\alpha}) = \alpha_s(1 - \alpha_s),$$

and `applyPutinar(r)` uses

$$S^{(c)} = S_0 + \sum_{s=1}^{\ell} g_s S_s, \quad S_0 = Z_{\mathbf{r}}^T Q_0 Z_{\mathbf{r}}, \quad S_s = Z_{\mathbf{r}-\mathbf{e}_s}^T Q_s Z_{\mathbf{r}-\mathbf{e}_s},$$

where every present $Q_s \succeq 0$. In this multivariate branch the direction-wise order $\mathbf{r}$ is absolute and satisfies $\mathbf{r} \geq \lceil \mathbf{M}/2 \rceil$ componentwise. Exact identities match the sign-normalized target, elevated without changing it, at tensor degree $2\mathbf{r}$. A multiplier block whose basis degree has a negative component is omitted. Each physical cell and stored rate row owns independent Gram and equality blocks; no generator products, implicit margin, or solver call are introduced.

A Schmüdgen full preordering includes every square-free generator product:

$$p = \sum_{\beta \in \{0,1\}^q} \sigma_{\beta} \prod_{j=1}^q g_j^{\beta_j}, \quad \sigma_{\beta} \text{ SOS.}$$

Strict positivity on a compact basic closed semialgebraic set has such a representation [13]. A finite truncation is still a sufficient certificate, and up to 2^q multiplier blocks may be required.

B.8 The Implemented Multivariate FullBox State

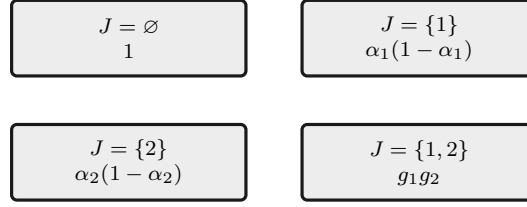
Basic idea. The mathematical construction is a full-box preordering: for several parameters, use every square-free product of the box generators. Each subset selects one weighted Gram block, and exact coefficient identities make the sum equal the sign-normalized target $S^{(c)}$ on one physical cell and stored rate row. The software state is `FullBox`.

Formula. For the hypercube, use

$$g_s(\boldsymbol{\alpha}) = \alpha_s(1 - \alpha_s), \quad s = 1, \dots, \ell.$$

Worked example. For two parameters, the four subset products are $1, g_1, g_2, g_1g_2$.

Visual conclusion.



For $\ell \geq 2$ and direction-wise absolute order $\mathbf{r}$, use the already defined tensor-product Bernstein lift $Z_{\mathbf{a}} = \mathbf{b}_{\mathbf{a}} \otimes I_n$. The implemented form is

$$S^{(\mathbf{c})} = \sum_{J \subseteq \{1, \dots, \ell\}} \left(\prod_{s \in J} g_s \right) Z_{\mathbf{r} - \chi_J}^T Q_J Z_{\mathbf{r} - \chi_J}, \quad Q_J \succeq 0.$$

Terms with negative tensor degree are omitted. A present block has dimension

$$n \prod_{s=1}^{\ell} (r_s + 1 - \mathbf{1}_{s \in J}).$$

Every physical cell and stored rate row gets independent dense Gram blocks and exact coefficient identities. The minimum order is componentwise $\lceil \mathbf{M}/2 \rceil$ for $\ell \geq 2$. This is a specific dense Bernstein realization of the full-box preordering, not a full-space SOS, Putinar-only, sparse, or general-domain parser. A fixed multivariate order is sufficient rather than exact.

Finite SDP example: $\ell = 2$, $\mathbf{M} = (3, 3)$. Let $(\alpha_1, \alpha_2) \in [0, 1]^2$ be the local coordinates of one cell, and suppose the sign-normalized target is

$$S^{(\mathbf{c})}(\boldsymbol{\alpha}) = \sum_{i_1=0}^3 \sum_{i_2=0}^3 B_{i_1}^3(\alpha_1) B_{i_2}^3(\alpha_2) C^{(\mathbf{c})}[i_1, i_2] \succeq 0.$$

This is the uniform specialization $\mathbf{r} = (2, 2) = \lceil \mathbf{M}/2 \rceil$. After exact elevation to degree $(4, 4)$, the implementation introduces four independent PSD Gram matrices and enforces the identity

$$\begin{aligned} S^{(\mathbf{c})}(\boldsymbol{\alpha}) &= Z_{(2,2)}^T Q_{\emptyset} Z_{(2,2)} + \alpha_1(1 - \alpha_1) Z_{(1,2)}^T Q_1 Z_{(1,2)} \\ &\quad + \alpha_2(1 - \alpha_2) Z_{(2,1)}^T Q_2 Z_{(2,1)} \\ &\quad + \alpha_1(1 - \alpha_1) \alpha_2(1 - \alpha_2) Z_{(1,1)}^T Q_{12} Z_{(1,1)}, \\ Q_{\emptyset} &\succeq 0, \quad Q_1 \succeq 0, \quad Q_2 \succeq 0, \quad Q_{12} \succeq 0. \end{aligned}$$

For an n -by- n residual, these blocks have dimensions $9n$, $6n$, $6n$, and $4n$. If

$$S^{(\mathbf{c})}(\boldsymbol{\alpha}) = \sum_{i_1=0}^4 \sum_{i_2=0}^4 B_{i_1}^4(\alpha_1) B_{i_2}^4(\alpha_2) \tilde{C}^{(\mathbf{c})}[i_1, i_2],$$

then $\tilde{C}^{(\mathbf{c})}[0, 0] = C^{(\mathbf{c})}[0, 0]$ and $\tilde{C}^{(\mathbf{c})}[1, 0] = C^{(\mathbf{c})}[0, 0]/4 + 3C^{(\mathbf{c})}[1, 0]/4$. The finite SDP has the four PSD-LMI constraints above and one exact matrix equality for each of the $5 \times 5 = 25$ elevated

Bernstein labels. With $[Q]_{\mathbf{i},\mathbf{j}}$ denoting the n -by- n Gram block indexed by labels $\mathbf{i}, \mathbf{j}$, two illustrative equalities are

$$C^{(\mathbf{c})}[0, 0] = [Q_{\emptyset}]_{(0,0),(0,0)},$$

$$\frac{1}{4}C^{(\mathbf{c})}[0, 0] + \frac{3}{4}C^{(\mathbf{c})}[1, 0] = \frac{1}{2}[Q_{\emptyset}]_{(0,0),(1,0)} + \frac{1}{2}[Q_{\emptyset}]_{(1,0),(0,0)} + \frac{1}{4}[Q_1]_{(0,0),(0,0)}.$$

The remaining 23 are generated by the same weighted Bernstein coefficient map. Thus the identities are affine equalities, not 25 additional LMIs. A $\preceq 0$ comparison instead matches the negatives of the elevated target coefficients.

```
yalmp('clear')
P = pdvar(1, {[0 1]}, Degree=2);
direct = P >= 0;
polya = direct.applyPolya(1);
putinar = direct.applyPutinar();
box = direct.applyFullBoxPreorder();
directConstraintCount = numel(direct.Constraints)
polyaState = [polya.PolyaDegree, numel(polya.Constraints)]
putinarState = [putinar.PutinarOrder, numel(putinar.Constraints)]
fullBoxState = [box.FullBoxOrder, numel(box.Constraints)]
```

```
directConstraintCount = 3
polyaState = 1×2 double
    1     4
putinarState = 1×2 double
    1     5
fullBoxState = 1×2 double
    1     5
```

Selections rebuild from the stored original **Residual** and **Relation**. Reapplying Pólya, Putinar, SparseFullBox, or FullBox replaces the previous selection; the certificates never compound, and the four opt-in families are mutually exclusive.

B.9 The Implemented SparseFullBox Band-Matrix State

Basic idea. SparseFullBox retains every FullBox parity or generator-mask term and every exact target-coefficient identity, but replaces each dense Gram matrix Q_J by a symmetric block-band Gram matrix $Q_J^{(w)}$ of bandwidth w . Only the admissible Gram entries change.

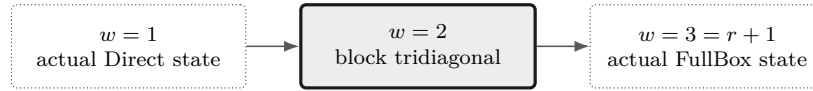
Formula. For $\ell \geq 2$, use the paper notation $\mathbf{a}_J = \mathbf{r} - \chi_J$ with the FullBox generators:

$$S^{(\mathbf{c})} = \sum_{J \subseteq \{1, \dots, \ell\}} \left(\prod_{s \in J} g_s \right) (\mathbf{b}_{\mathbf{a}_J}(\boldsymbol{\alpha}) \otimes I_n)^\top Q_J^{(w)} (\mathbf{b}_{\mathbf{a}_J}(\boldsymbol{\alpha}) \otimes I_n), \quad Q_J^{(w)} \succeq 0.$$

Each $Q_J^{(w)}$ is symmetric block-band with bandwidth w ; terms with negative tensor degree are omitted. The one-parameter parity forms make the same replacement $Q_J \mapsto Q_J^{(w)}$. Every coefficient of the sign-normalized, degree-elevated target is still matched exactly.

Worked example. For one parameter, even assembled residual degree $M = 4$, and absolute order $r = 2$, $w = 1$ makes each $Q_J^{(1)}$ block diagonal and is canonicalized to Direct. At $w = 2$, each $Q_J^{(2)}$ is block tridiagonal and the result retains SparseFullBox state. At $w = 3$, each $Q_J^{(3)}$ is block pentadiagonal; because $w = r + 1$, the implementation canonicalizes this endpoint to FullBox.

Visual conclusion.

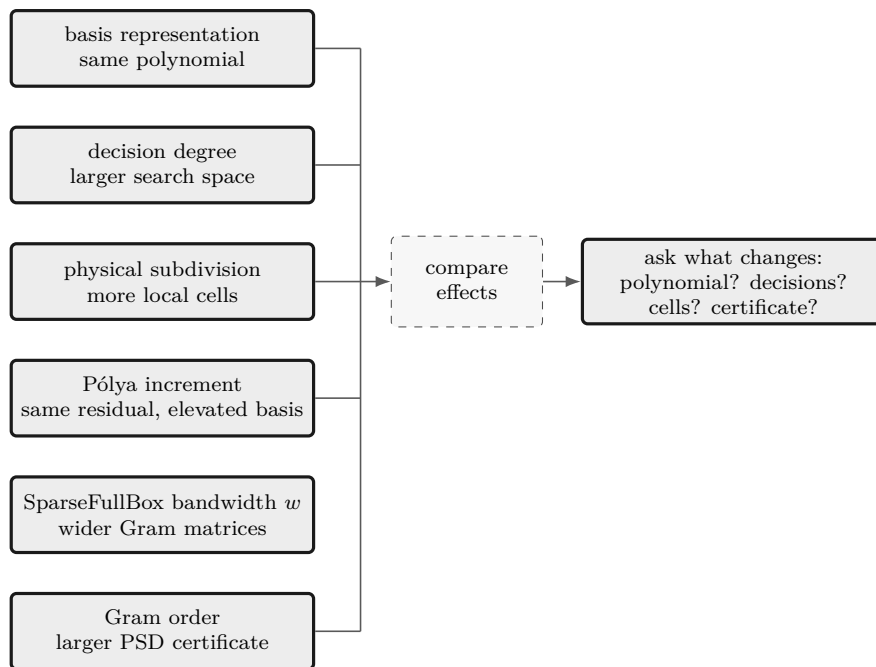


Increasing the Gram-matrix bandwidth gives nested sufficient certificates; a wider cone need not yield a strict improvement on a particular residual.

`applySparseFullBoxPreorder(w,r)` uses one scalar bandwidth w , default two, and the same dimension-dependent absolute-order minimum as FullBox: $\lfloor M/2 \rfloor$ in one parameter and componentwise $\lceil \mathbf{M}/2 \rceil$ otherwise. The order accepts scalar shorthand or a direction-wise row $\mathbf{r}$. Width one canonicalizes to actual Direct state after order validation. Width at least $\max_s(r_s + 1)$ canonicalizes to actual FullBox state. Intermediate widths retain SparseFullBox state. Every physical cell, stored rate row, and entry-wise column-major matrix entry owns an independent certificate; the method adds no hidden margin and makes no solver call.

B.10 Six Different Modeling Choices

These choices act on different parts of the problem.



Choice	Changes	Does not mean
basis conversion	coefficient coordinates	more decision freedom
decision degree	independent decision coefficients	certificate order
subdivision	physical cells and local hulls	higher polynomial degree
Pólya increment d	elevated residual coefficients	Gram variables
Putinar/full-box order	Gram bases and PSD block sizes	decision degree
SparseFullBox bandwidth w	admissible entries of the block-band Gram matrices	decision degree or physical grid

When a certificate fails, do not infer continuous infeasibility. Decide separately whether the model needs a richer direction-wise decision degree $\mathbf{m}$, a finer physical grid, a larger direction-wise Pólya increment $\mathbf{d}$, a larger scalar SparseFullBox bandwidth w , or a higher direction-wise absolute Gram order $\mathbf{r}$.

B.11 Implemented Boundary

Implemented public behavior	Background only
direct coefficient assembly	general full-space SOS parser
<code>applyPolya([d])</code> with scalar or direction-wise increment	automatic Pólya search
<code>applyPutinar([r])</code> with scalar or direction-wise absolute order	general-domain/general-generator Putinar interface
<code>applySparseFullBoxPreorder([w],[r])</code> with scalar w and scalar or direction-wise r	automatic or adaptive sparse-pattern selection
<code>applyFullBoxPreorder([r])</code> with scalar or direction-wise absolute order	general-domain Schmüdgen parser
exact one-dimensional parity forms	automatic or adaptive Gram-order search
all cells, active rate rows, and column-major entries	cross-cell/rate/entry Gram sharing
explicit user margins	package-owned strictness or solver policy

B.12 Further Reading

The primary papers by Putinar [12] and Schmüdgen [13] remain the mathematical authority for their respective theorems. For implementation-oriented background, YALMIP maintains an [SOS tutorial](#), and the [Positivstellensatz overview](#) gives a compact map of the surrounding result family. Neither link changes the fixed-order, box-specific package boundary described above.

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